

CoCURVE: CROSS-MODULE CO-PRUNING CURVATURE FOR STRUCTURED LLM PRUNING

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ABSTRACT

Structured pruning removes whole attention heads and feed-forward channel groups, but training-free methods rank those units independently, as if the damage from removing a set were the sum of the damage from removing each one. Transformer sublayers are coupled through a shared residual stream, so two units that look individually weak can be jointly indispensable, and independent scoring deletes them together. We introduce CoCurve (Cross-Module Co-Pruning Curvature), which prunes attention and FFN units under one shared budget. Expanding the token-level KL between the frozen model and its masked copy to second order yields a single Fisher matrix whose diagonal is classical node saliency and whose off-diagonal is the *co-pruning curvature* between units. We estimate that matrix as a Gram product of single-unit ablation features, so it follows from M ablations rather than $O(M^2)$ pairwise sweeps — no gradients, no weight update — and pruning becomes one budgeted quadratic program solved in a single shot. How far to trust the off-diagonal is measured rather than tuned: λ indexes a finite family of prunings whose breakpoints we enumerate exactly, and we keep the member with the lowest held-out calibration risk, using no labels and no benchmarks. Across five models from 3B to 24B at three ratios, CoCurve leads on 38 of 45 perplexity cells against a row holding, in each cell, whichever of seven training-free baselines wins it, and on 666 of 721 cells against those baselines one at a time; the strongest of them backpropagates through the whole model and so needs 94 GB to select at 24B, more than an 80 GB card has. Read along the other axis, at equal perplexity CoCurve removes 2.2 to 6.6 more points of the network than the strongest baseline at $\rho=30\text{--}40\%$. The off-diagonal also acts on a population we can name and delete on demand: forcing out the low-saliency, high-connectivity *bridge* units against a cost- and type-matched control costs nineteen of twenty capability-group cells across five models, while on three of the five perplexity moves the *other* way. Nothing in the construction is specific to a decoder-only stack: it transfers unchanged to seven vision-language models under one budget shared across both towers, leading every training-free baseline on five of them at $\rho=20\%$ with no baseline ahead of it by as much as a point on any. CoCurve reframes structured LLM pruning: *prune the edges, not just the nodes*.

1 INTRODUCTION

Large language models (LLMs) deliver their capabilities at a parameter and latency cost that is prohibitive for most deployments (Vaswani et al., 2017; Dubey et al., 2024). Structured pruning removes whole computational units, such as attention heads and feed-forward (FFN) channel groups. It is the most deployment-friendly route to compression: the pruned model stays dense and runs faster on commodity hardware, without the specialized kernels that unstructured or semi-structured sparsity needs (Frantar & Alistarh, 2023; Sun et al., 2024). The most practical setting is training-free, pruning one-shot from a small unlabeled calibration set, which avoids the cost and the data dependence of retraining (Ma et al., 2023; An et al., 2024).

The blind spot: nodes without edges. Structured pruners score each removable unit and delete the lowest-ranked ones, using weight (Han et al., 2015), activation (Sun et al., 2024; An et al., 2024) or local loss statistics (Ma et al., 2023; Frantar & Alistarh, 2022). Ranking assumes that set-level damage is additive, $\Delta\mathcal{L}(S) \approx \sum_{u \in S} \Delta\mathcal{L}(u)$. In a Transformer it is not. Attention and FFN units

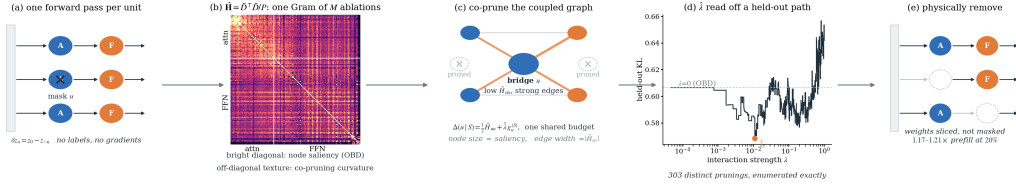


Figure 1: **Overview.** (a) One forward pass per removable unit records how the output distribution moves when that unit alone is masked; no labels and no gradients. (b) The Gram product of those M features estimates the Fisher matrix of the pruning risk: its diagonal is classical node saliency, its off-diagonal is co-pruning curvature. Shown is the measured curvature correlation on Llama-3.1-8B-Instruct, units ordered attention-then-FFN. (c) One budgeted greedy on that matrix, attention units and FFN groups drawing on a single budget; a low-saliency unit with strong edges — a *bridge* — is kept, because removing it alongside its partners is what costs. (d) How far to trust the off-diagonal is measured rather than tuned: the family indexed by λ is finite, we enumerate its breakpoints exactly, and keep the member with the lowest held-out KL. Shown is that path, all 303 members of it, on the same model. (e) The mask is realised by slicing weight matrices, so the speedup is measured on a smaller model and not on a masked one.

read and write one shared residual stream, so pruning is a joint operation on a coupled system: two units that look weak alone can be jointly indispensable, and two salient-looking ones can be partly redundant when their perturbations overlap. A ranking is blind to this even when attention and FFN units share a single global order, and it will delete a low-saliency *bridge* unit together with the units that depend on it. The interaction has to be modeled. *Prune the edges, not just the nodes.*

One matrix, read two ways. Interactions could be added heuristically — by correlating activations, say — but that measures the wrong thing, and not by a little. Activation correlation is a property of a layer’s inputs; pruning dependence is a property of how two removals *jointly* move the model’s output. Nothing in the former refers to the output at all, and two perfectly correlated units may be jointly harmless, because each covers for the other, or jointly critical, because whatever consumes them needs both — the correlation is identical in the two cases. An interaction term worth having must therefore be derived from the quantity being protected, so we derive it from the pruning objective. Define pruning risk as the token-level KL divergence between the frozen model and its masked copy over unlabeled text. The masked model equals the frozen one at zero pruning, so both the value and the gradient of this risk vanish there, and the local expansion begins at second order as a quadratic form $\frac{1}{2} s^T H s$ in the structured mask s . That single matrix has a direct reading (Figure 1):

its diagonal H_{uu} is classical node saliency; its off-diagonal H_{uv} is a co-pruning curvature edge, the mixed curvature governing the predicted marginal risk of removing u and v together.

These are not hand-designed information-flow arrows but symmetric interactions in the same Fisher geometry that defines the saliency, spanning attention $\leftrightarrow$ FFN pairs across all layers. Pruning becomes global co-pruning-risk minimisation rather than node ranking, and it inherits a classical floor: dropping the off-diagonal recovers an Optimal-Brain-Damage saliency (LeCun et al., 1989; Hassibi & Stork, 1993) lifted to structured units, so CoCurve is its cross-module generalisation.

Why the off-diagonal is affordable. Ablate each removable unit once and record how the output distribution moves. The inner product of two such recorded shifts estimates the curvature entry for that pair, so the full $M \times M$ estimator is a Gram product of M ablations rather than $O(M^2)$ pairwise sweeps, and no gradient is ever taken. Units are then removed greedily under one shared budget, each candidate charged its own cost plus its accumulated interaction with what has already gone, so the attention-versus-FFN split follows from the objective instead of being fixed by hand. Ablations replace the derivative columns, and Section B.2 measures what that substitution costs. Nothing is trained. The one quantity that must be chosen, how far to trust the pairwise term, is measured rather than tuned: it indexes a family of prunings that is finite and whose breakpoints we enumerate exactly, in the spirit of LARS (Efron et al., 2004), and we keep the member with the lowest held-out risk, consulting no labels and no benchmarks.

What this buys. We evaluate five dense LLMs from 3B to 24B across the Llama-3, Falcon3 and Mistral lineages against seven training-free baselines, under one shared guard and harness, on three language-modelling corpora and twelve multiple-choice benchmarks scored on their full test sets at five ratios from 10% to 50%, all five of them on Mistral-Small-24B. Against the per-column best of every baseline — an envelope, not one competitor — CoCurve takes 70 of the 105 cells this grid contains: 38 of the 45 perplexity cells and 32 of the 60 capability-group cells. Nothing in the construction is specific to a decoder-only stack: pointed at seven vision-language models, where the unit inventory spans two towers and one budget is shared across both, the same estimator and the same selector lead every training-free baseline on all five headline columns (Section 3.6). The edge term also acts through a population we can name and delete on demand. Low-saliency, high-connectivity *bridge* units are what node-first scoring removes first; forcing them out, against a control matched unit-for-unit on saliency, unit type and cost and completed by the same shared fill, costs nineteen of twenty capability-group cells across five models, by up to 23.7 points — while on three of the five perplexity moves the other way. Given one budget and no per-type rule, the same objective independently arrives at the same allocation across five models and five ratios: the first half of a network’s depth pruned entirely out of FFN width, attention touched only beyond it, in thirteen of the seventeen solutions that remove any attention at all — and eight of the twenty-five solutions, on two of the models, remove none anywhere.

2 METHOD

We target dense decoder-only LLMs and prune two kinds of structured units: attention units (a head in multi-head attention, or a KV-aligned query-head group in grouped-query attention (Ainslie et al., 2023)) and FFN groups (contiguous blocks of intermediate channels). Let $\mathcal{U} = \{u_1, \dots, u_M\}$ collect all such units across all layers, with a removable parameter cost $c_u = \text{Param}(u)$ each. Each unit carries a keep gate m_u scaling its write into the residual stream, and we write $s_u = 1 - m_u$ for the pruning variable: $s_u=0$ keeps unit u and $s_u=1$ zeroes its write. The deployed mask is binary, $s \in \{0, 1\}^M$. The gate is real-valued, so $\mathcal{R}$ below is defined on a neighbourhood of $\mathbf{0}$ in $\mathbb{R}^M$ and $\mathbf{0}$ is an interior point rather than a vertex of the cube, which is what makes its expansion well posed. Speedups are measured on models whose weight matrices are physically sliced, not on masked full-size ones (Section A.2).

2.1 PRUNING RISK AS SECOND-ORDER SELF-DISTILLATION

Let f_0 be the frozen full model (the teacher) with next-token distribution $p_0(\cdot \mid x_{\leq t}) = \text{softmax}(z_0(x, t))$ over a vocabulary of size V , and let f_s be the masked model with distribution p_s . Given a small unlabeled calibration set $\mathcal{C}$ and a deterministic set of token positions $\mathcal{T}(x)$ per sequence, we define the pruning risk as the token-level self-distillation KL divergence

$$\mathcal{R}(s) = \mathbb{E}_{x \in \mathcal{C}} \left[\frac{1}{|\mathcal{T}(x)|} \sum_{t \in \mathcal{T}(x)} D_{\text{KL}}(p_0(\cdot \mid x_{\leq t}) \parallel p_s(\cdot \mid x_{\leq t})) \right]. \quad (1)$$

This objective is label-free and ties directly to output behaviour. Its defining property is its local geometry. The masked model equals the teacher at $s = \mathbf{0}$, so the two distributions coincide there. Since D_{KL} is non-negative and vanishes only at equality, $\mathbf{0}$ is an interior minimiser of $\mathcal{R}$, and both the value and the gradient vanish there:

$$\mathcal{R}(\mathbf{0}) = 0, \quad \nabla_s \mathcal{R}(\mathbf{0}) = 0. \quad (2)$$

A second-order Taylor expansion around the full model has no first-order term, and the leading behaviour is a pure quadratic:

$$\mathcal{R}(s) \approx \frac{1}{2} s^\top \mathbf{H}^* s, \quad \mathbf{H}^* = \nabla_s^2 \mathcal{R}(s)|_{s=\mathbf{0}} \in \mathbb{R}^{M \times M}. \quad (3)$$

For binary masks $s_u^2 = s_u$, so expanding Eq. 3 gives the central decomposition

$$\mathcal{R}(s) \approx \underbrace{\frac{1}{2} \sum_u \mathbf{H}_{uu}^* s_u}_{\text{node saliency (diagonal)}} + \underbrace{\sum_{u < v} \mathbf{H}_{uv}^* s_u s_v}_{\text{co-pruning edges (off-diagonal)}}. \quad (4)$$

Remark 1 (Node saliency is the diagonal special case). Dropping the off-diagonal terms reduces Eq. 4 to a sum of independent saliencies $\frac{1}{2} \sum_u \mathbf{H}_{uu}^* s_u$, i.e. an Optimal-Brain-Damage-style criterion (LeCun et al., 1989) lifted to structured units (formalized in Remark 3). The cross-module

edges $\mathbf{H}_{uv}^*$ encode how the removal of u and v interacts: $\mathbf{H}_{uv}^* > 0$ means co-pruning is more harmful than the units’ individual terms predict; $\mathbf{H}_{uv}^* < 0$ means their perturbations partly cancel under the Fisher metric; $\mathbf{H}_{uv}^* \approx 0$ means they are second-order independent. A low-saliency *bridge* unit (small $\mathbf{H}_{uu}^*$) with many strong edges is thus protected by an accurate estimator, because pruning it together with its partners is costly, exactly the failure mode node-first scoring misses.

2.2 ESTIMATING THE EDGE MATRIX FROM SINGLE-UNIT ABLATIONS

The exact Hessian $\mathbf{H}^*$ in Eq. 3 is the Fisher curvature of Eq. 1, not a Gauss–Newton approximation of it: the KL gradient in logit space vanishes at $s = \mathbf{0}$ (Proposition 1), so the term a Gauss–Newton step would drop is exactly zero there. And it has a closed form we can evaluate. For one token position the KL Fisher in logit space is $F_{x,t} = \text{diag}(p_0) - p_0 p_0^\top$, and to second order the KL between teacher and masked model is the Fisher-weighted norm of the logit perturbation $\Delta z_s := z_0 - z_s$, that is $D_{\text{KL}}(p_0 \| p_s) \approx \frac{1}{2} \Delta z_s^\top F \Delta z_s$ (Proposition 2). The key practical step is that this perturbation is, to leading order, additive across units: if $\delta z_u(x, t) = z_0(x, t) - z_{-u}(x, t)$ is the logit shift from masking unit u alone, then $\Delta z_s \approx \sum_u s_u \delta z_u$. Writing $J_u = \partial z_s / \partial s_u|_{s=\mathbf{0}}$, the exact entry is $\mathbf{H}_{uv}^* = \mathbb{E}_{x,t}[J_u^\top F_{x,t} J_v]$. Replacing J_u by the observed full-ablation displacement δz_u gives the finite-ablation estimator used by the algorithm,

$$\mathbf{H}_{uv} := \mathbb{E}_{x,t}[\delta z_u(x, t)^\top F_{x,t} \delta z_v(x, t)]. \quad (5)$$

Whitening the perturbation into its probability-weighted, mean-centred form $\tilde{\delta z}_u = \sqrt{p_0} \odot (\delta z_u - \mathbb{E}_{y \sim p_0}[\delta z_{u,y}])$ turns the Fisher inner product into a plain dot product (Section A.3), giving

$$\mathbf{H}_{uv} = \frac{1}{P} \bar{D}_u^\top \bar{D}_v, \quad \bar{D}_u = \text{vec}(\{\tilde{\delta z}_u(x, t)\}_{x,t}), \quad (6)$$

where $\bar{D}$ is the matrix with columns $\bar{D}_u$ and P is the total number of calibration positions: we keep a fixed number of positions per sequence, so P is that count times the number of sequences (128×128 here) and pooling the positions is the same average as the nested one in Eq. 1. The entire $M \times M$ edge matrix is therefore a Gram matrix of *single-unit* feature vectors. It estimates $\mathbf{H}^*$ with M single-unit ablation passes, one per unit, rather than $O(M^2)$ pairwise ablations, and needs no gradients or Hessian-vector products. From here on $\mathbf{H}$ denotes this estimated matrix, and Section B.2 separates the two approximations involved and measures both. The $V \times V$ Fisher is never materialised: features live on the teacher’s top- r logit coordinates, renormalised to sum to one and shared across units, so the Gram remains valid and F is the Fisher of that truncated distribution (Remark 2). Section A.3 gives the implementation and its complexity.

2.3 ONE-SHOT COST-NORMALISED GREEDY CO-PRUNING

Write $\hat{\mathcal{R}}_\lambda(s) = \frac{1}{2} \sum_u \mathbf{H}_{uu} s_u + \lambda \sum_{u < v} \mathbf{H}_{uv} s_u s_v$ for the surrogate risk of Eq. 4, with the interaction strength λ of Section 2.4. Given a target ratio ρ , pruning solves the budgeted binary quadratic program

$$\min_{s \in \{0,1\}^M} \hat{\mathcal{R}}_\lambda(s) \quad \text{s.t.} \quad \sum_u c_u s_u \geq \rho \sum_u c_u. \quad (7)$$

Attention units and FFN groups share one budget pool, so how the budget divides between the two types emerges from the optimisation rather than being fixed. Exact solution is intractable. We use a one-shot greedy solver on the fixed surrogate, with no weight updates and no re-estimation of $\mathbf{H}$. Maintaining the cumulative perturbation of the current set S , the marginal risk of adding unit u is

$$\Delta R(u | S) = \frac{1}{2} \mathbf{H}_{uu} + \sum_{v \in S} \mathbf{H}_{uv} = \frac{1}{2} \mathbf{H}_{uu} + g_u^{(S)}, \quad (8)$$

where $g_u^{(S)} = \sum_{v \in S} \mathbf{H}_{uv}$ is a running interaction accumulator updated in $O(M)$ by $g \leftarrow g + \mathbf{H}_{:,u^*}$ after each selection. At each step we pick the unit with the smallest cost-normalised marginal risk, $u^* = \arg \min_{u \notin S} \Delta R(u | S) / c_u$, and stop once the budget is met. The solver is $O(M|S|)$ for $|S|$ selected units, and Algorithm 2 (Section A.4) summarises the full pipeline. A per-layer cap keeps removal from concentrating; it is shared by every method we compare, at every ratio, and specified in Section C.2.

2.4 INTERACTION STRENGTH: A PATH, NOT A HYPERPARAMETER

The diagonal follows from single-unit ablations alone. The off-diagonal additionally assumes that simultaneous removals perturb the logits additively (Eq. 5), so we weight it by a single *interaction strength* $\lambda \in [0, 1]$,

$$\hat{\mathcal{R}}_\lambda(s) = \frac{1}{2} \sum_u \mathbf{H}_{uu} s_u + \lambda \sum_{u < v} \mathbf{H}_{uv} s_u s_v, \quad (9)$$

so that $\lambda=0$ recovers pure node saliency and $\lambda=1$ the full edge model. The $\lambda=0$ endpoint is a floor rather than an arbitrary limit. It is exactly the OBD diagonal selector (Section B.3), so whatever λ we end up with, the method degrades at worst to a strong classical second-order baseline.

Choosing it is not a fitting problem. The solver consumes only the *order* the marginals induce, so λ has to be chosen by what the selections it produces actually cost — and that quantity turns out to be exactly computable.

The family of solutions is finite. What the solver does consume is the ordering of the conditional marginals $\Delta_\lambda(u \mid S) = \frac{1}{2} \mathbf{H}_{uu} + \lambda g_u^{(S)}$. Normalised by cost, each candidate contributes an affine function of λ ,

$$\Delta_\lambda(u \mid S)/c_u = \mu_u + \lambda \nu_u, \quad \mu_u = \frac{\mathbf{H}_{uu}}{2c_u}, \quad \nu_u = \frac{g_u^{(S)}}{c_u}, \quad (10)$$

so within a single greedy step the arg-min can change only where two of these lines cross. Every step of the solver therefore switches at finitely many values of λ , and the solution S_λ is *piecewise constant*. The family $\{S_\lambda : \lambda \in [0, 1]\}$ is a finite set of prunings, a few hundred per model rather than a continuum.

The family can be enumerated exactly. Because Eq. 10 is affine, the crossings are available in closed form. Starting at $\lambda=0$, the earliest crossing over all greedy steps is the right endpoint of the interval on which the current solution is optimal. Stepping just past it and re-solving moves to the next member, and repeating walks the whole family; the step past each crossing is 10^{-9} , nearly three orders of magnitude below the narrowest interval on which any member of the ten enumerated families is optimal, 7.6×10^{-7} (Section B.3.1). The construction is the same critical-value walk that makes the LASSO path enumerable by LARS (Efron et al., 2004), borrowed for its mechanics rather than its guarantees: LARS traces the exact optimum of a convex problem and its path is piecewise linear, while S_λ is the path of a greedy heuristic and is piecewise constant. The walk is CPU-only, one greedy solve per breakpoint and no forward passes, and what the selection then costs is one masked forward pass per distinct *member* — 303 of them on Llama-3.1-8B-Instruct. Sampling λ is not a substitute at any price. The median interval width is 1.4×10^{-3} across the ten models, so a grid at step 0.002 still reaches a median of only 61% of the members and as few as 35%: it misses solutions rather than costing more (Section B.3.1).

Selection is then a held-out measurement. With the family in hand there is nothing left to search over; we keep the member with the lowest measured held-out risk,

$$\hat{\lambda} = \arg \min_{\lambda \in [0, 1]} \mathcal{R}(S_\lambda; \mathcal{C}'), \quad (11)$$

where $\mathcal{R}(\cdot; \mathcal{C}')$ is the same token-level KL of Eq. 1, measured on a calibration split $\mathcal{C}'$ held out from the one that built $\mathbf{H}$. No labels and no benchmark enter, and each S_λ is judged by measured risk rather than by the surrogate that proposed it. Every member is scored on the same sequences, so differences across λ are paired and we report them in paired standard errors. The cost is one masked forward pass per member over a small held-out set, against the M ablations $\mathbf{H}$ already requires.

Two properties of this rule matter for how the results below should be read. First, the setting is fixed before any benchmark is seen: $\hat{\lambda}$ is chosen on held-out risk, so no test set enters the selection, and we deliberately do not pick the member that scores best downstream — a family member tuned on perplexity would be a different and much weaker claim. Second, the criterion is flat near its optimum. Neighbouring members sit within its own noise: on the headline model the member at $\lambda=0.15$ carries 2.1% more held-out risk, 1.3 paired standard errors, and lands at a comparable operating point. The rule therefore does not need to be delicate to be useful, and Section E.7 confirms that resampling the held-out set and changing the selection protocol move it very little.

What the resulting curve says. The diagonal contributes one term to the marginal while the off-diagonal contributes $|S|$ of them, so a balanced combination would sit near $\lambda \sim 1/(2|S|\bar{\rho}_S)$, with

$\bar{\rho}_S$ the mean curvature correlation over in-set pairs (Proposition 5) — of order 10^{-2} at the ratios we prune. The criterion lands at or below that scale: $\hat{\lambda} < 0.1$ on nine of the ten models and exactly 0 on two. The off-diagonal is worth consulting and is damped hard, and at $\hat{\lambda}$ it still supplies a median 12% to 46% of the winning score. $\mathcal{R}(S_\lambda)$ has an *interior* minimum on eight of the ten models we measure. Both $\lambda = 0$ (no edges) and $\lambda = 1$ (full trust) are worse than the calibrated combination, which is direct evidence that the off-diagonal carries signal and that it must nonetheless be damped. On the other two the selected segment begins at $\lambda = 0$ and the procedure returns the diagonal selector unchanged. That is what Eq. 11 buys: the diagonal solution is itself a member of the family being minimised over, so the criterion cannot return something with higher held-out risk than node saliency alone. We measure $\mathcal{C}'$ at the deployment context length, so the risk we minimise is the risk the pruned model will incur; the selection is insensitive to that choice either way (Section B.3.1).

One matrix, two uses. The couplings that say what not to remove together also say how the survivors should absorb what was removed. The same blocks of $\mathbf{H}$ give a coupling-aware compensation on the kept FFN units in closed form, from one small ridge solve (Section A.5). Every number outside Section 3.5 is selection-only.

3 EXPERIMENTS

3.1 SETUP

Models. Five dense decoder-only LLMs spanning the Llama-3, Falcon3 and Mistral lineages and the 3B–24B range: **Llama-3.2-3B** (Dubey et al., 2024), **Falcon3-7B** (Falcon-LLM Team, 2024), **Llama-3.1-8B-Instruct** (Dubey et al., 2024), **Mistral-Nemo-12B** (Mistral AI Team, 2024) and **Mistral-Small-24B** (Mistral AI Team, 2025). Four are base pretrained checkpoints and one is instruction-tuned, so the comparison spans both regimes; all use grouped-query attention, so attention units are KV-aligned query-head groups.

Benchmarks. We follow the evaluation protocol established for training-free structured pruning (Ma et al., 2023; An et al., 2024; Ling et al., 2024; Guo et al., 2025): language-modelling perplexity together with zero-shot multiple-choice accuracy. Perplexity is measured on WikiText-2 (Merity et al., 2017), PTB (Marcus et al., 1993) and C4 (Raffel et al., 2020) at a 2048-token context. Accuracy is measured on twelve benchmarks covering commonsense reasoning (PIQA (Bisk et al., 2020), HellaSwag (Zellers et al., 2019), WinoGrande (Sakaguchi et al., 2021), CommonsenseQA (Talmor et al., 2019)), science question answering (ARC-easy/challenge (Clark et al., 2018), OpenBookQA (Mihaylov et al., 2018)), reading comprehension (BoolQ (Clark et al., 2019), RACE and RACE-high (Lai et al., 2017), QuAIL (Rogers et al., 2020)) and world knowledge (MMLU (Hendrycks et al., 2021)). Every task is scored on its *full* test set through one shared harness (Section C.2).

We report both families, and it is worth saying at the outset which one carries which claim. Perplexity is how this literature compares methods, and we report it so our numbers are readable against published ones; it is also the family our objective is closest to. But a language-modelling loss can miss capability damage — Section 4 exhibits a mask that improves WikiText while losing double digits on every capability group — so we never let a perplexity win stand in for a capability claim. Where the two families disagree we say so and let the accuracy columns decide. The result is that the lead does not rest on the weaker witness: it is present in both families at once.

Baselines. Seven training-free baselines under identical budgets and harness: **Random**, **Magnitude** (Han et al., 2015), **Wanda-sp** (Sun et al., 2024; An et al., 2024), **FLAP** (An et al., 2024), **LLM-Pruner** (Ma et al., 2023), **SlimGPT** (Ling et al., 2024) and **SlimLLM** (Guo et al., 2025). Section C.3 adds two more on the headline model: **AMP** (Mugnaini et al., 2025) inside the same guard, and the two-stage **2SSP** (Sandri et al., 2025) under its own per-layer budget, because its two stages are the method. The regime is *selection-only*: the question is which units to remove, so post-pruning weight repair and fine-tuning are disabled for every method including ours, namely LLM-Pruner’s LoRA (Hu et al., 2022), SlimLLM’s regression update, SlimGPT’s OBS update and FLAP’s bias compensation. Section 3.5 adds an identical LoRA stage to every selection at once.

Protocol. Calibration is 128 sequences of C4 (the Wanda/SparseGPT convention), from which we build $\mathbf{H}$ with top- $r=256$ logits and 16 FFN groups per layer; every calibration-based baseline

Table 1: **Llama-3.1-8B-Instruct, all reported tasks.** Eight training-free methods under one guard and one harness, each realising its target ratio to within 0.2 points, on full test sets. Shading is rank within a column among the pruned methods, darkest best.

ρ	Method	LM perplexity ↓			Commonsense ↑				Science QA ↑			Reading ↑			Know. ↑	
		Wiki	PTB	C4	PIQA	HellaS	WinoG	CSQA	ARC-e	ARC-c	OBQA	BoolQ	RACE	RACE-h		QuAIL
—	Dense	6.40	10.36	10.67	80.9	77.4	66.2	59.2	78.1	54.4	47.6	85.0	68.8	54.1	62.7	47.8
20%	SlimLLM	17.36	30.59	29.36	75.2	60.0	56.2	47.6	65.2	39.5	39.8	65.8	51.3	39.4	47.8	37.2
	SlimGPT	19.24	35.52	29.19	73.4	58.8	56.6	44.6	61.9	37.4	35.4	64.3	47.8	36.6	44.5	35.1
	LLM-Pruner	13.23	19.36	21.98	75.0	64.6	57.4	51.8	60.4	37.2	36.8	79.1	57.2	45.2	52.4	36.8
	FLAP	20.88	36.77	29.06	73.2	59.0	54.9	46.7	62.5	37.7	38.2	70.9	49.9	39.2	45.6	35.3
	Wanda-sp	20.02	35.29	30.09	73.7	57.4	57.0	48.9	62.5	38.8	36.4	72.2	52.8	38.7	46.6	35.6
	Magnitude	20.49	31.38	34.93	68.9	55.5	55.9	41.7	54.6	31.4	32.4	65.2	50.2	39.1	49.4	32.8
	Random	21.20	30.44	36.77	68.2	53.8	54.5	38.5	50.3	29.4	31.4	65.4	47.1	36.1	47.8	31.8
	CoCurve	12.88	29.63	19.58	75.8	67.5	61.0	50.4	62.5	41.9	39.6	78.3	58.1	44.0	54.1	38.5
30%	SlimLLM	78.23	100.68	109.42	69.5	37.8	52.0	36.9	53.0	28.8	28.2	62.5	35.9	29.4	37.2	30.4
	SlimGPT	196.79	317.47	208.42	62.1	38.9	51.2	30.4	42.8	25.4	27.4	51.3	35.4	29.4	38.0	27.9
	LLM-Pruner	26.64	39.70	40.31	71.0	54.4	52.4	43.2	51.0	33.2	30.4	73.4	48.4	38.8	47.8	31.9
	FLAP	79.37	177.67	102.06	65.7	41.0	53.6	33.5	43.8	24.8	26.8	51.2	37.8	32.0	37.2	29.0
	Wanda-sp	61.15	94.85	87.17	65.7	38.5	53.4	33.7	45.8	27.8	28.2	47.7	41.9	31.0	40.1	29.6
	Magnitude	324.46	642.26	273.86	57.0	32.2	51.0	24.7	31.5	25.0	25.6	51.3	33.1	28.8	34.8	27.3
	Random	52.90	84.29	87.88	60.6	40.2	53.5	29.6	41.0	24.1	26.6	47.8	41.4	31.8	42.6	28.6
	CoCurve	21.59	36.31	31.99	70.5	55.8	55.5	41.0	55.6	34.4	35.2	73.1	46.5	37.3	50.1	33.4
40%	SlimLLM	183.53	211.36	217.27	62.6	31.2	50.4	29.2	41.6	24.0	25.0	51.2	31.6	27.3	32.1	27.6
	SlimGPT	26002	32398	13079	51.9	26.2	49.2	21.0	27.9	23.9	29.2	38.2	25.3	26.7	21.1	26.9
	LLM-Pruner	103.35	165.07	125.68	65.3	42.7	52.2	33.4	40.0	28.1	26.8	65.1	36.0	32.2	40.2	29.7
	FLAP	49940	29989	13823	52.1	26.0	49.2	22.0	28.0	23.5	22.6	41.6	26.6	25.4	23.3	25.6
	Wanda-sp	1015	1288	860.91	54.8	26.3	48.2	25.6	33.1	22.4	22.8	38.7	27.8	26.0	24.8	26.6
	Magnitude	27101	23937	13834	51.1	25.4	50.8	19.0	25.7	25.9	27.8	48.2	25.6	25.8	24.4	26.2
	Random	4168	3201	3152	54.0	29.0	50.5	24.1	31.3	23.4	24.0	42.5	33.3	26.4	35.4	26.7
	CoCurve	59.13	108.37	76.05	64.0	41.0	51.4	31.9	41.6	28.7	29.6	62.3	41.9	32.2	42.8	29.3

receives that same set. Cost is non-embedding parameters, and the same cost vector defines the budget for every method. Every method is pruned under one guard, a per-layer cap of 1.5ρ with the first and last two layers protected, and every mask must realise its target ratio to within 0.2 points to be reported: a run outside that band is not at the operating point its column names and is excluded from the comparison. $\hat{\lambda}$ is re-derived at each model and ratio from 16 held-out sequences by Eq. 11, with no benchmark consulted. Every configuration is fixed in advance and shared by all methods (Section C.2).

3.2 MAIN RESULTS

The gap opens as compression bites. Table 1 reports the headline model task by task. Across the three budgets it is first or second in *all forty-five* columns, and first in twenty-seven of them; the best competitor reaches thirteen of fifteen in its strongest block and no other method exceeds five. At 30% CoCurve holds ten of the fifteen columns outright — all three perplexities and seven of the twelve accuracy tasks, including world knowledge and both science benchmarks. Our WikiText margin over the best baseline widens monotonically with the budget on this model, from 2.6% at 20% to 19.0% at 30% and 42.8% at 40%. A coupling account predicts exactly this. While a model is over-parameterised, almost any 20% of its units can go without the survivors having to compensate. Once the budget forces removals that genuinely depend on one another, the term that models that dependence decides whether the remaining network still computes anything. Nothing about the method changes across the sweep: the same objective, estimator and solver, with $\hat{\lambda}$ re-derived at each ratio by Eq. 11 rather than scheduled. On Falcon3-7B it returns 0.218 at 20% and 0.149 at 30% without ever being shown the ratio.

The advantage holds across five models and three scales. Table 2 compares each model against a strict bar: the per-column best of every baseline run for it — a bar that in ten of the fifteen blocks no single competitor attains, and that in the other five LLM-Pruner attains alone. The envelope carries all seven baselines on thirteen of the fifteen model–ratio blocks, Mistral-Small-24B included at every ratio; the two that carry fewer are itemised with their reasons in Section C.3. Against that

envelope CoCurve leads on 38 of the 45 perplexity cells and 32 of the 60 capability-group cells. The envelope is a hard bar on the second of those: it is attained by LLM-Pruner in 70% of them, so beating it there is essentially beating the one baseline in the set that backpropagates through the whole model. It is also one of the two that do not run at 24B in published form: LLM-Pruner needs 94.3 GB of device memory against an 80 GB card, and SlimGPT 344 GB of host memory against 128 GB, and both are in the table there only because we re-implemented them to stream (Table 19). Against the seven taken one at a time, CoCurve leads in 666 of 721 cells. Five of the seven take under one cell in twenty; the two that do better are SlimLLM at 13% and LLM-Pruner at 27%.

Perplexity margins are easier to read along the ratio axis than the quality axis. Interpolating each model’s baseline envelope over the five measured ratios, the perplexity CoCurve delivers at $\rho=30\text{--}40\%$ is the perplexity the strongest baseline reaches 2.2 to 6.6 points of compression earlier. That holds on nine of the ten model-ratio points; the exception is Llama-3.2-3B at $\rho=40\%$. Figure 2 extends this to the full sweep.

Mistral-Small-24B makes the point most sharply, and it makes it along the ratio axis. At $\rho=20\%$ a 24B model has slack to spare, and the field is correspondingly tight: we take C4 and the envelope takes WikiText and PTB, all three within a few tenths of a point on a model whose dense WikiText is 4.70. That is the same observation as the widening margin above, seen from the end where there is not yet anything to choose. At $\rho=30\%$ its perplexity is 20.5 against the strongest baseline’s 78.8 — a $3.8\times$ margin — while four of the seven are above 470: at that ratio it is the only method on that model still holding a usable perplexity. At $\rho=40\%$ the margin is $11.9\times$, 277.1 against the best of all seven baselines, Random at 3299; the next two are SlimLLM at 3969 and LLM-Pruner at 5495, and SlimGPT lands at 9057. The direction reverses on the smallest model: at $\rho=40\%$ Llama-3.2-3B loses every cell to the envelope — its three perplexity columns to SlimLLM and three of its four capability groups to LLM-Pruner, though against LLM-Pruner alone we still lead on PTB and C4. That is the ratio at which a 3B model stops having an allocation worth choosing: the best of the eight methods there is already $19.7\times$ dense perplexity, so the block records which of two unusable models is nominally ahead.

The objective delivers what it optimises. The criterion is token-level KL to the dense model, so the pruned model should reproduce the dense model’s decisions more faithfully than the alternatives do. It does. At $\rho=30\%$ on Llama-3.1-8B-Instruct, CoCurve matches the dense prediction more often than any baseline on three of six tasks, and is a close second on the rest, with every baseline but one behind it everywhere. Section E.3.1 breaks this down and relates it to where the two metrics of Table 2 diverge.

The per-task columns show what an average hides. A pruned model collects a floor of 0.375 on the seven-task mean for answering nothing, so it can post a respectable average while having stopped answering several of its tasks. That is what has happened to the field at 30%: three of seven baselines are at or below chance on ARC-challenge, and five of seven are at or below the majority-class rate on BoolQ. CoCurve is above chance on all twelve tasks at every ratio in Table 1.

The result does not depend on the field we compare against, the prompt format, or the precision. Three checks, reported in full in the appendix, ask whether the lead is an artefact of the setup that produced it. Widening the field to nine competitors, including the two most recent selectors, AMP and 2SSP, leaves the ordering intact: CoCurve is first on seven of the eleven task columns and on the commonsense average, and second on two more (Table 15). Replacing the zero-shot prompt with five demonstrations — the regime in which damage to attention shows first, and the reason SlimGPT and 2SSP both report it — leaves it first on both models at both ratios (Table 24). Quantising the pruned model to 4 bits costs it about what the same step costs the dense model in accuracy, so the two compressions draw on different slack rather than the same slack twice (Section E.5). And the criterion’s own target is met: among the nine methods, ours reproduces the dense model’s predictions most faithfully, first or second on all six tasks measured (Section E.3.1).

3.3 ABLATIONS: WHAT THE EDGE TERM BUYS

The gain is the off-diagonal, not the diagonal it sits on and not the module split it happens to move. Table 3 tests all three readings at once on ten models spanning six families: one measurement, paired over the same sixteen held-out sequences, in three pairs of columns.

Table 2: **Five models, three shared ratios.** “Best base.” is the per-column best of the training-free baselines run for that model: each baseline’s own group average is taken first, then the best baseline for that column. **Bold** is the better of the two rows, in whichever row it falls; the blue band marks our row, and its intensity is the size of our margin where we lead. Groups average PIQA/HellaSwag/WinoGrande/CSQA, ARC-e/ARC-c/OBQA, and BoolQ/RACE/RACE-h/QuAIL. Per-method numbers: Table 16.

Model	ρ		Perplexity ↓			Accuracy ↑			
			Wiki	PTB	C4	Cmn.	Sci.	Read.	Know.
Llama-3.2-3B	20%	Best base.	16.90	29.84	24.64	55.4	40.8	45.3	33.5
		CoCurve	16.66	27.34	23.24	56.6	42.4	49.7	33.9
	30%	Best base.	42.69	54.64	50.48	48.5	33.7	43.5	29.5
		CoCurve	34.78	49.40	55.02	48.2	35.3	44.2	30.0
	40%	Best base.	136.1	159.8	130.0	43.7	30.3	39.7	28.0
		CoCurve	193.7	256.5	196.5	40.2	28.5	37.9	26.8
Falcon3-7B	20%	Best base.	7.43	12.62	16.49	60.0	48.3	55.0	39.8
		CoCurve	7.07	11.72	15.65	59.4	48.1	53.8	37.8
	30%	Best base.	10.07	17.65	22.34	55.3	41.0	50.6	34.8
		CoCurve	8.92	15.05	19.47	56.3	42.6	49.2	35.8
	40%	Best base.	18.36	28.74	37.94	49.7	35.2	46.5	31.1
		CoCurve	13.83	23.11	29.21	49.5	36.1	45.3	31.6
Llama-3.1-8B-It	20%	Best base.	13.23	19.36	21.98	62.2	48.2	58.5	37.2
		CoCurve	12.88	29.63	19.58	63.7	48.0	58.6	38.5
	30%	Best base.	26.64	39.70	40.31	55.2	38.2	52.1	31.9
		CoCurve	21.59	36.31	31.99	55.7	41.7	51.8	33.4
	40%	Best base.	103.3	165.1	125.7	48.4	31.6	43.4	29.7
		CoCurve	59.13	108.4	76.05	47.1	33.3	44.8	29.3
Mistral-Nemo-12B	20%	Best base.	11.44	175.7	17.83	60.1	46.7	51.4	35.6
		CoCurve	8.63	83.37	14.31	62.1	49.7	56.7	36.9
	30%	Best base.	17.75	164.5	24.84	57.2	42.9	50.5	33.0
		CoCurve	15.75	160.4	23.27	56.1	40.8	49.4	31.7
	40%	Best base.	73.06	499.3	86.26	48.8	35.0	45.9	30.0
		CoCurve	52.20	455.4	71.01	48.2	32.4	44.5	29.3
Mistral-Small-24B	20%	Best base.	8.71	69.46	15.35	67.0	55.2	59.8	43.7
		CoCurve	8.97	131.2	14.88	67.8	56.3	60.9	43.5
	30%	Best base.	78.78	464.7	116.4	58.5	49.8	51.5	36.5
		CoCurve	20.49	419.1	29.67	59.3	48.2	54.1	36.5
	40%	Best base.	3299	4747	1271	45.1	37.6	34.1	30.0
		CoCurve	277.1	4037	231.6	44.3	33.1	41.8	30.0

Removing the edges costs, and never gains. Setting $\lambda = 0$ deletes the off-diagonal and reduces CoCurve to an OBD-style diagonal saliency (Eq. 4) that shares everything else: the same $\mathbf{H}$ diagonal, cost normalisation, budget and guard. The path rejects it on eight of ten models, by up to 9.6% of held-out risk and 8.2 paired standard errors, and the differences are paired over the same held-out sequences, so sequence difficulty cancels rather than being averaged away. The gap carries downstream: on Llama-3.1-8B-Instruct at the same cap and realised ratio the diagonal selector reaches MMLU .373 against .385, a gap of 1.2 points, against the at most 0.81 that redrawing the calibration set moves any capability group (Section E.8). On the remaining two models the selected segment

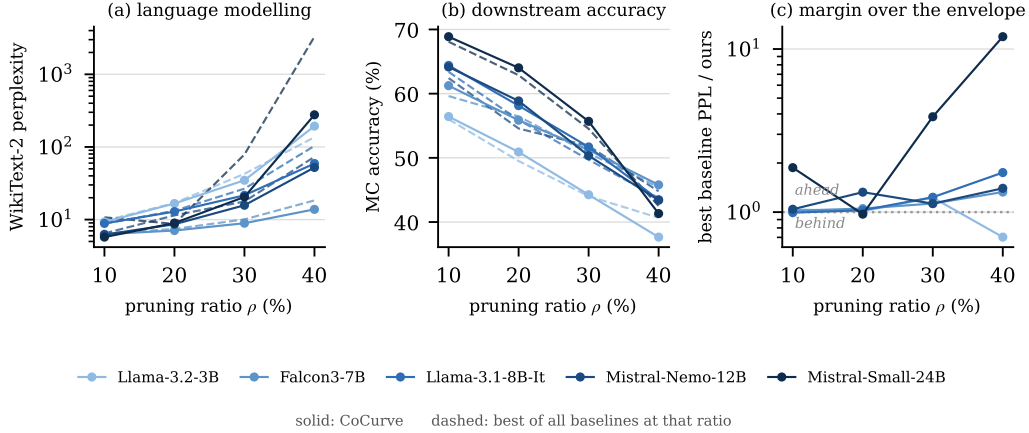


Figure 2: **Across ratios and scales.** Solid: CoCurve. Dashed: the best of all baselines at that ratio, an envelope rather than one competitor. (c) is (a) as a ratio; above one is ahead. The margin widens with compression on Falcon3-7B, Llama-3.1-8B-Instruct and Mistral-Small-24B, reaching $11.9\times$ at 40% on the last. The panels stop at $\rho=40\%$ because at 50% the axis stops separating methods rather than favouring one: in Table 17 every method measured at $\rho=50\%$ is at least $7.6\times$ dense perplexity, and on four of the five models even the best of them is past $100\times$.

Table 3: **The off-diagonal is what pays, and how much to trust it has to be measured** ($\rho=20\%$, 16 held-out sequences at the 2048-token deployment length). $|\mathcal{F}|$ is the number of distinct prunings in the enumerated family; $\hat{\lambda}$ is the selected segment. Δ is how much *worse* each alternative is than $\hat{\lambda}$, with σ its paired standard errors. Right-hand block: the attention/FFN split is frozen at the diagonal selector’s own allocation, so the two arms remove identical cost of each type and only the ordering can differ. Rows are ordered by size; on the two Qwen models the selected segment begins at exactly $\lambda=0$, so the criterion keeps the diagonal solution and σ is undefined.

Model	$ \mathcal{F} $	$\hat{\lambda}$	$\lambda=0$		$\lambda=1$		split frozen	
			$\Delta\%$	σ	$\Delta\%$	σ	$\Delta\%$	σ
Llama-3.2-3B	288	0.0911	+9.6	3.3	+12.4	3.8	+11.5	4.2
Falcon3-7B	181	0.218	+6.6	8.2	+29.7	23.1	+7.6	11.4
Mistral-7B	318	0.0165	+5.8	5.9	+32.0	11.0	–	–
Qwen2.5-7B	115	0	+0.0	–	+6.7	4.7	–	–
Llama-3.1-8B-It	303	0.0108	+6.7	4.6	+13.0	6.0	+8.5	5.4
Yi-1.5-9B	400	0.0141	+3.8	2.1	+746.9	39.1	+4.4	3.1
Falcon3-10B	402	0.0126	+2.5	3.9	+30.1	8.3	–	–
Mistral-Nemo-12B	371	$2.4\cdot 10^{-4}$	+0.4	3.1	+32.2	17.1	+0.9	2.7
Llama-2-13B	553	$1.2\cdot 10^{-5}$	+0.1	1.4	+94.7	21.5	–	–
Qwen2.5-14B	716	0	+0.0	–	+22.2	13.9	–	–

begins at $\lambda = 0$, and the procedure returns the diagonal selector unchanged rather than something worse.

Trusting them fully costs more. Taking the whole off-diagonal at face value, $\lambda = 1$, is rejected on *every* model, by 6.7% to 747% and at 3.8 to 39.1 paired standard errors. How far to trust the off-diagonal is therefore a measurement, and a method that asserts $\lambda=1$ without measuring is wrong on ten models out of ten, and one that asserts $\lambda=0$ is wrong on eight of them.

And the gain is a reordering, not a re-split. λ also moves how the shared budget divides between attention and FFN. We therefore freeze that division at the *diagonal* selector’s own per-type budget, a quantity any OBD-style method produces without our matrix, and re-derive $\hat{\lambda}$ inside the constraint. Both arms then remove identical attention and FFN cost by construction, and the $\lambda = 0$ member of the constrained family *is* the diagonal selector. An interior $\hat{\lambda}$ still wins on every model we ran it on,

Table 4: **What compression buys**, measured on physically removed units rather than masks (one RTX 6000 Ada, bf16, eager attention, on an otherwise idle machine; the 24B row is timed the same way on one A100-80GB, which is the card that holds it dense). Throughput is relative to dense, and *Params* is the ratio of total parameters, so a 20% non-embedding removal reads below 1 but not at 0.80. At a fixed removed fraction every structured method lands in the same band: prefill $1.17\text{--}1.21\times$ at $\rho=20\%$ across the four architectures timed here, and on the headline model the eight methods sit within 0.07 GB of each other on peak memory. The axis on which methods differ is therefore accuracy at a given speedup, not the speedup.

Model	Removed	Params↓	Prefill↑	Decode↑	Peak mem↓
Llama-3.2-3B	20%	$0.82\times$	$1.17\times$	$1.10\times$	$0.83\times$
Falcon3-7B	20%	$0.82\times$	$1.19\times$	$1.16\times$	$0.83\times$
Llama-3.1-8B-Instruct	20%	$0.83\times$	$1.17\times$	$1.06\times$	$0.83\times$
Mistral-Nemo-12B	20%	$0.82\times$	$1.21\times$	$1.18\times$	$0.82\times$
Llama-3.2-3B	40%	$0.65\times$	$1.40\times$	$1.09\times$	$0.66\times$
Llama-3.1-8B-Instruct	40%	$0.65\times$	$1.49\times$	$1.22\times$	$0.66\times$
Mistral-Small-24B	40%	$0.62\times$	—	—	$0.63\times$

by 2.7 to 11.4 paired standard errors. The same conclusion survives coarser and finer interaction granularity (Table 21), a resampled held-out set and three alternative selection rules (Section E.7), and rebuilding $\mathbf{H}$ from disjoint calibration text (Section E.8). Restricting the eligible units to a single module type costs wherever the restriction binds: on Llama-3.2-3B both restrictions cost, and an attention-only budget cannot reach 20% at all on Falcon3-7B, which has four attention units per layer. On that model the objective spends nothing on attention of its own accord, so the FFN-only arm *is* the joint arm (Table 20).

3.4 EFFICIENCY

Compression is worth its accuracy cost only if it buys speed, so we measure it on physically removed units rather than masks (Table 4): every timing runs on sliced weight matrices rather than masked full-size ones (Section A.2). Removing 40% of the non-embedding parameters gives a $1.40\times$ prefill speedup at $0.66\times$ peak memory on Llama-3.2-3B and $1.49\times$ at $0.66\times$ on Llama-3.1-8B-Instruct, and leaves $0.62\times$ the parameters at $0.63\times$ peak memory on Mistral-Small-24B. At a fixed removed fraction, structured methods land in one narrow band — Table 18 times all eight methods at $\rho=20\%$ on one machine and their prefill speedups span 1.13 to 1.19 $\times$. What a method is worth is therefore the accuracy it delivers at that speedup (Table 1, Table 2). Selection cost does differ between methods, and Section D.4.1 reports it.

3.5 WITH A LIGHT RECOVERY STAGE

Past roughly 40%, no training-free method leaves a model within reach of dense accuracy. A practitioner then asks what a short recovery stage buys, and whether a better selection still matters once weights are allowed to move. One experiment answers both. We apply an identical LoRA stage, at the same rank, token budget, schedule and corpus, to CoCurve and to four baselines at each of three ratios. The corpus is disjoint from the calibration set. These models are no longer training-free, so we report them separately from every other number in this paper (Table 5).

Recovery compresses the field, as it should — a random selection, once tuned, finishes above at least two tuned baselines at every ratio, which is a caution against reading tuned numbers as evidence about selection. What survives the compression is the ordering: CoCurve holds the best value in 15 of the 18 cells, and after tuning it has the highest seven-set commonsense average at all three ratios, by 0.2, 1.8 and 1.5 points. A short recovery stage therefore does not substitute for choosing well; it starts from wherever the selection left the model.

Table 5: **Selection quality survives recovery** (Llama-3.1-8B-Instruct). The same five selections before and after one identical LoRA stage — same rank, token budget, schedule and corpus, on a corpus disjoint from the calibration set. Cmn_7 is the seven-set commonsense average conventional in the pruning literature (ARC-e/c, HellaSwag, WinoGrande, PIQA, OBQA, BoolQ) — a wider set than the four-task Cmn group of Table 1 — and $\text{Ret.}\%$ is the fraction of the dense model’s Cmn_7 retained. Shading is rank within a column.

ρ	Method	w/o tune (training-free)			w/ tune (LoRA)		
		Wiki↓	$\text{Cmn}_7 \uparrow$	$\text{Ret.}\%$	Wiki↓	$\text{Cmn}_7 \uparrow$	$\text{Ret.}\%$
–	Dense	6.40	70.2	100.0	–	–	–
20%	SlimLLM	17.36	57.5	81.9	10.17	62.0	88.3
	SlimGPT	19.24	55.5	79.0	10.88	61.9	88.1
	LLM-Pruner	13.23	58.7	83.7	9.86	61.3	87.4
	Random	21.20	50.5	72.0	11.32	62.4	89.0
	CoCurve	12.88	61.0	87.0	10.04	62.6	89.2
30%	SlimLLM	78.23	47.4	67.6	13.93	56.3	80.2
	SlimGPT	196.79	42.8	61.0	14.69	57.1	81.4
	LLM-Pruner	26.64	52.2	74.4	13.01	55.2	78.6
	Random	52.90	42.1	60.0	15.29	56.8	80.9
	CoCurve	21.59	54.3	77.4	12.64	58.9	83.9
40%	SlimLLM	183.53	40.8	58.2	19.61	51.9	74.0
	SlimGPT	26002	35.2	50.2	24.27	50.4	71.8
	LLM-Pruner	103.35	45.7	65.1	17.43	52.1	74.2
	Random	4168	36.4	51.8	20.04	52.2	74.4
	CoCurve	59.13	45.5	64.8	16.33	53.7	76.5

3.6 BEYOND ONE TOWER: VISION–LANGUAGE MODELS

Nothing in Section 2 is specific to a decoder-only stack. The risk is a KL divergence between a frozen model and a masked copy of it; the estimator is a Gram matrix of single-unit ablation features; the selector is a cost-normalised greedy under one budget. Point those three objects at a vision–language model and the only thing that changes is that the unit inventory grows a second tower. Everything else is held fixed: the same M single-unit ablations, the same $\mathbf{H}_{uv} = \bar{D}_u^\top \bar{D}_v / P$, the same λ protocol of Section 2.4, the same guard, no gradients, no fine-tuning and no adapter. One piece of bookkeeping is new — the layer index is composite, so the per-layer cap treats a vision layer and a language layer as distinct layers rather than pooling them — and it introduces no hyperparameter.

Protocol. The table below reports five vision–language models spanning 2B to 8B and three model families — Qwen2-VL (Wang et al., 2024a), Qwen2.5-VL (Bai et al., 2025), Qwen3-VL (Qwen Team, 2025), InternVL3 (Zhu et al., 2025) and LLaVA-1.5 (Liu et al., 2024a); the full grid of seven models at three ratios, adding LLaVA-NeXT (Liu et al., 2024b), is in Section E.9, on the seven benchmarks this literature reports as a matter of course — MMBench-EN (Liu et al., 2024c), SEED-Bench (Li et al., 2024), ScienceQA-IMG (Lu et al., 2022), AI2D (Kembhavi et al., 2016), MMStar (Chen et al., 2024b), POPE (Li et al., 2023) and MME (Fu et al., 2023) — at 1000 items each, with the dense model measured under our own harness rather than quoted. Calibration is a fixed draw of Flickr30k image–caption pairs (Young et al., 2014), whose images are disjoint from the COCO-derived suites by construction; Section E.9 gives the draw for each model. The baselines are the training-free structured selectors this field returns to: Magnitude (Han et al., 2015), Wanda-sp

Table 6: Vision–language models, mean accuracy over seven benchmarks at 1000 items each. Every selector is training-free and runs under one shared budget and one shared guard. Retention is floor-corrected against the dense model measured here.

	Qwen2-VL-2B	Qwen2.5-VL-7B	Qwen3-VL-8B	InternVL3-8B	LLaVA-1.5-7B
Selector	20%	20%	20%	20%	20%
Dense (no pruning)	73.0	81.6	83.4	83.6	64.1
CoCurve	59.3	74.4	71.1	75.9	57.1
diagonal only ($\lambda=0$)	53.7	73.5	66.5	73.9	50.0
LLM-Pruner	57.0	72.2	65.8	71.8	53.7
Wanda-sp	51.5	62.9	40.5	49.8	36.0
FLAP	36.6	73.3	68.4	74.8	41.1
ShortGPT (layer drop)	35.5	35.3	35.2	43.4	34.6
Magnitude	40.3	72.9	57.0	70.7	48.1
Random	48.0	64.4	53.8	67.9	52.8
<i>CoCurve retention of dense</i>	66%	86%	76%	85%	78%

(Sun et al., 2024), FLAP (An et al., 2024), LLM-Pruner’s gradient Taylor score (Ma et al., 2023), a ShortGPT-style layer drop (Men et al., 2024), and Random. Every arm receives the identical budget, guard, calibration draw and scoring code; the only thing that differs between rows is which units get removed. Section E.9 gives the per-benchmark breakdown and every configuration measured.

Result. CoCurve leads every baseline in all five columns, by 1.1 to 3.4 points of seven-benchmark mean, and is first outright in 22 of the 35 per-benchmark cells against a standard error of 0.5 points. No single baseline is the one being beaten: the runner-up is FLAP in three columns and LLM-Pruner in two columns. Because the budget is shared and the objective is cost-normalised, the split between towers is an output rather than a hyperparameter: on the five models whose vision tower holds under a tenth of the removable parameters it spends nothing there at all, and on Qwen2-VL-2B, where the vision tower holds a third of them, it takes a fifth of the budget from the vision side. Nothing in the objective was told to do either.

Along the ratio. Table 7 carries Qwen2.5-VL-7B and Qwen3-VL-8B to $\rho=30\%$. The margin grows with the budget rather than shrinking — 1.1 to 4.9 points on Qwen2.5-VL-7B and 2.6 to 10.8 on Qwen3-VL-8B — which is what a second-order objective should do if the interactions it models are real: the more units leave together, the more the coupling between them matters. Section E.9 carries the same axis out to $\rho=40\%$ on all seven models of the grid.

Across the grid. Section E.9 tables all 21 model–ratio cells — every model, every arm, every benchmark. At $\rho=20\%$ the result is not confined to the columns above: on all seven models no baseline is ahead of CoCurve by as much as 1 point, and CoCurve is ahead by more than 2 standard errors on five of them. At $\rho=40\%$ the axis has closed — no arm on any model retains a third of its dense score above chance — and a seven-benchmark mean no longer orders selectors that all sit within a few points of one another. What separates methods is the budget in between, which is where Table 7 sits.

What the off-diagonal is worth. Setting $\lambda=0$ keeps the budget, the guard and the saliency term and deletes only the off-diagonal. Table 8 runs that on every vision–language model we built a curvature matrix for, at one ratio and one item count: it costs between 1.9 and 7.6 points of seven-benchmark mean, 3 to 13 times the standard error of the measurement, and it costs something on all seven models. This is the same effect the text-only half of the paper reports, in an architecture the estimator was never specialised for — and because both arms are ours, run under one budget, one guard and one calibration draw, it isolates the off-diagonal without routing the conclusion through anyone else’s implementation.

Table 7: The ratio axis on Qwen2.5-VL-7B and Qwen3-VL-8B, 1000 items per benchmark. Same seven benchmarks, same arms, same guard as Table 6.

Selector	Qwen2.5-VL-7B		Qwen3-VL-8B	
	20%	30%	20%	30%
Dense (no pruning)	81.6	81.6	83.4	83.4
CoCurve	74.4	63.6	71.1	57.7
diagonal only ($\lambda=0$)	73.5	61.1	66.5	50.1
LLM-Pruner	72.2	57.2	65.8	47.0
Wanda-sp	62.9	52.2	40.5	35.9
FLAP	73.3	58.5	68.4	46.5
ShortGPT (layer drop)	35.3	35.2	35.2	35.3
Magnitude	72.9	58.7	57.0	37.7
Random	64.4	46.2	53.8	41.2
<i>CoCurve retention of dense</i>	86%	64%	76%	50%

Table 8: What the off-diagonal buys, per model, at $\rho=30\%$ and 1000 items per benchmark. $\lambda=0$ keeps the budget, the guard and the diagonal saliency term and removes only the coupling; Δ is what that costs.

Model	Dense	CoCurve	$\lambda=0$	Δ
Qwen3-VL-8B	83.4	57.7	50.1	+7.6
LLaVA-1.5-7B	64.1	43.6	36.4	+7.2
Qwen2.5-VL-3B	77.4	50.0	42.9	+7.1
InternVL3-8B	83.6	62.4	55.5	+6.9
Qwen2.5-VL-7B	81.6	63.6	61.1	+2.5
LLaVA-NeXT-7B	70.0	53.0	50.7	+2.3
Qwen2-VL-2B	73.0	39.7	37.8	+1.9

4 THE ANATOMY OF CO-PRUNING CURVATURE

The edge term is not a regulariser that happens to help. It acts through a structure we can measure and a population of units we can name, and deleting exactly that population reproduces the damage on demand. Figure 3 is that argument end to end. The matrix has cross-module and long-range mass on every model we estimate (a–d), and that is not a property of a decoder-only stack: the same statistic on the 7 vision–language matrices of Section 3.6 puts the attention–FFN block at 0.93–0.99 of the within-module mean, against 0.74–0.98 on the text-only models, so panel (c) carries both families. A two-tower model also measures the quantity a single stack cannot: coupling between a language unit and a vision unit runs 0.58–0.89 of the coupling inside a tower, which is why one shared budget across both towers is a measurement rather than a preference. That mass singles out a minority of units node saliency cannot see (e). Forcing those units out costs nineteen of twenty capability cells across five models (f). Nothing in the top row depends on a selection, a λ , or a benchmark; it is a property of the estimated $\mathbf{H}$.

The interactions node-first scoring discards are the same size as the ones it keeps. The curvature correlation $|\rho_{uv}|$ puts every entry of $\mathbf{H}$ on one scale, so the blocks are directly comparable. The attention $\leftrightarrow$ FFN block is not a correction term. It runs at 0.74 to 0.98 times the mean of the two within-module blocks across five models (Figure 3c), and on Mistral-Nemo-12B it reaches 0.130 against 0.155 within attention. This is the quantitative content of treating pruning as a joint operation across module types. It is also what makes a *shared* attention/FFN budget meaningful at all: without a term coupling the two types, their saliencies live on incommensurable scales and any joint ranking is arbitrary.

And they are not local. Coupling peaks at a layer distance of zero or one, at 1.1 to 1.6 times the average pair, and decays slowly from there. It still stands at 0.33 to 0.53 of the average across the full depth of the network (Figure 3d), and the layer $\times$ layer view shows it massed in the first half of the depth (Figure 3b), where every cell in the top decile of off-diagonal mass lies. A curvature model

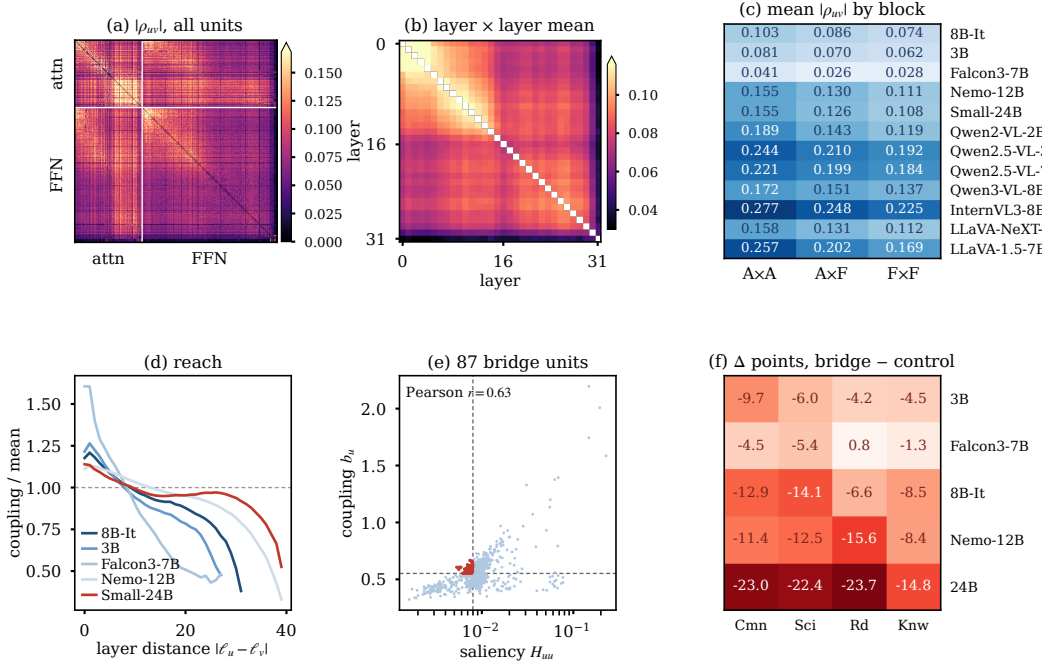


Figure 3: **The anatomy of the edge matrix, and what ignoring it costs.** Curvature correlation $|\rho_{uv}| = |\mathbf{H}_{uv}| / \sqrt{\mathbf{H}_{uu}\mathbf{H}_{vv}}$ puts every entry on one scale. (a) the estimated matrix, units ordered attention-then-FFN: the off-diagonal quadrants are populated, not residual. (b) the same matrix averaged into layer $\times$ layer cells: mass persists far from the diagonal band and concentrates in the first half of the depth. (c) the attention $\leftrightarrow$ FFN block is 0.74–0.98 times the mean of the two within-module blocks on all five text-only models, and 0.93–0.99 on the 7 vision–language models (rows marked VL), which the estimator was never specialised for. (d) coupling by layer distance, normalised so 1.0 is the average pair. (e) saliency against coupling, one point per unit; red is the bridge corner. (f) matched-pair test: points lost by forcing the bridge corner out against a control matched on saliency, unit type and cost, with one greedy fill shared by both arms. Twenty cells, five models, nineteen negative; the colour scale spans the full range, so the largest model’s cells dominate it and the exact value is printed in every cell.

that is block-diagonal per layer, or confined to one module group, is therefore not an approximation of this matrix that keeps most of its mass. It discards the part that spans the residual stream.

The structure names a minority of units. Write $d_u = \mathbf{H}_{uu}$ for a unit’s own saliency and $b_u = \sum_{v \neq u} |\mathbf{H}_{uv}|$ for its total coupling. The two do not move together at the bottom of the saliency range: 87 of 768 units on Llama-3.1-8B-Instruct sit in the low- d_u , high- b_u corner of Figure 3e, and that corner holds under 15% of units on every one of the ten models (3.5 to 14.5%). A diagonal selector sees only d_u and removes them early. CoCurve sees the second sum in Eq. 8 grow as their partners are taken, and holds them back. These are *bridge* units.

What the diagonal cannot see is where a unit sits. Coupling falls with depth on every one of the ten models: the rank correlation $\rho_s(\ell_u, b_u)$ between a unit’s layer index and its coupling runs from -0.54 to -0.91 , median -0.87 , and is negative on 10 of 10. Saliency carries no such signal. Its $|\rho_s|$ has median 0.18, and the *sign* is positive on five models and negative on five (Table 12). Depth is therefore not a weak feature of node saliency, it is absent from it, and it is present in the off-diagonal at close to four times the rank strength (median ratio $3.9\times$). That is what makes the bridge corner an early-layer population: coupling and depth are close to the same axis, and the corner is where low saliency meets high coupling. It is also why the allocation below is not a heuristic we imposed. The solver holds the early layers back because the second sum in Eq. 8 is largest there.

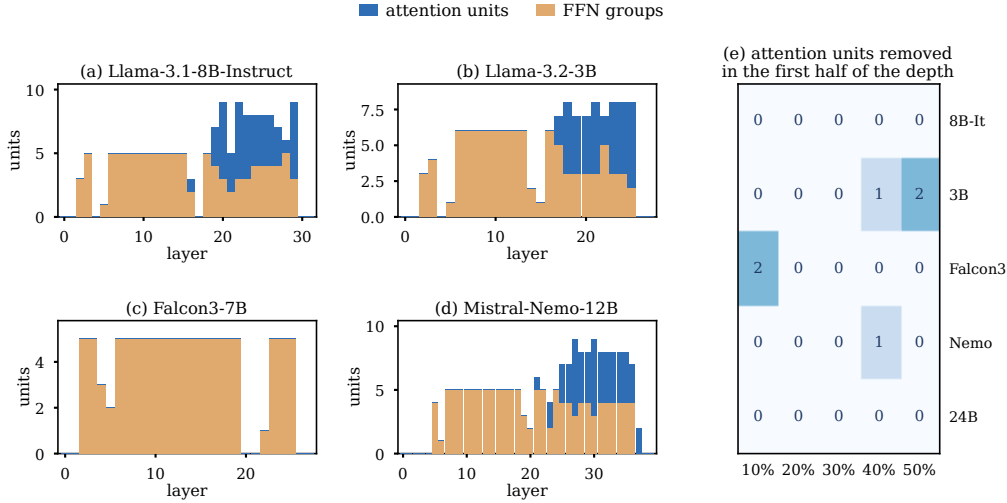


Figure 4: **What one shared budget decides on its own**, read off the deployed solutions. (a)–(d) per-layer, per-type removal at $\rho=20\%$: the first half of the depth is spent entirely on FFN width and attention is touched only beyond it; on Falcon3-7B the objective declines to remove attention anywhere. (e) the same quantity over five models and five ratios: eight of the twenty-five solutions remove no attention anywhere, and of the seventeen that do, thirteen remove *exactly zero* from the first half and no exception exceeds two units. Mistral-Small-24B is not on the machine that produced (a)–(d); its counts come from the same deployed masks. Nothing imposes this; the guard constrains only per-layer totals.

Deleting exactly those units reproduces the damage. Structure in $\mathbf{H}$ is not yet evidence about the model, so we force the issue, on every model we build $\mathbf{H}$ for. We prune the bridge units outright against a control matched unit-for-unit on saliency, unit type and parameter cost but drawn from the low-coupling population. One greedy fill, computed once and shared by both arms, completes them, so the two masks hold the same number of units, the same attention/FFN split and the same removed cost to the last bit, and differ only in which low-saliency population was forced out. On Llama-3.1-8B-Instruct commonsense falls 12.9 points, science 14.1, world knowledge 8.5 and reading 6.6 — and WikiText perplexity *improves* by 4.5.

That dissociation is the mechanism in its sharpest form, and it is not one model’s accident. Across five models, nineteen of the twenty capability cells fall, by up to 23.7 points, and the largest model gives the largest effect: on Mistral-Small-24B all four groups fall by 14.8 to 23.7 (Figure 3f, per-model numbers in Table 23). On three of the five, perplexity moves the other way while every capability group in that row falls, so the arm a next-token proxy prefers is the arm that has lost the most capability. The claim is not that perplexity always points the wrong way — on Mistral-Small-24B the bridge removal costs 425 perplexity as well, and there the two agree — but that nothing makes it point the right way, and on the majority of models here it does not. What node-first scoring mis-reads is not how important a unit is on its own but what it is holding together, and the proxy such scoring is built on cannot be relied on to notice. This is also why Section 3.2 reports the two families side by side rather than leading with the one our objective is closest to: the same caution applies to our own numbers, and the capability columns are the ones we ask to be believed.

Given one budget, the objective writes its own allocation, and writes the same one every time. Methods in this literature fix the attention/FFN split by hand, or prune the two types in separate stages. We fix nothing beyond the guard, and what comes out is not “some of each everywhere” (Figure 4). The first half of the depth is pruned *entirely* out of FFN width. Not one attention unit is removed before layer 17 of 32 on Llama-3.1-8B-Instruct, layer 18 of 28 on Llama-3.2-3B, or layer 22 of 40 on Mistral-Nemo-12B, counting from one. Attention then takes close to half of the removals over the second half (50%, 49%, 45%) — while on Falcon3-7B and Mistral-Small-24B the objective declines to touch attention at all. This is not a property of one operating point. Of the twenty-five solutions over five models and five ratios, eight remove no attention anywhere, and the

Table 9: **Where CoCurve sits.** Cross-module off-diag.: an explicit attention $\leftrightarrow$ FFN or cross-layer curvature term. SAViT carries one but collapses each block to a scalar, so it acts at component-type rather than unit-pair granularity. SOSP’s SOSP-I variant carries a full pairwise structure matrix, on homogeneous structures in convolutional networks (Section F.1). FLAP adds a compensation bias rather than changing any retained weight, and SlimLLM regresses each sublayer’s output matrix; both are post-pruning updates. Bonsai is the closest on estimation — forward passes only, no weight update — but fits an additive model with no pairwise term, over many rounds of re-estimation. Selection-only: no post-pruning weight update. Single-shot: no iterative re-estimation. Properties describe each method’s native form — ZipLM also has a one-shot mode, but its closed-form weight update keeps it off the selection-only column either way — and in our experiments recovery is disabled for every method (Section 3.1). CoCurve is the only row that holds all six.

Method	Structured	Cross-module off-diag.	Training-free	Selection-only	Gradient-free	Single-shot
AMP (Mugnaini et al., 2025)	✓	✗	✗	✗	✓	✓
2SSP (Sandri et al., 2025)	✓	✗	✓	✓	✓	✗
LLM-Surgeon (van der Ouderaa et al., 2024)	✓	✗	✓	✗	✗	✗
SlimGPT (Ling et al., 2024)	✓	✗	✓	✗	✓	✗
SlimLLM (Guo et al., 2025)	✓	✗	✓	✗	✓	✓
FLAP (An et al., 2024)	✓	✗	✓	✗	✓	✓
Bonsai (Kolawole et al., 2024)	✓	✗	✓	✓	✓	✗
ZipLM (Kurtic et al., 2023)	✓	✗	✗	✗	✓	✗
SAViT (Zheng et al., 2022)	✓	✓	✗	✗	✗	✓
Kwon et al. (2022)	✓	✗	✓	✗	✗	✗
SOSP (Nonnenmacher et al., 2022)	✓	✓	✗	✗	✗	✗
Opt. Brain Connection (Chen et al., 2025)	✓	✗	✗	✗	✗	✗
D ² Prune (Xiong et al., 2026)	✗	✗	✓	✗	✓	✓
CoCurve (ours)	✓	✓	✓	✓	✓	✓

other seventeen remove it only late: thirteen of those seventeen take *exactly zero* attention units from the first half of the depth, and the remaining four take one or two (Figure 4e). Early attention is what the residual stream is built on. A criterion that never sees the coupling between module types has no way to discover that, and here it falls out of a single minimisation in five models spanning three distinct attention geometries: (24, 8, 128), (12, 4, 256) and (32, 8, 128) query heads, KV heads and head dimension.

What this asks of the field. Structured pruning has spent its effort on better node scores, and a term none of them contain sets the ceiling on that programme. The pair term is measurable in any dense Transformer, it costs one ablation per unit and no gradients, and the units it protects can be located in a model before anything is pruned. The question to ask is not which units matter, but which units matter *together*. The object of study that follows is the matrix, not the ranking.

5 RELATED WORK

Training-free structured pruning of LLMs. The dominant paradigm scores each removable unit independently and deletes the lowest-ranked ones, using first/second-order Taylor importance (Ma et al., 2023), activation statistics (An et al., 2024; Sun et al., 2024), rotate-and-slice projections (Ashkboos et al., 2024), per-layer OBS (Ling et al., 2024; Frantar & Alistarh, 2023; 2022), or least-squares output reconstruction (Guo et al., 2025). A parallel line prunes whole layers (Men et al., 2024; Kim et al., 2024; Yang et al., 2024) or searches a global layer-wise budget (Li et al., 2025; Jansen et al., 2026). All share a node-first assumption. Set-level pruning damage is treated as additive, and even when several module types are scored they are merged into one independent ranking. These are our baselines and concurrent comparisons. CoCurve departs by making the cross-unit interaction a first-class, derived quantity. Table 9 positions it against interaction-aware and modern structured pruners.

Beyond the diagonal: interaction-aware pruning. CoCurve descends from classical second-order saliency (LeCun et al., 1989; Hassibi & Stork, 1993) and Fisher/Taylor importance for modern networks (Theis et al., 2018; Molchanov et al., 2019). Diagonal CoCurve is an OBD-style structured saliency (Eq. 4). Curvature beyond the diagonal is not new either. What CoCurve contributes is a

specific object, and it is worth naming its properties because they are what the rest of this section is measured against: an attention $\leftrightarrow$ FFN, cross-layer Fisher block at *unit-pair* granularity, recovered from M single-unit ablations with no gradients and no pairwise sweep, and used to decide *which individual units* go under a training-free selection-only solve.

Each of the four properties is where a near neighbour stops. LLM-Surgeon’s KFAC curvature is Kronecker-factored within a single weight matrix and explicitly disregards cross-layer interactions (van der Ouderaa et al., 2024): no cross-module block. SlimGPT models coupling inside a layer and compensates as it prunes (Ling et al., 2024), and ZipLM couples layers only through a runtime-aware sparsity profile rather than an explicit curvature term (Kurtic et al., 2023): no cross-layer curvature, and neither is selection-only. SAViT comes closest in spirit — an explicit cross-component second-order block for vision Transformers — but collapses each block to its scalar mean (Zheng et al., 2022), so its interaction sets *how much* of each component type to remove while the choice among individual units stays independent: no unit-pair granularity. Two works come at the object we build from opposite sides, and together they say why it has stayed out of reach. Kwon et al. (2022) write the pairwise Fisher between *unit* mask variables and note in as many words that the interaction between two heads that cover for one another lives in its non-diagonal elements — then approximate that matrix block-diagonally, one block per attention or FFN layer, with no interaction between mask variables in different layers. Yang et al. (2023) write the same object for vision Transformers, observe that a model has tens of thousands of structural groups, declare computing it directly infeasible, and retreat to a per-unit norm. The block CoCurve estimates is exactly what both of them excluded, and the Gram construction of Section 2.2 is why we can afford it: M forward ablations, no gradients, no pairwise sweep. Section F.1 does the same accounting for the remaining neighbours — SOSP (Nonnenmacher et al., 2022), the Combinatorial Brain Surgeon (Yu et al., 2022), whose weight-level *cancellation* argument is the ancestor of ours, and Optimal Brain Connection (Chen et al., 2025), which assumes the block-diagonal form we discard.

Recovering what compression costs. Restoring capability after pruning belongs to pipelines that retrain, such as the distillation and continued pretraining of Minitron (Muralidharan et al., 2024) or evolutionary search (Tang et al., 2025). That is a different regime from the selection-only one studied here, where the question is which units to remove before any weight moves (Jaiswal et al., 2024). Section 3.5 reports what a light recovery stage adds when it is enabled for every method at once.

Compressing a vision-language model. The dominant axis there is the visual token sequence rather than the weights: FastV (Chen et al., 2024a), SparseVLM (Zhang et al., 2025) and MADTP (Cao et al., 2024) shorten the sequence a decoder attends to and leave every parameter in place. That is orthogonal to removing structure — one shortens the input, the other shrinks the network — and the two compose. Where vision-language work does remove weights it trains: UPop (Shi et al., 2023) searches subnets and retrains progressively, and EfficientLLaVA (Liang et al., 2025) searches a per-component pruning policy while optimising the vision projector. Section 3.6 is the training-free structural setting in that picture, and the property that makes it a test of this paper rather than a second paper is that one $\mathbf{H}$ spans both towers, so the budget split between them is an output of the objective.

6 CONCLUSION

Structured LLM pruning has been operating at the wrong granularity: it ranks units as independent nodes, while Transformer computation couples them through a shared residual stream. CoCurve replaces that ranking with global co-pruning-risk minimisation, from a single Fisher matrix that is node saliency and cross-module co-pruning curvature at once, recovered from single-unit ablations and solved in one shot under a shared budget. How much to trust the off-diagonal is read off a finite path of solutions by a held-out, label-free criterion, using the same rule at every compression ratio. Across five models from 3B to 24B and three compression ratios, CoCurve leads on 38 of 45 perplexity cells and 32 of 60 capability-group cells against the per-column best of every baseline, and on 666 of 721 cells against those baselines one at a time. The margin is largest where the field breaks: on Mistral-Small-24B at $p=40\%$ its perplexity is 277.1 against the best baseline’s 3299. Our ablations attribute the gain to the off-diagonal rather than to node saliency, and it survives a control that fixes the attention-FFN split at the diagonal selector’s own allocation. The term acts through a

population we can name and delete on demand, and it carries positional structure the diagonal does not: coupling falls with depth on all ten models, while saliency does not even keep its sign. Under one shared budget the objective then recovers, without being told to, an allocation that spends the first half of the depth on FFN width alone: of the twenty-five solutions across five models and five ratios, seventeen remove attention at all, and thirteen of those take none of it from the first half.

None of this is tied to one modality. The same three objects — a KL risk, a Gram matrix of single-unit ablations, a budgeted greedy — transfer to vision–language models unchanged, where they also measure something a single stack cannot: the curvature coupling a language unit to a vision unit, which runs 0.58–0.89 of the coupling inside a tower and is why one budget shared across both towers is a measurement rather than a preference.

Limitations and outlook. Estimating the edge matrix costs hours rather than minutes, but only once per model. It is independent of the ratio and of λ , so one estimate serves the whole path. Turning a mask into a smaller model needs a slicing path per operator family. What a removable unit *is* is the open question: under mixture-of-experts routing the units are already sparsely activated, and in stacks that interleave attention with linear-recurrent layers some layers have no heads to remove. The risk of Eq. 1 is defined in both cases; the inventory it is minimised over is not, and extending it there is the natural next step. The inventory changes; the object does not. Wherever units share a residual stream, the damage from removing a set is not the sum of the damages, and the term that says so is measurable without a single gradient: *prune the edges, not just the nodes*.

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APPENDIX OVERVIEW

The appendix is organised into six parts that follow the paper’s evidence chain. Parts A–B complete the method and theory; Part C fixes the shared experimental protocol; Part D provides the full result record; Part E collects mechanism, ablation and robustness evidence; and Part F records scope and reproducibility information. Parts are top-level sections; the heavy ones group their material into subsections.

- **Section A, method and algorithmic details.** Notation, the structured-unit and mask semantics with the physical-removal equivalence test, the estimator and its Gram construction, the solver with its guards and complexity, and the closed-form compensation the same matrix affords.
- **Section B, theoretical analysis.** The risk geometry that makes pruning purely second order, the Gram identity and the additivity bound behind the surrogate, the reduction of diagonal CoCurve to structured Optimal Brain Damage, why the calibrated interaction strength is small, and the solver’s marginal-risk and overshoot guarantees.
- **Section C, experimental setup.** Datasets and splits, models and per-model configurations, baseline descriptions, the evaluation protocol, and the scope of the suite.
- **Section D, complete results.** The per-task record behind every main-text table, the ratio sweep from 10% to 50%, the recovery stage, and efficiency.
- **Section E, mechanism, ablations, and robustness.** Extended ablations, surrogate fidelity, the mechanism evidence (edge spectra, bridge units, layer structure), the robustness of the selection under resampling and independent calibration, and how the method behaves where the edges carry least signal.
- **Section F, discussion.** Extended related work, limitations, broader impact, and the reproducibility checklist.

A METHOD AND ALGORITHMIC DETAILS

A.1 NOTATION

Table 10 collects the symbols used throughout the paper.

Table 10: **Summary of notation.**

Symbol	Meaning
<i>Units, masks, and budget</i>	
$\mathcal{U} = \{u_1, \dots, u_M\}$	the M removable structured units (attention heads, FFN channel groups)
s_u, s	pruning variable ($s_u=1$ removes unit u) and the full mask vector; the deployed mask is binary, and s is real-valued near $\mathbf{0}$ so that Eq. 3 is well posed
$m_u = 1 - s_u$	keep gate scaling unit u 's write into the residual stream
$c_u = \text{Param}(u)$	removable parameter cost of unit u
ρ	target pruning ratio (removed non-embedding parameter fraction)
$S, S $	selected (pruned) unit set and its size; $\mathcal{K}$: kept FFN groups (compensation)
<i>Curvature matrix (the central object)</i>	
$\mathbf{H} \in \mathbb{R}^{M \times M}$	co-pruning curvature (Fisher) matrix, as estimated from the M ablations (Eq. 6)
$\mathbf{H}^* = \mathbb{E}[J^\top F J]$	the exact curvature $\mathbf{H}$ estimates, with $J = \partial z_s / \partial s _{s=\mathbf{0}}$ (Remark 2)
$\mathbf{H}_{uu}$	node saliency (diagonal): an OBD-style importance
$\mathbf{H}_{uv}$	co-pruning curvature edge (off-diagonal): the joint-removal cost of u and v
$\rho_{uv} \in [-1, 1]$	curvature correlation $\mathbf{H}_{uv} / \sqrt{\mathbf{H}_{uu}\mathbf{H}_{vv}}$; signed, since cross terms cancel. Figure 3 reports its magnitude $ \rho_{uv} $
<i>Objective and estimation</i>	
$\mathcal{R}(s)$	pruning risk: token-level KL between teacher and masked model (Eq. 1)
$\hat{\mathcal{R}}_\lambda(s)$	the surrogate the solver minimises, $\lambda=1$ written $\hat{\mathcal{R}}(s) = \frac{1}{2}s^\top \mathbf{H}s$ (Eq. 7)
$\Delta z_s = z_0 - z_s$	logit perturbation of the masked model, $\approx \sum_u s_u \delta z_u$
$\Delta R(u S), g_u^{(S)}$	marginal risk of adding u to S , and its interaction accumulator (Eq. 8)
$p_0, z_0 / p_s, z_s$	teacher (frozen) and masked-model next-token distributions / logits
F	per-token Fisher $\text{diag}(p_0) - p_0 p_0^\top$ in logit space
$\delta z_u, \tilde{\delta z}_u$	single-unit logit perturbation from ablating u ; its whitened feature ($\mathbf{H}$ is their Gram)
$\mathcal{C}, \mathcal{C}'$	calibration set building $\mathbf{H}$, and the disjoint split $\hat{\lambda}$ is measured on (Eq. 11)
r, n_f, P	top- r logit support, FFN groups per layer, total calibration positions
<i>Solver and compensation</i>	
$\lambda \in [0, 1], \hat{\lambda}$	interaction strength scaling the off-diagonal, and the value selected on held-out risk (Eq. 9, Eq. 11)
$\mathcal{F} = \{S_\lambda\}, \mathcal{F} $	the finite family of prunings the path enumerates, and its size
$\kappa, \mathcal{L}_p$	per-layer removed-cost cap and the set of protected layers; one rule, $\kappa=1.5\rho$ with two layers protected at each end (Section C.2)
μ_u, ν_u	intercept and λ -slope of the cost-normalised marginal (Eq. 10); the break-points of the path are their crossings
$b_u = \sum_{v \neq u} \mathbf{H}_{uv} $	a unit's total coupling; with $\mathbf{H}_{uu}$ it defines the <i>bridge</i> corner (Section E.3)
$\bar{\rho}_S$	the $\sqrt{\mathbf{H}_{uu}\mathbf{H}_{vv}}$ -weighted mean of ρ_{uv} over in-set pairs (Proposition 5)
γ, τ, α	compensation ridge strength, clip radius, and rescale coefficients (Eq. 12)

A.2 STRUCTURED UNITS, MASK SEMANTICS, AND PHYSICAL PRUNING

Attention units. For multi-head attention (MHA), one head is one unit; pruning head i removes the corresponding rows of W_Q, W_K, W_V and the input columns of W_O . For grouped-query attention (GQA) (Ainslie et al., 2023) with $n_g = n_q/n_{kv}$ query heads per KV head, the unit is the KV-aligned query-head group $\mathcal{I}_g = \{gr_{kv}, \dots, (g+1)r_{kv}-1\}$; pruning group g removes the rows of

Algorithm 1 CoCurve matrix estimation (single-unit, top- r Fisher Gram)

Require: frozen f_0 , calibration $\mathcal{C}$, units $\mathcal{U}$, top- r , token budget P_{tok}
Teacher pass: for each $x \in \mathcal{C}$, select token positions $\mathcal{T}(x)$; store top- r indices $\mathcal{V}_r(x, t)$ and renormalised probs p_0
for each unit $u \in \mathcal{U}$ **do** *// M passes; embarrassingly parallel*
 run f_{-u} (mask only u); gather z_{-u} on the teacher support $\mathcal{V}_r(x, t)$
 $\delta z_u \leftarrow z_0 - z_{-u}$; $\tilde{\delta z}_u \leftarrow \sqrt{p_0} \odot (\delta z_u - \mathbb{E}_{p_0}[\delta z_u])$
 append $\tilde{\delta z}_u$ to feature store $\bar{D}_u$ (CPU/memmap)
end for
 $\mathbf{H} \leftarrow \frac{1}{P} \bar{D}^\top \bar{D}$ (blockwise Gram); $\mathbf{H} \leftarrow \frac{1}{2}(\mathbf{H} + \mathbf{H}^\top)$
 assert $\min_u \mathbf{H}_{uu} \geq -\epsilon$ *// non-negativity check (Proposition 3)*
return $\mathbf{H}$

W_Q for $i \in \mathcal{I}_g$, the rows of W_K, W_V for KV head g , and the matching input columns of W_O . The head dimension d_h is kept fixed (we never re-infer $d_h = d/n_q$ after pruning, which would corrupt models that derive d_h from the head count); a patched attention module stores the reduced n'_q, n'_{kv} explicitly.

FFN groups. The intermediate dimension is partitioned into n_f contiguous groups. Because the map from the (gated) intermediate activation to the FFN output is linear, the output is exactly the sum of per-group contributions, so pruning group j removes the rows of $W_{\text{gate}}, W_{\text{up}}$ (or W_1) and the columns of W_{down} (or W_2) for that group. The runtime mask zeros the gated product before the down projection; equivalently it zeros those up/gate rows, since the SwiGLU/GELU gate satisfies $\sigma(0) \cdot (\cdot) = 0$.

Mask boundary and equivalence. A runtime mask zeros only a unit’s residual-stream contribution and never alters normalisation, RoPE, residual addition, causal/padding masks, GQA repeat rules, or dropout (the model is in eval mode). Two things follow, and we check them separately. Zeroing a unit’s weight slices is bit-identical to masking it at runtime, which we verify on a held-out batch, so a masked evaluation and a zeroed one are the same measurement. Slicing the weight matrices instead is what the timings run on, and its correctness is checked by the realised parameter count of every timed model: on Llama-3.1-8B-Instruct at $\rho=20\%$ the sliced model holds 6.633B parameters against the dense 8.030B, and the logit equivalence of the slicing path is verified on a synthetic grouped-query Llama to 10^{-4} . Reported speedups therefore come from a smaller model, not from zeroed weights in a full-size one.

A.3 THE COCURVE ESTIMATOR

Algorithm 1 details the estimator. The crux (Proposition 3) is that gathering every unit’s ablated logits on the same teacher top- r support $\mathcal{V}_r(x, t)$ places all single-unit features in one coordinate system, so $\mathbf{H}$ is a Gram matrix and the off-diagonal edges cost nothing beyond the M diagonal ablations. We never materialize the $V \times V$ Fisher.

Top- r coordinate consistency. Using a per-unit top- r would place different units in different coordinate systems and invalidate the Gram; the shared teacher support is essential. Defaults: $r=256$, 128 C4 sequences of length 2048, deterministic token subsampling ($P_{\text{tok}}=128$, so $P = 16384$ stored positions). Increasing r beyond 256 leaves $\mathbf{H}$ nearly unchanged (Section E.1).

Optional prefix-cache replay. An exact speedup caches residual states before the masked computation and replays everything after: attention ablations replay from the pre-attention residual, FFN ablations from the pre-FFN residual. Multiple masks can share an input batch via the batch dimension. This changes wall-clock, not the estimate.

A.4 THE GREEDY CO-PRUNING SOLVER

Algorithm 2 summarises the full CoCurve pipeline end to end (calibrate, ablate, build the Gram edge matrix, greedily co-prune, physically remove). Algorithm 3 gives the solver. By Proposition 6,

Algorithm 2 CoCurve: end-to-end co-pruning pipeline.

Require: frozen model f_0 , calibration set $\mathcal{C}$, units $\mathcal{U}$ with costs $\{c_u\}$, ratio ρ , interaction strength $\hat{\lambda}$ (Eq. 11), per-layer cap κ , protected layers $\mathcal{L}_p$
run f_0 on $\mathcal{C}$; cache token positions, teacher top- r indices and probabilities
for each unit $u \in \mathcal{U}$ **do**
mask only u ; gather ablated logits on the teacher’s top- r indices; form $\tilde{\delta}z_u$ // M ablations
end for
 $\mathbf{H} \leftarrow \frac{1}{P} \bar{D}^\top \bar{D}$; symmetrize $\mathbf{H} \leftarrow \frac{1}{2}(\mathbf{H} + \mathbf{H}^\top)$ // Gram, Eq. 6
 $S \leftarrow \emptyset$, $g \leftarrow \mathbf{0}$, $C_S \leftarrow 0$
while $C_S < \rho \sum_u c_u$ **do**
 $\mathcal{A} \leftarrow \{u \notin S : \text{layer}(u) \notin \mathcal{L}_p, c_u \text{ fits the layer cap } \kappa\}$; **if** $\mathcal{A} = \emptyset$ **then break**
 $u^* \leftarrow \arg \min_{u \in \mathcal{A}} (\frac{1}{2} \mathbf{H}_{uu} + \hat{\lambda} g_u) / c_u$ // Eq. 8
 $S \leftarrow S \cup \{u^*\}$; $g \leftarrow g + \mathbf{H}_{:,u^*}$; $C_S \leftarrow C_S + c_{u^*}$
end while
slice the weight matrices of S ; verify the realised parameter count
return compressed model f_S and $\rho_{\text{actual}} = C_S / \sum_u c_u$

Algorithm 3 One-shot cost-normalised greedy co-pruning

Require: $\mathbf{H}$, costs $\{c_u\}$, ratio ρ , interaction strength $\hat{\lambda}$, per-layer cap κ , protected layers $\mathcal{L}_p$
 $S \leftarrow \emptyset$; $g \leftarrow \mathbf{0}$; $C_S \leftarrow 0$
while $C_S < \rho \sum_u c_u$ **do**
 $\mathcal{A} \leftarrow \{u \notin S : \text{layer}(u) \notin \mathcal{L}_p, C_{\text{layer}(u)} + c_u \leq \kappa C_{\text{layer}(u)}^{\text{tot}}\}$; **if** $\mathcal{A} = \emptyset$ **then break**
 $u^* \leftarrow \arg \min_{u \in \mathcal{A}} (\frac{1}{2} \mathbf{H}_{uu} + \hat{\lambda} g_u) / c_u$
 $S \leftarrow S \cup \{u^*\}$; $g \leftarrow g + \mathbf{H}_{:,u^*}$; $C_S \leftarrow C_S + c_{u^*}$
end while
slice the weight matrices of S ; verify the realised parameter count
return f_S , $\rho_{\text{actual}} = C_S / \sum_u c_u$

maintaining the running interaction vector $g_u = \sum_{v \in S} \mathbf{H}_{uv}$ makes each step $O(M)$. Two guards keep pruning distributed without changing the objective: a per-layer cost cap and first/last-layer protection. Both are fixed by one rule, $\text{cap} = 1.5\rho$ with the first and last two layers protected, applied identically to every model, every ratio and every method, with no per-model choice and no reference ratio (Section C.2).

A.4.1 COMPLEXITY AND COST

Estimation is M forward-only ablation passes over $|\mathcal{C}|$ sequences (no backward pass, Hessian-vector product, or pairwise ablation); the Gram is $O(M^2 Pr)$ multiply-adds, accumulated blockwise. The solver is $O(M|S|)$ for $|S|$ selected units. With FFN grouping ($M \sim 10^3$) the whole pipeline is one-shot per (model, ratio) and fits commodity hardware; wall-clock is reported in Section D.4.

A.5 EDGE-DERIVED COUPLING-AWARE COMPENSATION

Selection alone discards every removed unit’s contribution, but the same edge matrix that scored the interactions tells us how to absorb that lost contribution. Removing S shifts the residual stream by $\Delta z_S \approx \sum_{u \in S} \delta z_u$; among the kept FFN groups $\mathcal{K}$ we choose a per-group output rescale $1 + \alpha_v$ that cancels Δz_S in the Fisher geometry, minimizing $\mathbb{E}_{x,t} [(\sum_{v \in \mathcal{K}} \alpha_v \delta z_v - \Delta z_S)^\top F (\sum_{v \in \mathcal{K}} \alpha_v \delta z_v - \Delta z_S)] + \gamma \|\alpha\|_2^2$ over the coefficient vector α . Taking the expectation over positions turns the quadratic and linear terms into blocks of $\mathbf{H}$ itself (Eq. 5). The normal equations are then a small ridge system in edge blocks we have already computed,

$$(\mathbf{H}_{\mathcal{K}\mathcal{K}} + \gamma I) \alpha = \mathbf{H}_{\mathcal{K}S} \mathbf{1}_S, \quad \text{gain}_v = \text{clip}(1 + \alpha_v, 1 - \tau, 1 + \tau), \quad (12)$$

where $\mathbf{1}_S$ is the all-ones vector indexed by S , and the ridge strength γ and clip radius τ are the released defaults of Section C.2. This is one $|\mathcal{K}| \times |\mathcal{K}|$ CPU solve, with no gradients, no data beyond the calibration used to build $\mathbf{H}$, and, unlike OBS-style reconstruction, no per-layer weight inverse. No number reported anywhere in this paper uses it: every result in the main tables and in Table 17 is selection-only, so the comparison stays about which units are removed. We include the derivation because it is the same matrix used a second way. The couplings that say what *not* to remove together also say how the survivors should absorb what was removed, and because it costs one small ridge solve rather than the per-layer weight inverse that OBS-style reconstruction needs.

B THEORETICAL ANALYSIS

We collect the formal results behind Section 2 and, for each, state how it is validated empirically. Notation: p_0 is the teacher distribution, z_0 its logits, $F = \text{diag}(p_0) - p_0 p_0^\top$ the KL Fisher, and $\delta z_u = z_0 - z_{-u}$ the single-unit logit perturbation. We use local additivity $z_0 - z_s \approx \sum_u s_u \delta z_u$; Proposition 4 bounds the third-order remainder of the exact quadratic, and the additivity substitution itself is measured rather than bounded (Section B.2).

B.1 GEOMETRY OF THE PRUNING RISK

Why pruning is purely second order.

Proposition 1 (Vanishing zeroth/first order). *With s ranging over a neighbourhood of $\mathbf{0}$ in $\mathbb{R}^M$ (Section 2), so that $\mathbf{0}$ is interior and $\mathcal{R}$ is differentiable there, $\mathcal{R}(\mathbf{0}) = 0$ and $\nabla_s \mathcal{R}(\mathbf{0}) = 0$; the Taylor expansion at the full model begins at second order.*

Proof. At $s = \mathbf{0}$, $p_0 = p_0$, so every term is $D_{\text{KL}}(p_0 \| p_0) = 0$. For the gradient, with $g(s) = D_{\text{KL}}(p_0 \| p_s)$ and p_s a softmax, $\partial_{s_u} g = -\mathbb{E}_{p_0}[\partial_{s_u} z_s] + \mathbb{E}_{p_s}[\partial_{s_u} z_s]$; at $s = \mathbf{0}$, $p_s = p_0$ and the two expectations cancel. Averaging over positions preserves equality. $\square$

Significance. The absence of a first-order term is what makes the off-diagonal curvature, not a gradient, the carrier of interaction information; it also means saliency itself is second order, unifying our diagonal with classical second-order pruning.

Proposition 2 (Fisher quadratic). *To second order, $D_{\text{KL}}(p_0 \| p_s) \approx \frac{1}{2} \Delta z_s^\top F \Delta z_s$ with $\Delta z_s = z_0 - z_s$; hence under local additivity $\mathbf{H}_{uv} = \mathbb{E}_{x,t}[\delta z_u^\top F \delta z_v]$ (Eq. 5).*

Proof. For $A(z) = \log \sum_y e^{z_y}$ we have $\nabla A = p_0$ and $\nabla^2 A = F$ at z_0 . Writing $\Delta = \Delta z_s = z_0 - z_s$, so that $z_s = z_0 - \Delta$, $A(z_0 - \Delta) = A(z_0) - p_0^\top \Delta + \frac{1}{2} \Delta^\top F \Delta + o(\|\Delta\|^2)$. Since $D_{\text{KL}}(p_0 \| p_s) = -p_0^\top (z_s - z_0) + A(z_s) - A(z_0) = p_0^\top \Delta + A(z_0 - \Delta) - A(z_0)$, the linear terms cancel and $D_{\text{KL}}(p_0 \| p_s) = \frac{1}{2} \Delta^\top F \Delta + o(\|\Delta\|^2)$. Local additivity gives $\Delta z_s \approx \sum_u s_u \delta z_u$ with $\delta z_u = z_0 - z_{-u}$; substituting and taking expectations gives the matrix. $\square$

B.2 TRACTABLE ESTIMATION

The full edge matrix from M ablations.

Proposition 3 (Gram equivalence). *With $\tilde{\delta z}_u = \sqrt{p_0} \odot (\delta z_u - \mathbb{E}_{p_0}[\delta z_u])$, $\delta z_u^\top F \delta z_v = \tilde{\delta z}_u^\top \tilde{\delta z}_v$. Hence $\mathbf{H} = \frac{1}{P} \bar{D}^\top \bar{D} \succeq 0$, where $\bar{D}$ has columns $\bar{D}_u$, with $\mathbf{H}_{uu} = \frac{1}{P} \|\bar{D}_u\|^2 \geq 0$, recoverable from M single-unit ablations with no pairwise computation.*

Proof. $a^\top F b = \sum_y p_{0,y} a_y b_y - (\sum_y p_{0,y} a_y)(\sum_y p_{0,y} b_y) = \sum_y p_{0,y} (a_y - \bar{a})(b_y - \bar{b}) = (\sqrt{p_0} \odot (a - \bar{a}))^\top (\sqrt{p_0} \odot (b - \bar{b}))$. Set $a = \delta z_u, b = \delta z_v$; stack over positions to get the Gram, which is PSD with non-negative diagonal. $\square$

Significance. A model with $O(M^2)$ edges costs only $O(M)$ ablations because each edge is an inner product of two single-unit features. This is what makes edge-aware structured pruning practical, and the symmetrisation and non-negative-diagonal steps of Algorithm 1 are guards on float accumulation, not conditions the estimator could fail: Proposition 3 gives both exactly.

Exact curvature vs. the estimator we use. It is worth separating what is exact from what is approximated, since our claims rest only on the former being a good surrogate, not on equality.

Remark 2 (What is exact, what is a surrogate). Two statements are exact (to the stated order): (a) the KL between teacher and masked model is the Fisher-weighted logit norm to second order, $D_{\text{KL}}(p_0 \| p_s) = \frac{1}{2} \Delta z_s^\top F \Delta z_s + o(\|\Delta z_s\|^2)$ (Proposition 2); and (b) the Gram identity $\delta z_u^\top F \delta z_v = \tilde{\delta z}_u^\top \tilde{\delta z}_v$ (Proposition 3). What we estimate rather than compute exactly is the curvature matrix itself. The exact Hessian, which here coincides with the Gauss–Newton matrix because the logit-space gradient vanishes at $s = \mathbf{0}$ (Proposition 1), is $\mathbf{H}^* = \mathbb{E}[J^\top F J]$ with $J = \partial z_s / \partial s|_{s=0}$; we replace the infinitesimal column J_u by the finite single-unit ablation $\delta z_u = z_0 - z_{-u}$, and we take F from the teacher’s top- r support renormalised to sum to one, so F throughout is the Fisher of that truncated distribution rather than of the full vocabulary. Hence our $\mathbf{H}$ is a finite-difference, renormalised top- r Fisher-Gram surrogate for $\mathbf{H}^*$, exact in the linear-network, infinitesimal-mask, full-vocabulary limit and an empirical proxy otherwise. The truncation behaves the same way at both ends of our scale range: at $r=256$ the retained teacher mass has median 0.993 on Llama-3.2-3B and 0.992 on Mistral-Small-24B, over 16,384 sampled calibration positions each. The tail is worth stating rather than hiding — the fifth percentile is 0.90 and 0.87, the first 0.77 and 0.71, and about two positions in five retain under 0.99 — because the claim that carries weight is not that every position is covered but that the estimator does not depend on how much of the tail is kept, and that is measured directly: r over a fourfold range moves nothing (Table 22). What we rely on is $\mathbf{H}$ ranking co-pruning damage, which we validate directly (Section E.2, Section 4).

Three approximations, separated and measured. Three distinct steps separate the token-level KL from the quadratic the solver minimises. Only the first is asymptotic in $\|s\|$; the other two are fixed properties of the estimator, which is why we measure them rather than bound them.

Proposition 4 (Third-order remainder). *Let $\mathbf{H}^* = \mathbb{E}[J^\top F J]$ with $J = \partial z_s / \partial s|_{s=0}$ be the exact Gauss–Newton curvature of Eq. 1. If the logit map is twice differentiable in the mask variables near $s = \mathbf{0}$ with second derivatives bounded on $\|s\|_1 \leq \delta$, then there is a constant C , depending on that bound and not on s , with $|\mathcal{R}(s) - \frac{1}{2} s^\top \mathbf{H}^* s| \leq C \|s\|_1^3$ for all $\|s\|_1 \leq \delta$.*

The masks we deploy are binary with $\|s\|_1 = |S|$ in the hundreds, well outside any radius on which this is a useful numerical bound. It is stated because it identifies *which* term is missing — third order in the mask, not the two estimator steps below — and the size of that term at the ratios we prune is settled by measurement, not by the constant.

Proof sketch. The KL expansion of Proposition 2 drops an $o(\|\Delta\|^2)$ term whose leading part is the cubic log-partition derivative, $O(\|\Delta\|^3)$ with $\Delta = \Delta z_s$; smoothness bounds $\|\Delta\|$ by $O(\|s\|_1)$. The same smoothness makes $\Delta z_s = J s + O(\|s\|_1^2)$, whose cross term with the quadratic contributes at the same cubic order. $\square$

This bounds $\mathbf{H}^*$, the exact curvature. Two further steps carry us from $\mathbf{H}^*$ to the matrix we build, and we measure both.

(i) Finite ablation in place of the derivative column. We replace J_u by $\delta z_u = z_0 - z_{-u}$, a secant over the whole mask interval rather than a tangent. Scaling a unit’s residual contribution by $1 - \epsilon$ and

forming $\delta z_u^{(\epsilon)} = (z_0 - z_{u,\epsilon})/\epsilon$ for $\epsilon \in \{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}$ lets us compare the two directly. On a stratified sample of units the hard ablation recovers the derivative direction to a median cosine of 0.949–0.981 across four models, with $\|\delta z_u\|/\|J_u\| = 1.03$ –1.07: the secant is the tangent to within a few degrees and a few percent. This comparison requires fp32: in bf16 the small- ϵ difference falls to the logit noise floor and the ϵ -ladder no longer resolves.

(ii) Local additivity. The off-diagonal additionally assumes $\Delta z_S \approx \sum_{u \in S} \delta z_u$. Comparing the predicted quadratic $\frac{1}{2} \mathbb{E}_{x,t} \|\sum_{u \in S} \tilde{\delta z}_u\|^2$ (which is exactly $\frac{1}{2} s^\top \mathbf{H} s$ by Proposition 3) against the same form evaluated on the true joint perturbation, $\frac{1}{2} \mathbb{E}_{x,t} \|\tilde{\delta z}_S\|^2$ with $\tilde{\delta z}_S = \sqrt{p_0} \odot (\Delta z_S - \mathbb{E}_{p_0}[\Delta z_S])$, isolates this step. The additive reconstruction drifts in *direction* as the set grows (cosine $0.997 \rightarrow 0.53$ from $|S| = 2$ to $\rho = 25\%$), but the quadratic *value* is preserved throughout, the ratio staying in 0.67–1.18. Both diagnostics rebuild the features in fp32 on eight held-out sequences of 512 tokens, because the ϵ -ladder of (i) does not resolve in bf16; they therefore isolate the two substitutions rather than characterising the shipped matrix, which is accumulated in bf16 over 128 sequences of 2048.

What this calibrates, and what it does not. Both substitutions leave the quadratic’s *value* stable across ρ , so neither is what degrades with the budget; what grows with ρ is the third-order term in the mask, the one Proposition 4 names and the one no second-order surrogate carries. Neither substitution, however, calibrates the surrogate in nats. On the deployed Llama-3.1-8B-Instruct mask at $\rho=20\%$, $\frac{1}{2} s^\top \mathbf{H} s = 7.21$ against a measured held-out risk of 0.57: the surrogate over-predicts the absolute KL by about an order of magnitude, and by a near-uniform factor across ratios. This is immaterial to the use we put it to. By Lemma 1 the solver reads $\mathbf{H}$ only up to a positive scalar, and $\hat{\lambda}$ is selected on measured risk rather than on the surrogate’s value, so no positive rescaling — from the bf16 accumulation, the top- r renormalisation or the finite ablation — can move a single selection. What the surrogate must do is *rank* prunings at a fixed budget, and that is measured directly in Section E.2.

What makes the surrogate an object one can select over is therefore the measurement, not the remainder, and it is validated directly in Section E.2: predicted surrogate risk and measured post-pruning KL correlate strongly at low-to-moderate ρ and diverge as ρ grows, matching the cubic gap.

B.3 CONNECTION TO CLASSICAL PRUNING

Remark 3 (Diagonal CoCurve is the structured KL-Fisher analogue of OBD). Dropping off-diagonals from the implemented surrogate reduces it to $\frac{1}{2} \sum_u \mathbf{H}_{uu} s_u$ with $\mathbf{H}_{uu} = \mathbb{E}[\delta z_u^\top F \delta z_u] \geq 0$, which is an Optimal-Brain-Damage saliency (LeCun et al., 1989) with structured units in place of weights and a self-distillation KL Fisher in place of the training-loss Hessian; the first-order term vanishes here exactly at $s = \mathbf{0}$ (Proposition 1) rather than by an assumption of convergence. CoCurve adds the signed cross-module curvature $\mathbf{H}_{uv}$, which the diagonal cannot express.

Significance. This places CoCurve on a continuum with classical second-order pruning: $\lambda=0$ is structured OBD and $\lambda=1$ is the full edge model, so the path of Table 3 measures, at every point, exactly the value added by edges over the classical baseline.

What a diagonal (cross-term-dropping) method omits. We can quantify exactly the error that a diagonal saliency, or any method that discards the off-diagonal terms as higher-order negligible, incurs relative to the second-order surrogate $\frac{1}{2} s^\top \mathbf{H} s$ that both are approximations of (Remark 2 separates that surrogate from the true risk).

Proposition 5 (The diagonal omission is the off-diagonal co-pruning mass). *For any pruning set S with indicator mask $s_S \in \{0, 1\}^M$ ($s_{S,u}=1$ iff $u \in S$), the second-order co-pruning risk and its diagonal (OBD) approximation differ by exactly the off-diagonal mass of $\mathbf{H}$ on S :*

$$\Delta_{\text{off}}(S) := \underbrace{\frac{1}{2} s_S^\top \mathbf{H} s_S}_{\text{full (CoCurve)}} - \underbrace{\frac{1}{2} \sum_{u \in S} \mathbf{H}_{uu}}_{\text{diagonal (OBD)}} = \frac{1}{2} \sum_{\substack{u,v \in S \\ u \neq v}} \mathbf{H}_{uv} = \frac{1}{2} \sum_{\substack{u,v \in S \\ u \neq v}} \rho_{uv} \sqrt{\mathbf{H}_{uu} \mathbf{H}_{vv}},$$

with $\rho_{uv} = \mathbf{H}_{uv} / \sqrt{\mathbf{H}_{uu} \mathbf{H}_{vv}}$. Consequently, writing $\bar{\rho}_S$ for the $\sqrt{\mathbf{H}_{uu} \mathbf{H}_{vv}}$ -weighted mean of ρ_{uv} over off-diagonal pairs in S , $\Delta_{\text{off}}(S) = \frac{1}{2} \bar{\rho}_S [(\sum_{u \in S} \sqrt{\mathbf{H}_{uu}})^2 - \sum_{u \in S} \mathbf{H}_{uu}]$.

Significance. The error of dropping the cross terms is not a vanishing higher-order remainder. It is a second-order quantity of the same order as the diagonal itself, scaled by the mean off-diagonal correlation $\bar{\rho}_S$. It also explains the Table 3 gap between $\lambda=0$ and $\hat{\lambda}$: the off-diagonal mass the diagonal ignores is what re-ranks bridge units onto the keep list.

B.3.1 SELECTING THE INTERACTION STRENGTH

This section reports the enumeration, the protocol and the per-model outcome of the path construction of Section 2.4.

Why λ is not fitted. The obvious criterion is to choose λ so that $\hat{\mathcal{R}}_\lambda$ best matches the risk we measure. It optimises the wrong thing, because the solver never reads the value of $\hat{\mathcal{R}}_\lambda$ — only the order it induces over candidates, and that order is invariant to the units $\hat{\mathcal{R}}_\lambda$ is expressed in.

Lemma 1 (Selection is scale-invariant). *For any $a > 0$, $\arg \min_u a \Delta R(u \mid S)/c_u = \arg \min_u \Delta R(u \mid S)/c_u$, where $\Delta R(u \mid S)$ is the marginal risk of Eq. 8.*

So a criterion may fit the magnitude of $\hat{\mathcal{R}}_\lambda$ perfectly and still select badly, and it does: magnitude matching returns $\lambda \approx 0.1$ – 0.4 on Llama-3.2-3B, and the pruning it produces is clearly worse than the one Eq. 11 selects. Minimising the risk of the selections themselves is the criterion that matches what the solver consumes.

The family is small, and a sweep does not find it. Enumerating the breakpoints exactly (Section 2.4) yields a few hundred distinct prunings per model over the whole of $[0, 1]$: 288 on Llama-3.2-3B, 303 on Llama-3.1-8B-Instruct, 181 on Falcon3-7B, 400 on Yi-1.5-9B. The intervals are narrow and very unevenly sized (Llama-3.2-3B: minimum 5.6×10^{-6} , median 1.8×10^{-3} , maximum 0.061), which is why sampling λ is not an adequate substitute. A sweep at step 0.002 costs 501 solves and still reaches only 199 of the 288 members, missing 31% of the family and with it the minimum. The miss rate is set by how unevenly the breakpoints fall rather than by the step size alone: the same sweep misses 13% on Falcon3-7B and 57% on Yi-1.5-9B, so no step size fixes it across models. The two costs are in different currencies and the walk is not uniformly cheaper: on Llama-3.2-3B it spends roughly 3×10^3 CPU greedy solves against the sweep’s 501, and 288 masked forward passes against the sweep’s 199. Forward passes dominate the wall clock, so the walk costs about $1.4\times$ the sweep — and returns the whole family instead of 69% of it. Completeness is what is being bought, not speed.

The criterion is stable to the calibration split. λ moves the attention/FFN split of the removed set substantially — on Llama-3.1-8B-Instruct attention falls from 38% of removed units at $\lambda=0$ to 2% at $\lambda=1$ — so it is worth asking how much the split $\mathcal{C}'$ the criterion is measured on decides the answer. Over both context lengths and both corpora, at $\rho=20\%$ with everything else fixed:

$\mathcal{C}'$ length	$\mathcal{C}'$ corpus	$\hat{\lambda}$	J vs deployed
1024	C4	0.0120	0.967
1024	WikiText	0.0040	0.877
2048	C4	0.0108	<i>deployed</i>
2048	WikiText	0.0213	0.844

$\hat{\lambda}$ spans $5.3\times$ across the four splits, and the prunings they select agree with the deployed one at Jaccard 0.84 to 0.97. The criterion returns the same pruning whichever split it is measured on, so we measure it at the deployment context length because that is the risk the pruned model will incur, not because the number moves: at fixed corpus, doubling the length moves $\hat{\lambda}$ by 11%.

Figure 5 shows the path itself on four models. **The path on ten models.** The two studies in this paper have different costs, and that is what sets their breadth. The path needs **H** and a held-out set and nothing else, so we report it on all ten models we estimate **H** for. The cross-model comparison additionally needs every baseline re-selected under the shared guard and re-scored on the full suite — eight methods, five ratios, twelve tasks per model — and we ran that for the five of Table 2; on the other five no baseline has been run under this harness, so no comparison is available on them either way. Table 3 applies the protocol unchanged across families and scales. Two patterns hold across them. First, at this ratio $\hat{\lambda}$ is small on every model, at most 0.22 and below 0.05 on eight of

the ten, so the edge term wants to be damped hard. It stays small at the other ratios too, including on the model whose path was the one apparent exception: on Mistral-Small-24B the criterion returns 0.034, 0.044, 0.043 and 0.073 at $\rho=20, 30, 40$ and 50%. The one point above that scale anywhere in this paper is $\rho=10\%$ on the same model, where it returns 0.71. We report it and do not explain it: the $1/(2|S|\bar{\rho}_S)$ scale would predict roughly a factor of two between $\rho=10\%$ and 20% and the observed ratio is twenty, so whatever moves it is not the size of the removed set alone. It is also the least consequential ratio in the sweep — at $\rho=10\%$ that model still holds 96.7% of its dense twelve-task average — and the mask it selects there leads on every column it is compared on. Second, and much more sharply, $\lambda=1$ is rejected on *every* model, by 6.7% to 747% and at 3.8 to 39.1 paired standard errors. The minimum is interior on eight of ten, with a margin over $\lambda=0$ of at least two paired standard errors on seven. On Qwen2.5-7B and Qwen2.5-14B the selected segment begins at exactly $\lambda=0$: the protocol declines the edges and returns the diagonal selector. That is the property that makes $\hat{\lambda}$ safe to apply without knowing in advance whether a model’s edges are informative.

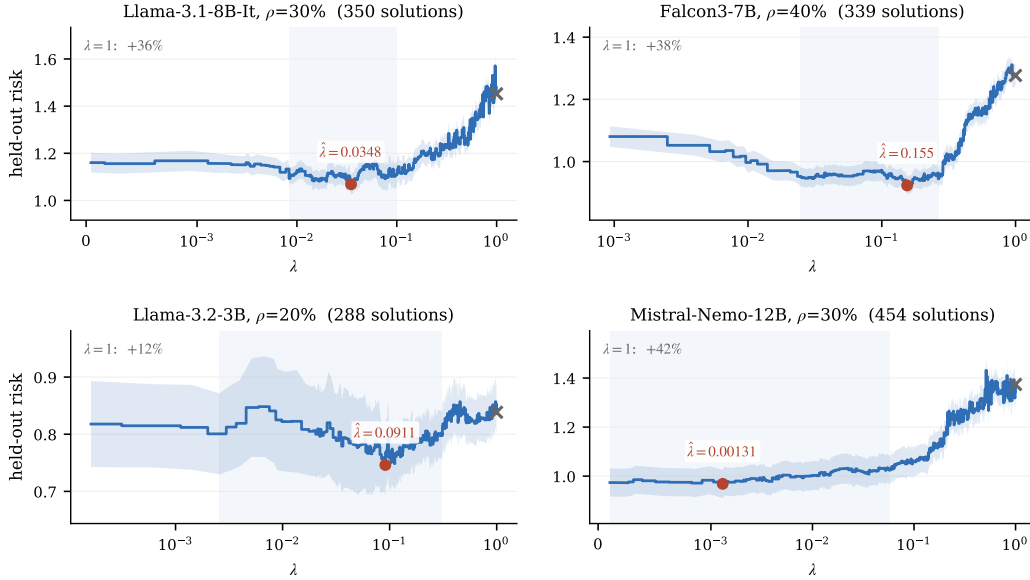


Figure 5: **The regularisation path.** Held-out risk against λ on four models. The curve is a step function, since the solution changes only at breakpoints, with the selected point in red, its 1-SE band shaded, and the risk at $\lambda=1$ marked. The minimum is interior and $\lambda=1$ is far above it everywhere.

B.3.2 STABILITY OF THE ESTIMATED MATRIX

We rebuild $\mathbf{H}$ independently on random halves of the calibration set, resampling over *sequences* (positions within a sequence are strongly dependent, so treating the stored positions as i.i.d. would understate the error). Across ten models spanning six families and 3B–14B the off-diagonal entries reproduce across halves at Pearson 0.885–0.996 and the diagonal at 0.994–1.000 (Table 11). The edges are estimated, not noise. The finite-ablation substitution and the additivity step behind them are measured in Section B.2.

B.4 SOLVER GUARANTEES

Proposition 6 (Running-sum marginal). $\Delta_\lambda(u \mid S) = \frac{1}{2}\mathbf{H}_{uu} + \lambda g_u$ with $g_u = \sum_{v \in S} \mathbf{H}_{uv}$; maintaining g by $g \leftarrow g + \mathbf{H}_{:,u^*}$ makes each step $O(M)$ and the solver $O(M|S|)$, for any fixed λ .

Proof. $\hat{\mathcal{R}}_\lambda(S) = \frac{1}{2} \sum_{u \in S} \mathbf{H}_{uu} + \lambda \sum_{u < v \in S} \mathbf{H}_{uv}$; adding u contributes $\frac{1}{2}\mathbf{H}_{uu} + \lambda \sum_{v \in S} \mathbf{H}_{uv}$. The vector g stores exactly that sum and is updated by one column per selection, independently of λ . $\square$

Proposition 7 (Overshoot bound). *The stopping rule yields $\rho_{\text{actual}} - \rho \leq \max_u c_u / \sum_u c_u$.*

Table 11: **Split-half stability of $\mathbf{H}$** , resampled over calibration sequences, mean of four random halvings. $\text{Jac}(\lambda)$ is the Jaccard overlap of the two independently estimated 20% selections. The edges reproduce nearly as well as the diagonal on every model.

Model	diagonal (Pearson)	off-diagonal (Pearson)	$\text{Jac}(\lambda=0)$	$\text{Jac}(\lambda=1)$
Llama-3.2-3B	0.996	0.885	0.661	0.569
Falcon3-7B	0.998	0.979	0.919	0.582
Mistral-7B	1.000	0.962	0.764	0.418
Qwen2.5-7B	1.000	0.991	0.819	0.362
Llama-3.1-8B-It	0.994	0.905	0.672	0.509
Yi-1.5-9B	1.000	0.966	0.853	0.729
Falcon3-10B	0.997	0.965	0.920	0.596
Mistral-Nemo-12B	0.999	0.982	0.598	0.456
Llama-2-13B	1.000	0.996	0.666	0.585
Qwen2.5-14B	0.999	0.950	0.661	0.455

Table 12: **Depth is invisible to the diagonal and monotone in the off-diagonal.** Spearman rank correlation between a unit’s layer index ℓ_u and its own saliency $\mathbf{H}_{uu}$, and between ℓ_u and its total coupling $b_u = \sum_{v \neq u} |\mathbf{H}_{uv}|$. One $\mathbf{H}$ per model, the same build Table 11 measures. Coupling is negative on all ten; the saliency column changes sign across models.

Model	L	$\rho_s(\ell_u, \mathbf{H}_{uu})$	$\rho_s(\ell_u, b_u)$
Llama-3.2-3B	28	−0.082	− 0.874
Falcon3-7B	28	+0.143	− 0.543
Mistral-7B	32	+0.107	− 0.880
Qwen2.5-7B	28	−0.115	− 0.878
Llama-3.1-8B-It	32	−0.229	− 0.867
Yi-1.5-9B	48	+0.331	− 0.806
Falcon3-10B	40	+0.031	− 0.745
Mistral-Nemo-12B	40	−0.225	− 0.910
Llama-2-13B	40	−0.384	− 0.884
Qwen2.5-14B	48	+0.403	− 0.821
median $ \rho_s $		0.184	0.871

Proof. Before stopping, $C_{S^-} < \rho \sum_u c_u$; one unit of cost $\leq \max_u c_u$ is added, so $C_S < \rho \sum_u c_u + \max_u c_u$. Divide by $\sum_u c_u$. $\square$

Remark (cost-normalised selection). Choosing u by $\Delta R(u \mid S)/c_u$ is the density-ordering heuristic for the underlying min-quadratic knapsack, the standard surrogate when an exact solve is intractable, and the shared budget lets the attention-vs-FFN split emerge from the objective rather than a hand-set ratio. Density ordering is the right ordering while the marginals are non-negative, which holds in every run we inspect: $\mathbf{H}_{uu} \geq 0$ exactly (Proposition 3) and all 11,325 in-set edges are positive on Llama-3.1-8B-Instruct at $\rho=20\%$. On a model with sign-mixed edges a negative marginal should be ordered by Δ rather than by Δ/c_u , since dividing a negative gain by cost prefers the cheaper unit.

C EXPERIMENTAL SETUP

C.1 DATASETS

Table 13 lists all evaluation benchmarks with task type, metric, and split. Calibration is 128 C4 sequences (length 2048); no task labels are used for pruning. Every method is evaluated on this same fixed suite and on nothing outside it.

Table 13: **Evaluation suite.** Every multiple-choice task is scored on its full split through one shared harness; n is the number of items actually scored, and for the perplexity corpora the number of target tokens under the headline model’s tokenizer (the token count, not the corpus, varies across tokenizers). Multiple choice is length-normalised log-likelihood over the candidate answers, zero-shot. The capability groups are the ones Table 2 averages over.

Group	Benchmark	Split	n
Language modelling	WikiText-2 (Merity et al., 2017)	test	288,936
	PTB (Marcus et al., 1993)	test	98,914
	C4 (Raffel et al., 2020)	validation, 500 docs	229,527
Commonsense	PIQA (Bisk et al., 2020)	validation	1,838
	HellaSwag (Zellers et al., 2019)	validation	10,042
	WinoGrande (Sakaguchi et al., 2021)	validation	1,267
	CommonsenseQA (Talmor et al., 2019)	validation	1,221
Science QA	ARC-easy (Clark et al., 2018)	test	2,376
	ARC-challenge (Clark et al., 2018)	test	1,172
	OpenBookQA (Mihaylov et al., 2018)	test	500
Reading	BoolQ (Clark et al., 2019)	validation	3,270
	RACE (Lai et al., 2017)	middle, test	1,436
	RACE-high (Lai et al., 2017)	high, test	3,498
	QuAIL (Rogers et al., 2020)	validation	2,164
Knowledge	MMLU (Hendrycks et al., 2021)	test	14,042

C.2 MODELS AND EXPERIMENTAL PROTOCOL

All models use GQA (attention units are KV-aligned query-head groups); FFN groups are $n_f=16$ per layer; cost is removable parameter count. The unit definition is a property of the comparison rather than of any method: every selector here, ours included, chooses among the same units under the same budget.

Keeping the removal distributed. We prevent a selector from concentrating its budget in a few layers, using a per-layer cap κ on removed cost together with protection of the first two and last two layers. This is not a device we introduced: it is a weaker version of what the strongest baseline in our set already does in its published form. LLM-Pruner reports that the first and last layers dominate performance, and its released configuration accordingly leaves the first three layers and the final layer untouched and applies a *uniform* ratio to every layer in between — for a 20% target, 25% from layer 5 to layer 30 of 32 (Ma et al., 2023). Our guard protects four layers instead of four-and-a-half and then lets the objective allocate freely up to κ inside the rest, rather than fixing one ratio per layer. It is the looser constraint, and it is applied to every method equally. It applies to *every* method compared and to every ratio, by one rule used throughout,

$$\kappa = 1.5 \rho, \quad \text{first and last two layers protected.}$$

The rule has two properties. First it has no free parameter and no reference ratio, so nothing is tuned per model or extrapolated from one setting to another. Second, and more important for the comparison, it guarantees the target is attainable: with $|\mathcal{L}_p|$ protected layers of L the reachable ratio is at most $(1 - |\mathcal{L}_p|/L) \kappa$, which for $|\mathcal{L}_p|=4$ and $\kappa=1.5\rho$ exceeds ρ whenever $L > 12$, true of every model here. A guard chosen without that check silently prevents a method from reaching the requested ratio: a per-layer cap that ignores the protected layers leaves a shortfall of up to 6 points at $\rho=0.5$, which flatters the affected method, since it is then compared while less compressed. We

therefore verify the realised ratio of every mask before evaluating it and report only masks within 0.2 points of their target. That tolerance is not a round number: Proposition 7 bounds the overshoot of the stopping rule by the largest single unit’s share of the cost, which is 0.13 to 0.20 points on the models here, so a mask that misses its target by more than that has missed it for some reason other than unit granularity.

Switching the guard off. The rule is one line, but it is still a choice, so we removed it and measured: no cap, no protected layers, and $\hat{\lambda}$ re-derived in the unguarded family, on Llama-3.1-8B-Instruct at $\rho=20\%$. Three things follow. It does not interact with the calibration: $\hat{\lambda}$ moves from 0.0108 to 0.0098. It is a protection rather than a handicap: unguarded, FLAP and Wanda-sp delete the first seven layers outright and collapse, to WikiText 41,189 and 30,742 with every capability group at or below chance, so the cap and the first-layer protection are what let them be compared at all. And it is not shaped to suit us: our own solution concentrates too without it — eight layers past 30% of their removable cost and one at 76% — and loses capability, while LLM-Pruner, whose unguarded solution instead takes nothing before layer 18, gains on multiple choice, the direction reported for depth pruning (Men et al., 2024; Kim et al., 2024). What the guard changes is therefore how concentrated a solution may be, and the selectors do not agree on whether concentration helps; what it does not do is give any of them a protocol the comparison could have used instead. The layer protection in particular is not load-bearing for CoCurve: with the guard off the objective still takes nothing from the first two layers, and 5% and 2% from the last two.

Table 14 gives per-model structure. The compensation of Section A.5 runs at $\gamma = \overline{\text{diag}(\mathbf{H}_{\mathcal{K}\mathcal{K}})}$ and $\tau = 0.35$ in the released implementation; no result in this paper uses it, and it is described because the same edge blocks that select also support repair.

Table 14: **Unit inventory** for the models pruned under our own harness. The guard is one rule for every row and every ratio; nothing here is configured per model.

Model	Family	Layers	Units M	Attn/FFN
Llama-3.1-8B-Instruct	Llama-3	32	768	256/512
Llama-3.2-3B	Llama-3	28	672	224/448
Falcon3-7B	Falcon3	28	560	112/448
Mistral-Nemo-12B	Mistral	40	960	320/640
Mistral-Small-24B	Mistral	40	960	320/640

C.3 BASELINE METHODS

All baselines are training-free, run under identical budgets and harness with no recovery.

Every baseline is given our allocator, not only our budget. A baseline here is its own importance metric driving *our* global cost-normalised greedy with our per-layer cap, free to spend the budget across layers and across module types exactly as CoCurve does. Several of these methods do not allocate globally in their published form — FLAP and Wanda-sp threshold per layer, LLM-Pruner works over coupled groups — so the harness gives them allocation freedom they do not otherwise have, and any remaining difference is the importance metric alone. Two arms keep their own allocation because ours would misrepresent them: SlimLLM, whose head and FFN scores are not comparable across module types, and 2SSP, whose two stages are the method. Nothing here is tuned per model.

And a finer unit definition, where that is what a baseline wants. The shared $n_f=16$ is a property of the comparison, but three baselines — Wanda-sp, FLAP and Magnitude — score channels natively, so the shared definition could be costing them resolution while suiting us: our own selection costs M single-unit ablations, so we need M in the thousands and they do not. We therefore re-ran those three at $n_f=64$, four times finer than the shared definition and four times finer than our own method could afford, at the same ratio, guard and harness. Resolution was indeed part of what held two of them back: Wanda-sp improves from 20.02 to 14.86 WikiText perplexity and on all four capability groups, and FLAP from 20.88 to 16.23 and on three of the four, while Magnitude gets worse on four of the five measures. Even so, at four times our resolution the best of the three is still 2.0 perplexity and 0.6 to 3.8 points per capability group behind our $n_f=16$ selection, so resolution

is not what the gap is made of. **Random** prunes structured units uniformly (lower bound). **Magnitude** (Han et al., 2015) scores units by weight norm. **Wanda-sp**, the structured generalisation of Wanda (Sun et al., 2024) introduced and named by An et al. (2024), is the structured input-norm-weighted magnitude. **FLAP** (An et al., 2024) scores units by activation-fluctuation importance, $\text{Var}(a) \cdot \|W\|^2$ per column. **LLM-Pruner** (Ma et al., 2023) applies a Taylor (first-order with diagonal Fisher) group importance over coupled weight slices; its gradient pass reads the same calibration text at 512 tokens per sequence, the short-sequence regime the method is defined in. **SlimGPT** (Ling et al., 2024) and **Which blocks carry fewer than seven baselines**. Two of the fifteen model-ratio blocks of Table 2 compare against fewer than the full set, and in both cases the reason is a property of the baseline rather than a choice of ours. On Mistral-Nemo-12B at $\rho=20\%$ LLM-Pruner exceeds the memory of the card it was run on (Section D.4.1). On Falcon3-7B at $\rho=30\%$ SlimLLM keeps its own per-layer, per-type allocation, which cannot spend an attention budget smaller than one of that model’s four units per layer, so its mask lands 0.8 points off target and the ratio band of Section C.2 excludes it. Every other block compares against all seven, Mistral-Small-24B at all three reported ratios included. In the wider sweep of Table 17 that model carries one baseline at $\rho=10\%$ and five at $\rho=50\%$, ratios outside the three the main comparison reports.

SlimLLM (Guo et al., 2025) scores heads by the Pearson similarity of the sublayer output with and without each head and FFN channels in a PCA eigenbasis, then restores performance by least-squares regression on each sublayer’s output matrix — a reconstruction step, but not an OBS-style weight surgery. It keeps its own per-layer ratio allocation and per-module ranking rather than being forced into a global one, since its head and FFN scores are not comparable across module types; that allocation is per layer *and* per unit type, so on Falcon3-7B, which has four attention units per layer, it cannot spend an attention budget of less than one unit and two of its runs fall outside the ratio band and are excluded from the comparison. **AMP** (Mugnaini et al., 2025) prunes a uniform fraction of attention heads and MLP neurons in every layer, scoring units by the ℓ_1 norm of calibration activations projected through the output weights. **2SSP** (Sandri et al., 2025) is a two-stage structured pruner that removes FFN width in a first stage and whole attention submodules by depth in a second; it argues that pruning *within* attention at neuron or width granularity is impractical under the scaled dot-product. We additionally report two single-module variants of our pipeline (head-only, FFN-only) and the diagonal-only ($\lambda=0$) variant. For methods originally including LoRA recovery, it is disabled for the no-recovery comparison. The two most recent methods, AMP and 2SSP, are compared on the headline Llama-3.1-8B-Instruct, where their released configurations apply directly; Table 15 carries both, AMP inside the shared guard and 2SSP with its own two-stage budget. 2SSP is the strongest of the nine on ARC-easy and ahead of CoCurve on PTB perplexity, where LLM-Pruner is ahead of both. The cross-model study (Table 2, Section D.1) uses the seven established baselines, which port across every architecture studied here under the shared harness without method-specific re-tuning. On Mistral-Small-24B the comparison carries all seven at $\rho=30\%$, at $\rho=20\%$ and at $\rho=40\%$; SlimGPT’s and LLM-Pruner’s numbers there come from the streamed re-implementation of Section D.4.1, which is numerically identical to the published form.

C.4 EVALUATION PROTOCOL

Multiple-choice tasks are scored by length-normalised log-likelihood over the candidate answers (zero-shot, `acc_norm`); BoolQ uses the full validation split and MMLU is zero-shot log-likelihood. Perplexity uses a 2048-token context with stride 512 over the full corpus. All scores come from one shared harness.

A sliding window gives every predicted token more left context than a non-overlapping pass, so these perplexities are not comparable to numbers computed at a different stride; every row here, the dense reference included, uses the same one.

D COMPLETE RESULTS

Table 15 is the per-task record on the headline model: nine training-free baselines, one checkpoint, three perplexity corpora and the eight multiple-choice tasks this literature reports on it; Table 16 carries all twelve. Table 16 is then the same record for every model and ratio.

Table 15: **Per-task results on Llama-3.1-8B-Instruct at 20%**, full test sets. **Bold** = best among pruned, underline = second, so a baseline that beats us is marked where it does. **Avg** is the seven-task commonsense mean, the set this literature reports under that name. Every row is the same instruction-tuned checkpoint, and every row but one is selected under the shared guard of Section C.2; ‡2SSP sets its own per-layer budget, because its two stages *are* the method (Section C.3).

Method	Perplexity ↓			Commonsense (acc) ↑							Know. ↑	
	Wiki	PTB	C4	ARC-c	ARC-e	HellaS	WinoG	PIQA	OBQA	BoolQ	Avg	MMLU
Dense (0%)	6.40	10.36	10.67	.544	.781	.774	.662	.809	.476	.850	.700	.478
Random	21.20	30.44	36.77	.294	.503	.538	.545	.682	.314	.654	.504	.318
Magnitude	20.49	31.38	34.93	.314	.546	.555	.559	.689	.324	.652	.520	.328
Wanda-sp	20.02	35.29	30.09	.388	.625	.574	.570	.737	.364	.722	.569	.356
FLAP	20.88	36.77	29.06	.377	.625	.590	.549	.732	.382	.709	.566	.353
AMP	20.39	47.27	33.26	.388	.639	.581	.567	.747	.388	.694	.572	.357
LLM-Pruner	<u>13.23</u>	19.36	<u>21.98</u>	.372	.604	<u>.646</u>	.574	.750	.368	.791	<u>.586</u>	.368
SlimGPT	19.24	35.52	29.19	.374	.619	.588	.566	.734	.354	.643	.554	.351
SlimLLM	17.36	30.59	29.36	.395	<u>.652</u>	.600	.562	<u>.752</u>	.398	.658	.574	.372
2SSP‡	16.29	<u>24.07</u>	29.00	<u>.413</u>	.668	.619	<u>.597</u>	.736	.386	.643	.580	<u>.379</u>
CoCurve (ours)	12.88	29.63	19.58	.419	.625	.675	.610	.758	<u>.396</u>	<u>.783</u>	.609	.385

D.1 FULL PER-MODEL RESULTS

Table 16 is the complete record: every reported task, every model, every reported ratio, every baseline and CoCurve, from selection-only runs under the single guard rule. It exists so that the selection in the main tables can be checked against everything that was measured. Read against Table 2, the per-cell pattern behind the aggregate is visible: the strongest specialised baseline, usually LLM-Pruner, takes a number of commonsense and knowledge cells on Llama-3.2-3B at 40% and on Mistral-Nemo-12B at 30–40%, while CoCurve leads all three perplexities on nine of the twelve model–ratio combinations among the models pruned here.

Mistral-Small-24B is the clearest illustration of how the margin depends on available slack. At $\rho=20\%$ a 24B model is redundant enough that every method stays close to dense and the field is tight; by 30% CoCurve is the only method still holding a usable model, and by 40% its perplexity lead over the best baseline reaches $11.9\times$ (Table 17).

Table 16: **Every reported task, every model, every ratio, every method.** The complete record behind the selected main tables, under the single guard rule and the reported suite of Section C.1. Shading is rank within a column among the pruned methods of the same block, darkest best. A dash is a cell that was not evaluated.

Model	ρ	Method	Wiki	PTB	C4	ARC-c	ARC-e	HellaS	WinoG	PIQA	OBQA	BoolQ	MMLU	QuAIL	RACE	RACE-h	CSQA
3B	20%	SlimLLM	<u>16.90</u>	<u>29.84</u>	<u>24.64</u>	<u>32.3</u>	<u>54.2</u>	<u>52.9</u>	51.4	<u>70.3</u>	<u>32.8</u>	46.8	<u>32.0</u>	<u>44.1</u>	<u>43.5</u>	<u>34.4</u>	<u>40.7</u>
		SlimGPT	31.83	62.81	43.46	27.8	49.7	45.7	52.0	69.3	30.6	63.0	28.8	39.7	37.4	30.7	33.1
		LLM-Pruner	<u>20.84</u>	<u>31.32</u>	<u>27.12</u>	<u>32.7</u>	<u>55.2</u>	<u>55.3</u>	<u>53.4</u>	<u>70.6</u>	<u>34.6</u>	<u>60.8</u>	<u>33.5</u>	<u>42.4</u>	<u>43.2</u>	<u>35.0</u>	<u>42.3</u>
		FLAP	40.29	81.15	50.16	27.0	51.4	46.4	52.7	67.1	28.8	63.9	30.2	42.1	40.1	31.8	36.5
		Wanda-sp	35.15	<u>53.77</u>	<u>37.43</u>	30.1	53.5	48.6	51.6	68.9	31.6	59.3	30.7	43.0	41.4	34.0	37.7
		Magnitude	36.97	56.29	50.58	31.8	50.5	46.8	<u>54.2</u>	66.0	31.8	45.8	<u>31.6</u>	<u>44.1</u>	<u>42.8</u>	33.8	37.9
		Random	32.51	45.86	46.56	27.1	45.0	42.4	51.5	63.1	28.6	44.5	28.8	39.7	37.7	29.4	32.1
		CoCurve	<u>16.66</u>	<u>27.34</u>	<u>23.24</u>	<u>35.2</u>	<u>57.7</u>	<u>55.7</u>	<u>54.1</u>	<u>70.6</u>	<u>34.4</u>	<u>65.6</u>	<u>33.9</u>	<u>46.2</u>	<u>49.3</u>	<u>37.8</u>	<u>45.9</u>
3B	30%	SlimLLM	<u>42.69</u>	<u>78.97</u>	<u>50.48</u>	26.4	<u>45.3</u>	<u>40.0</u>	50.3	<u>64.6</u>	27.8	38.5	<u>28.8</u>	<u>38.7</u>	<u>35.3</u>	<u>29.9</u>	<u>33.3</u>
		SlimGPT	<u>111.25</u>	<u>169.18</u>	<u>141.70</u>	23.6	35.7	32.7	51.0	59.2	26.2	<u>56.3</u>	26.8	36.2	33.7	26.9	24.7
		LLM-Pruner	<u>44.57</u>	<u>54.64</u>	<u>52.08</u>	27.0	<u>44.7</u>	<u>42.5</u>	<u>52.6</u>	<u>64.4</u>	29.4	<u>62.3</u>	<u>29.5</u>	<u>40.7</u>	<u>39.3</u>	<u>31.6</u>	<u>34.4</u>
		FLAP	8305	10898	3835	23.6	32.1	27.3	49.3	54.2	24.8	39.7	26.8	26.5	27.8	25.8	24.7
		Wanda-sp	<u>103.32</u>	<u>130.75</u>	<u>109.87</u>	23.0	<u>42.2</u>	<u>33.1</u>	49.7	59.7	26.0	51.7	<u>27.3</u>	<u>35.8</u>	<u>32.9</u>	<u>27.1</u>	<u>29.5</u>
		Magnitude	318.16	427.77	472.14	25.4	34.3	31.5	51.2	57.5	23.8	48.3	27.2	33.6	30.2	26.8	28.4
		Random	147.94	170.18	190.30	23.9	34.7	31.0	49.7	58.8	24.8	39.0	27.1	36.9	29.3	27.4	24.0
		CoCurve	<u>34.78</u>	<u>49.40</u>	<u>55.02</u>	<u>27.7</u>	<u>45.8</u>	<u>42.3</u>	<u>51.2</u>	<u>63.7</u>	<u>32.2</u>	<u>61.1</u>	<u>30.0</u>	<u>44.0</u>	<u>39.7</u>	<u>31.9</u>	<u>35.6</u>
3B	40%	SlimLLM	<u>136.08</u>	<u>159.83</u>	<u>130.01</u>	22.5	<u>39.9</u>	<u>30.9</u>	49.5	<u>61.0</u>	24.8	41.0	<u>28.0</u>	<u>35.6</u>	<u>32.5</u>	<u>26.6</u>	<u>29.4</u>
		SlimGPT	8584	9960	6614	22.5	29.8	25.6	48.2	50.2	25.6	38.8	26.2	25.5	26.5	25.6	22.5
		LLM-Pruner	<u>164.85</u>	<u>316.25</u>	<u>204.40</u>	<u>25.6</u>	<u>37.8</u>	<u>35.0</u>	<u>51.7</u>	<u>59.0</u>	<u>27.6</u>	<u>60.6</u>	<u>28.0</u>	<u>36.6</u>	<u>32.8</u>	<u>28.8</u>	<u>29.0</u>

continued on the next page

Model	ρ	Method	Wiki	PTB	C4	ARC-c	ARC-e	HellaS	WinoG	PIQA	OBQA	BoolQ	MMLU	QuAIL	RACE	RACE-h	CSQA
		FLAP	11144	19076	6720	25.8	27.3	25.3	49.4	50.9	23.2	41.4	26.3	25.3	25.3	25.2	21.4
		Wanda-sp	700.93	592.09	655.43	20.2	32.3	27.0	48.4	54.2	22.8	45.4	26.2	27.7	28.4	26.4	24.2
		Magnitude	12676	8180	8512	25.6	27.6	24.9	48.2	51.3	23.2	38.1	27.0	24.2	26.2	25.9	21.0
		Random	13262	9552	4957	24.4	28.8	26.7	50.5	52.7	24.2	40.1	26.5	30.1	26.7	24.7	23.4
		CoCurve	193.66	256.52	196.49	24.4	34.3	30.1	49.6	55.9	26.8	53.3	26.8	36.4	33.6	28.2	25.2
F3-7B	20%	SlimLLM	11.30	17.85	20.04	41.2	63.9	62.4	54.2	74.3	33.4	65.7	36.8	42.3	45.1	37.8	42.9
		SlimGPT	7.43	12.62	16.49	38.3	66.2	62.9	59.3	74.0	33.6	73.1	38.8	48.8	55.4	41.9	40.7
		LLM-Pruner	7.72	12.86	16.94	42.7	67.6	62.9	56.9	74.8	31.8	69.9	39.8	50.1	55.1	43.0	43.5
		FLAP	7.90	13.24	16.63	42.3	67.5	64.2	56.8	74.7	35.2	72.8	38.2	48.4	56.1	41.7	43.6
		Wanda-sp	7.86	13.36	16.82	42.4	65.2	63.8	58.0	74.4	32.4	72.0	38.9	49.4	55.8	42.8	43.9
		Magnitude	378.46	460.52	554.41	22.1	31.1	27.1	51.0	53.5	23.8	39.9	26.7	28.7	28.8	26.0	21.8
		Random	8.32	14.20	18.54	41.6	67.6	61.5	58.1	74.6	32.0	73.8	38.1	48.3	52.6	40.7	43.3
		CoCurve	7.07	11.72	15.65	41.7	67.6	63.2	57.9	73.0	34.8	70.8	37.8	49.8	53.3	41.2	43.7
F3-7B	30%	SlimGPT	10.07	17.82	22.64	33.9	55.3	53.1	53.7	68.6	28.4	64.6	33.3	43.5	47.0	36.0	38.1
		LLM-Pruner	11.01	18.30	23.60	34.0	59.4	55.4	56.6	70.5	29.6	65.6	34.8	47.6	51.0	38.2	38.7
		FLAP	10.13	17.65	22.34	34.5	58.8	52.4	54.5	69.0	29.4	62.9	33.5	43.2	45.8	37.2	38.2
		Wanda-sp	10.07	17.71	22.63	32.0	53.3	52.3	54.9	66.7	28.8	65.1	33.2	42.8	46.0	36.8	33.6
		Magnitude	456.73	528.71	661.34	22.6	34.0	28.3	49.4	54.7	23.8	42.4	27.4	27.8	28.8	26.9	23.6
		Random	12.03	21.76	27.16	38.1	57.2	55.3	57.1	68.4	26.4	65.8	33.6	44.8	46.3	36.9	38.9
		CoCurve	8.92	15.05	19.47	35.2	61.3	57.1	56.1	69.3	31.2	64.0	35.8	46.0	48.5	38.5	42.5
F3-7B	40%	SlimLLM	58.44	101.69	57.48	29.3	47.5	43.5	50.6	63.5	27.6	59.3	30.0	30.5	34.5	29.8	27.8
		SlimGPT	22.89	40.66	46.37	25.7	45.0	40.4	51.5	61.7	25.6	50.3	29.7	38.8	36.8	31.3	30.5
		LLM-Pruner	18.36	28.74	37.94	28.8	49.7	46.1	52.6	65.2	27.0	62.9	31.1	44.4	44.1	34.7	34.9
		FLAP	31.42	51.77	55.38	24.8	41.3	39.3	50.5	57.8	23.0	48.3	28.7	33.3	34.5	29.5	28.3
		Wanda-sp	29.90	52.44	53.07	27.0	43.4	39.0	50.0	59.9	26.2	47.3	29.1	35.4	33.4	30.1	28.6
		Magnitude	484.89	778.64	618.41	22.4	32.0	27.1	49.8	52.9	23.6	47.6	27.3	23.4	29.7	26.7	24.4
		Random	20.40	36.92	45.62	29.7	45.9	43.4	51.9	62.7	23.6	49.0	29.8	42.8	39.6	33.3	31.4
		CoCurve	13.83	23.11	29.21	31.0	50.1	46.3	53.1	63.8	27.2	63.2	31.6	42.1	42.1	33.9	34.7
8B-It	20%	SlimLLM	17.36	30.59	29.36	39.5	65.2	60.0	56.2	75.2	39.8	65.8	37.2	47.8	51.3	39.4	47.6
		SlimGPT	19.24	35.52	29.19	37.4	61.9	58.8	56.6	73.4	35.4	64.3	35.1	44.5	47.8	36.6	44.6
		LLM-Pruner	13.23	19.36	21.98	37.2	60.4	64.6	57.4	75.0	36.8	79.1	36.8	52.4	57.2	45.2	51.8
		FLAP	20.88	36.77	29.06	37.7	62.5	59.0	54.9	73.2	38.2	70.9	35.3	45.6	49.9	39.2	46.7
		Wanda-sp	20.02	35.29	30.09	38.8	62.5	57.4	57.0	73.7	36.4	72.2	35.6	46.6	49.8	38.7	48.9
		Magnitude	20.49	31.38	34.93	31.4	54.6	55.5	55.9	68.9	32.4	65.2	32.8	49.4	50.2	39.1	41.7
		Random	21.20	30.44	36.77	29.4	50.3	53.8	54.5	68.2	31.4	65.4	31.8	47.8	47.1	36.1	38.5
		CoCurve	12.88	29.63	19.58	41.9	62.5	67.5	61.0	75.8	39.6	78.3	38.5	54.1	58.1	44.0	50.4
8B-It	30%	SlimLLM	78.23	100.68	109.42	28.8	53.0	37.8	52.0	69.5	28.2	62.5	30.4	37.2	35.9	29.4	36.9
		SlimGPT	196.79	317.47	208.42	25.4	42.8	38.9	51.2	62.1	27.4	51.3	27.9	38.0	35.4	29.4	30.4
		LLM-Pruner	26.64	39.70	40.31	33.2	51.0	54.4	52.4	71.0	30.4	73.4	31.9	47.8	48.4	38.8	43.2
		FLAP	79.37	177.67	102.06	24.8	43.8	41.0	53.6	65.7	26.8	51.2	29.0	37.2	37.8	32.0	33.5
		Wanda-sp	61.15	94.85	87.17	27.8	45.8	38.5	53.4	65.7	28.2	47.7	29.6	40.1	41.9	31.0	33.7
		Magnitude	324.46	642.26	273.86	25.0	31.5	32.2	51.0	57.0	25.6	51.3	27.3	34.8	33.1	28.8	24.7
		Random	52.90	84.29	87.88	24.1	41.0	40.2	53.5	60.6	26.6	47.8	28.6	42.6	41.4	31.8	29.6
		CoCurve	21.59	36.31	31.99	34.4	55.6	55.8	55.5	70.5	35.2	73.1	33.4	50.1	46.5	37.3	41.0
8B-It	40%	SlimLLM	183.53	211.36	217.27	24.0	41.6	31.2	50.4	62.6	25.0	51.2	27.6	32.1	31.6	27.3	29.2
		SlimGPT	26002	32398	13079	23.9	27.9	26.2	49.2	51.9	29.2	38.2	26.9	21.1	25.3	26.7	21.0
		LLM-Pruner	103.35	165.07	125.68	28.1	40.0	42.7	52.2	65.3	26.8	65.1	29.7	40.2	36.0	32.2	33.4
		FLAP	49940	29989	13823	23.5	28.0	26.0	49.2	52.1	22.6	41.6	25.6	23.3	26.6	25.4	22.0
		Wanda-sp	1015	1288	860.91	22.4	33.1	26.3	48.2	54.8	22.8	38.7	26.6	24.8	27.8	26.0	25.6
		Magnitude	27101	23937	13834	25.9	25.7	25.4	50.8	51.1	27.8	48.2	26.2	24.4	25.6	25.8	19.0
		Random	4168	3201	3152	23.4	31.3	29.0	50.5	54.0	24.0	42.5	26.7	35.4	33.3	26.4	24.1
		CoCurve	59.13	108.37	76.05	28.7	41.6	41.0	51.4	64.0	29.6	62.3	29.3	42.8	41.9	32.2	31.9
Nemo-12B	20%	SlimLLM	11.44	187.41	17.83	39.5	63.8	62.3	57.9	74.4	35.4	68.2	35.0	47.7	50.8	39.1	41.3
		SlimGPT	15.24	281.60	22.45	35.6	54.8	62.4	59.1	74.2	33.6	64.1	33.7	45.8	43.3	35.2	36.9
		FLAP	15.07	477.73	22.34	36.6	59.2	63.5	55.0	72.7	33.0	63.1	35.4	43.9	47.4	36.2	39.1
		Wanda-sp	12.10	175.69	19.23	41.8	62.8	63.8	56.5	74.8	35.4	65.7	35.5	47.3	48.4	38.0	41.2
		Magnitude	16.95	617.32	22.62	36.5	59.0	61.4	60.5	74.1	33.0	64.3	34.6	48.4	49.1	37.4	42.2
		Random	11.69	203.74	18.31	37.8	59.4	64.9	60.4	75.6	35.8	63.1	35.6	50.4	49.3	38.1	39.6
		CoCurve	8.63	83.37	14.31	43.3	66.5	66.5	62.9	76.3	39.4	79.1	36.9	52.6	54.1	41.0	42.7
Nemo-12B	30%	SlimLLM	33.31	460.25	39.47	31.3	53.6	46.3	54.1	68.2	28.8	62.1	30.8	40.1	40.3	32.0	34.2
		SlimGPT	57.98	714.78	61.04	28.7	41.6	44.5	53.2	64.4	31.4	56.9	29.0	39.3	34.1	30.6	30.9
		LLM-Pruner	17.75	164.52	24.84	37.3	59.9	57.6	58.7	72.0	31.4	64.9	33.0	48.0	50.7	38.5	40.6
		FLAP	37.34	1105	56.23	27.2	42.0	45.6	52.0	64.0	29.6	56.5	29.4	39.5	38.5	31.8	29.3
		Wanda-sp	34.77	571.74	47.03	28.6	48.6	46.5	51.5	66.2	29.4	62.1	30.2	42.1	42.1	32.8	31.0
		Magnitude	216.07	1441	145.43	24.3	34.0	34.9	50.4	58.1	27.2	58.8	27.3	33.6	32.2	28.8	24.4
		Random	30.25	1047	41.38	29.5	47.2	51.8	54.1	66.1	30.2	48.5	30.2	44.9	41.0	32.4	33.0
		CoCurve	15.75	160.35	23.27	35.3	56.7	56.7	57.5	71.4	30.4	62.9	31.7	48.7	48.7	37.3	38.9

continued on the next page

Model	ρ	Method	Wiki	PTB	C4	ARC-c	ARC-e	HellaS	WinoG	PIQA	OBQA	BoolQ	MMLU	QuAIL	RACE	RACE-h	CSQA
Nemo-12B	40%	SlimLLM	143.75	1132	145.63	24.1	38.7	30.8	50.0	58.7	24.0	53.4	27.5	32.6	31.7	26.8	26.7
		SlimGPT	84318	51181	2786	22.8	28.9	28.1	48.2	52.8	24.8	52.3	26.2	28.4	29.0	26.4	20.1
		LLM-Pruner	73.06	499.33	86.26	28.3	47.9	43.1	54.5	63.4	28.8	62.2	30.0	43.1	43.7	34.7	34.3
		FLAP	833.96	16257	545.07	23.4	29.1	29.8	48.9	53.8	24.6	55.1	26.3	27.8	28.2	27.3	24.4
		Wanda-sp	478.67	8909	550.44	23.3	35.7	29.3	49.9	56.3	22.8	59.1	27.2	28.8	28.8	27.9	26.0
		Magnitude	236919	163055	191695	25.4	25.0	26.0	48.2	49.4	25.6	38.4	26.7	23.8	24.5	26.3	21.0
		Random	197.13	6702	262.29	22.8	35.4	34.1	49.6	57.3	24.6	58.0	27.0	33.6	33.1	27.9	27.0
		CoCurve	52.20	455.37	71.01	29.4	43.8	42.1	53.7	62.7	24.2	62.1	29.3	41.6	41.9	32.5	34.2
24B	20%	SlimLLM	18.08	94.04	31.38	49.5	73.8	74.2	65.0	78.0	40.4	78.6	43.5	52.9	58.9	44.9	50.7
		SlimGPT	23.10	122.54	37.78	46.1	69.9	71.5	61.5	76.2	40.0	74.7	41.4	52.1	60.5	42.5	47.9
		LLM-Pruner	25.79	198.80	47.32	51.1	73.1	73.5	65.2	74.2	41.4	78.5	43.7	55.1	58.8	47.0	49.1
		FLAP	17.04	89.64	30.81	44.8	65.2	68.9	59.6	71.6	40.4	78.8	40.6	51.5	56.5	41.5	46.4
		Wanda-sp	14.61	274.90	29.29	45.5	66.7	70.8	59.9	73.2	41.4	81.2	41.0	53.7	58.1	41.2	46.8
		Magnitude	33.58	189.37	47.15	45.7	72.9	67.5	60.7	76.7	40.6	73.2	42.4	52.2	58.6	42.4	49.1
		Random	8.71	69.46	15.35	47.0	72.9	69.3	61.9	76.9	39.0	75.7	43.2	52.6	60.2	43.6	50.5
		CoCurve	8.97	131.23	14.88	52.7	74.1	76.0	66.3	77.8	42.0	79.7	43.5	55.9	63.6	44.3	51.2
24B	30%	SlimLLM	78.78	464.69	116.40	42.3	67.0	61.5	54.7	70.8	40.0	63.4	36.5	47.0	51.3	39.1	47.1
		SlimGPT	7854	3854	5390	30.7	48.9	48.9	52.5	63.1	32.4	59.6	31.4	40.5	39.2	30.6	33.0
		LLM-Pruner	150.63	6060	432.21	38.2	55.7	59.4	58.0	62.8	31.4	70.8	33.7	46.3	50.2	38.6	38.2
		FLAP	4846	6397	3159	24.2	25.8	26.8	49.4	52.3	23.0	37.8	26.5	20.0	27.3	26.8	20.0
		Wanda-sp	945.38	2307	242.81	36.0	55.6	50.5	51.5	63.2	33.6	59.4	33.7	37.2	41.8	33.3	37.3
		Magnitude	477.54	3159	380.16	34.9	52.6	50.3	54.1	62.6	29.4	66.1	31.7	43.6	42.9	34.7	33.5
		Random	328.16	1066	435.17	37.1	59.4	54.5	57.7	70.2	33.8	42.2	35.5	47.3	53.8	38.0	44.4
		CoCurve	20.49	419.05	29.67	42.2	63.8	63.8	58.8	72.0	38.4	69.9	36.5	51.4	55.2	40.0	42.4
24B	40%	SlimLLM	3969	11483	6273	32.3	47.7	39.0	52.6	58.9	32.8	46.9	30.0	25.2	34.3	30.0	30.1
		SlimGPT	9057	11141	7197	23.0	25.4	25.3	50.4	51.3	26.8	37.8	26.8	24.2	26.0	26.3	21.9
		LLM-Pruner	5495	8725	4551	27.0	35.9	34.4	52.5	55.0	26.4	38.3	28.5	27.9	30.0	28.6	23.5
		FLAP	5340	6506	4034	23.5	29.3	27.7	49.4	52.2	26.2	39.5	26.8	23.2	28.4	26.4	22.7
		Wanda-sp	4809	4747	1271	22.0	29.7	28.1	49.6	52.4	25.6	38.6	26.8	25.8	27.3	27.2	22.0
		Magnitude	7766	12239	5370	27.6	32.6	27.9	50.4	55.0	29.4	50.6	27.2	25.0	30.7	27.4	24.3
		Random	3299	5950	2847	28.4	34.9	36.3	51.6	57.2	26.2	38.2	27.9	31.6	31.2	28.3	25.6
		CoCurve	277.08	4037	231.60	30.1	41.0	39.9	54.1	56.0	28.2	51.3	30.0	42.2	40.5	33.1	27.2

D.2 PRUNING-RATIO SWEEP

This section widens the sweep to 10% and 50%. Table 17 gives, at every ratio and for every model, CoCurve against the best of the baselines run for it, on WikiText-2 perplexity and the multiple-choice mean; Table 16 above is the same comparison resolved method by method and task by task at the ratios the main text reports.

All methods are pruned under the identical guard of Section C.2 and every mask realises its target ratio to within 0.2 points, so the columns are comparable without the matched-realised-ratio caveats that a mismatched guard would force. Second, the sweep is not monotone in usefulness: beyond roughly 40% no training-free method leaves a model that is useful without recovery (Section 3.5), and the 50% column should be read as a stress test of the selection rather than as a deployment setting.

D.3 THE RECOVERY STAGE

Selection-only is the setting this paper studies, and every number outside this subsection has weight repair disabled for every method. Table 5 reports what happens when it is enabled: one LoRA stage, identical for all five arms: same rank, token budget, schedule and corpus (cleaned Alpaca, disjoint from the calibration set), the recipe LLM-Pruner introduced and SlimGPT and SlimLLM adopted, applied unchanged to each selection.

Recovery compresses the field. Random, at or near the bottom before tuning at every ratio, finishes above at least two tuned baselines at every ratio, so a large part of a recovered model’s quality comes from the recovery rather than from which units were removed, and the ordering survives it. After tuning, CoCurve has the best commonsense accuracy at all three ratios and the best perplexity at two of three. A better selection is therefore not something a short tuning stage makes irrelevant, which is the question a practitioner asks before paying for the selection at all.

Table 17: **Ratio sweep**, WikiText-2 perplexity / multiple-choice mean (%). “Best base.” is the per-ratio best of every baseline run for that model; bold marks the better of the two in each half of a cell. Selection-only throughout: no compensation, no recovery. The mean is over the eight multiple-choice tasks present at every ratio for every model, which is why it is comparable down the column as well as across it; a dash marks a cell with no run. Mistral-Small-24B carries one baseline at $\rho=10\%$ and five at $\rho=50\%$, so its “best base.” is over fewer methods at those two ratios than at the three the main text reports.

Model		10%	20%	30%	40%	50%
Llama-3.2-3B	Best base.	9.6/56	16.9/50	42.7/44	136/41	1199/37
	CoCurve	9.3/56	16.7/51	34.8/44	194/38	4617/35
Falcon3-7B	Best base.	6.4/60	7.4/56	10.1/51	18.4/45	42.6/39
	CoCurve	6.3/61	7.1/56	8.9/51	13.8/46	41.7/40
Llama-3.1-8B-It	Best base.	8.8/63	13.2/56	26.6/50	103/44	773/37
	CoCurve	8.9/64	12.9/58	21.6/52	59.1/43	956/38
Mistral-Nemo-12B	Best base.	6.5/62	11.4/55	17.8/52	73.1/45	513/40
	CoCurve	6.2/64	8.6/59	15.8/50	52.2/43	839/38
Mistral-Small-24B	Best base.	10.8/68	8.7/63	78.8/55	3299/43	4666/35
	CoCurve	5.7/69	9.0/64	20.5/56	277/41	2560/36

D.4 EFFICIENCY

Table 4 reports parameter reduction, prefill and decode throughput and peak memory from real physical removal (Section A.2). Table 18 resolves the same measurement method by method on the headline model, and the result is the one the body states: at iso-budget the gain is governed by the removed non-embedding parameter fraction, so it is shared across selectors rather than distinguishing them: prefill spans only $1.13\text{--}1.19\times$ and decode $1.06\text{--}1.17\times$ across the eight methods, with peak memory identical to within 0.07 GB. Prefill behaves this way on every model we time ($1.17\text{--}1.21\times$ for CoCurve over the four architectures timed here) because it is compute-bound and the removed non-embedding fraction is matched. Single-stream decode is memory-bandwidth-bound and is *not* a band: on the smallest model, Llama-3.2-3B, five of the eight selectors decode *slower* than dense (down to $0.70\times$) because their per-layer widths leave badly shaped kernels, while CoCurve keeps $1.10\times$ and is second of the nine rows. So throughput does not separate selectors where the hardware is compute-bound, and where it does separate them it favours us; either way the discriminator is accuracy at a given speedup (Figure 2). The pruning wall-clock is a one-time cost dominated by the single-unit ablation pass, which carries no gradients or labels and shards trivially across devices; it is paid once per model.

Two of Table 4’s cells are deliberately blank, and the two $\rho=40\%$ rows above them are what stands in for them: both were timed here, on an otherwise idle fleet, with the same physical removal and the same mask the accuracy tables report. The Mistral-Small-24B row’s throughput was timed on the second machine before that model’s mask was corrected, and prefill and decode depend on the attention/FFN split, which the correction moved on 64% of the removed set at that ratio; reprinting a throughput number measured against a mask this paper no longer reports would be worse than leaving it out. Its parameter and peak-memory columns are printed: at this scale the peak is weights, and the removed non-embedding fraction is set by the budget rather than by which units the solver chose. A throughput measurement is only as good as the quiet of the machine it runs on — the same fleet, timed while three neighbouring cards were busy, reported a dense reference 40% below its idle value and prefill speedups scattered from 1.20 to $1.70\times$ where an idle run gives the tight $1.13\text{--}1.19\times$ band above — so the numbers in this table were taken with the fleet otherwise idle, and any run whose method-to-method spread exceeds $0.25\times$ is discarded as contended rather than published.

D.4.1 WHAT THE SELECTION COST BUYS

Building **H** is a one-time forward-only pass: 6.0 GPU-hours on Llama-3.2-3B and 12.6 on Llama-3.1-8B-Instruct, sharded across devices. Two properties decide what that buys.

Table 18: **Inference efficiency of every timed method at 20%** (Llama-3.1-8B-Instruct, real physical removal, one RTX 6000 Ada). At iso-budget the throughput is a method-independent band, so the differentiator is accuracy at a given speedup.

Method	Param red. (%)	Prefill $\times\uparrow$	Decode $\times\uparrow$	Peak mem (GB) $\downarrow$
Dense	0.00	1.000	1.000	16.55
Random	17.44	1.169	1.121	13.72
Magnitude	17.41	1.132	1.091	13.75
Wanda-sp	17.41	1.167	1.174	13.69
FLAP	17.45	1.175	1.105	13.68
LLM-Pruner	17.44	1.153	1.093	13.72
SlimGPT	17.45	1.172	1.124	13.68
SlimLLM	17.52	1.186	1.141	13.69
CoCurve (ours)	17.40	1.170	1.064	13.71

It is quadratic information at linear cost. The object being estimated is the full $M \times M$ interaction matrix. Measuring the same thing directly, by ablating every pair and recording the joint damage, costs $M(M-1)/2$ sweeps: 294,528 on Llama-3.1-8B-Instruct against the 768 we actually run, a factor of 383. The comparison that carries information is therefore against the other method that models pairwise interactions rather than against one that computes a per-unit norm, and on that comparison this is the cheap end. LLM-Surgeon reports 17h08m on four H100s for a 7B model, 68.5 GPU-hours, and 267 GPU-hours at 13B (van der Ouderaa et al., 2024); we spend 12.6 GPU-hours on a single card at 8B.

One build serves every operating point. $\mathbf{H}$ depends on the model, the calibration set and the unit definition, not on ρ and not on λ , so one build serves the whole ratio sweep and the whole regularisation path. This distinguishes us from reconstruction-based selectors, not from activation-based ones: a per-unit norm is also ratio-independent and can be thresholded anywhere, so the amortisation argument is about SlimGPT, SlimLLM and SliceGPT rather than about Wanda-sp or FLAP. Against the former, SlimGPT’s own timings show the dependence: 678 s at 20% against 1074 s at 50% on a 7B model (Ling et al., 2024). The comparable unit is therefore cost per delivered operating point, and over the five ratios we report our build is 151 minutes per point against SlimGPT’s 82.

And one configuration serves the whole family. The guard has no free parameter ($\kappa=1.5\rho$, two protected layers at each end) and λ is selected by Eq. 11 rather than tuned, so the same configuration produces every model at every ratio in this paper: five models, five ratios, one setting. None of the timings above — ours or anyone else’s — include the cost of finding a per-model or per-ratio configuration, and that omission is the one that favours us.

It is forward-only, and at 24B that is the difference between running and not. No backward pass, no optimiser state, no labels: peak memory during selection is inference memory, so a model that fits for evaluation fits for pruning. The two strongest baselines do not have this property. Neither of the two strongest baselines is expensive at the scales their papers report, and both become infeasible at 24B for a reason that is structural rather than incidental. LLM-Pruner backpropagates through the whole model, so it holds the weights and a full set of gradients at once: $2 \times 47.1 = 94.3$ GB against the 79.3 GiB an A100-80GB makes available. SlimGPT is layer-wise and needs only about 7 GB of device memory at 7B, but its host-side accumulator is a $d \times d$ float64 covariance per layer, and at $d=5120$ over 40 layers that is $8.59 \times 40 = 344$ GB against a 128 GB host. Run in their published form at 24B, both abort, at measured peaks of 83.9 GB device memory and 125 GB resident host memory.

Their 24B numbers in this paper exist because we re-implemented both to stream. The group count is a run parameter set from the host budget: SlimGPT’s covariance is collected in eight groups of five layers, which brings its peak from 344 to 43 GB, and LLM-Pruner’s backward is split into three, keeping a third of the gradients resident at a time. Neither change touches what is computed, since a per-layer covariance and a per-weight gradient do not depend on how the layers are grouped, so the scores are identical to the published form. What changes is that one pass becomes several. That multi-pass overhead is the price of estimating curvature from gradients at this scale, and it is exactly what a forward-only estimator does not pay.

Nor is the ceiling confined to the largest model. At $\rho=20\%$ on Mistral-Nemo-12B, LLM-Pruner’s scoring pass reaches a measured peak of 47.35 GB on a 47.37 GB card with nothing else resident and aborts; the same pass completes at 30% and 40%, where fewer groups survive to be scored. A twelve-billion-parameter model on a 48 GB card is an ordinary configuration, and the selector that needs weights plus a full set of gradients is the one it excludes.

Table 19: **Where each selector stops, and what it costs per operating point.** Peak memory decides which methods run at 24B as published, and the 24B failures are measured rather than extrapolated (A100-80GB, 128 GB host). W is one copy of the weights. *Once* is the selection pass on Llama-3.1-8B-Instruct on one RTX 6000 Ada; *per point* divides it over the five compression ratios each method is reported at, which is the comparable quantity because H does not depend on the ratio while a layer-wise selector is re-run at every one.

Method	At 24B, as published	Peak	Per point	Once
Random	no limit	–	0.05 min	0.05 min
Magnitude	no limit	–	0.07 min	0.07 min
FLAP	no limit	$1 \times W$	1.2 min	1.2 min
Wanda-sp	no limit	$1 \times W$	1.2 min	1.2 min
LLM-Pruner	needs 94.3 GB on an 80 GB card; streams in 3 passes	$2 \times W$	2.3 min	2.3 min
SlimLLM	no limit	$1 \times W$	1.8 min	1.8 min
SlimGPT	needs 344 GB of host memory; streams in 8 passes	$1 \times W$	82 min	82 min
CoCurve	runs as published: forward passes only	$1 \times W$	151 min	12.6 h

E MECHANISM, ABLATIONS, AND ROBUSTNESS

E.1 EXTENDED ABLATIONS

Table 20: **Restricting the eligible units to one module type** at a 20% target, under the same guard, the same $\hat{\lambda}$ and the same layer protection as everywhere else. *attn/FFN* counts how the removed units divide between attention units and FFN groups. Attention alone cannot reach the budget on Falcon3-7B, which has four attention units per layer: it stops at 9.1%, and what it does remove is catastrophic. On Llama-3.2-3B it does reach the target, at $28\times$ the joint arm’s perplexity. The Falcon3-7B row pair is not a comparison: the joint objective spends nothing on attention there, so its mask is the FFN-only mask, which is why the two rows agree to the digit. The comparison lives on Llama-3.2-3B, where the joint arm removes 38 attention units and leads on the capability group. Best in column in bold.

Eligible units	Falcon3-7B				Llama-3.2-3B			
	ρ	attn/FFN	Wiki↓	Cmn.↑	ρ	attn/FFN	Wiki↓	Cmn.↑
Attention only	9.1%	96/0	851	.369	20.1%	180/0	465	.377
FFN only	20.2%	0/101	7.07	.594	20.1%	0/120	16.42	.550
Both (CoCurve)	20.2%	0/101	7.07	.594	20.1%	38/95	16.66	.566

The remaining design choices are isolated on Llama-3.1-8B-Instruct at 20% under the guard used everywhere else (cap 0.30, protect 2/2), so every row is directly comparable. We analyse three axes.

(i) Interaction granularity. The edges could act at a coarser level than unit pairs: keep the cross-component block but replace it with its scalar mean, and the interaction decides how much of each component type to remove and never which unit. That is the level SAViT (Zheng et al., 2022) interacts at, and Table 21 runs it. Because averaging the block rescales the edge mass that enters the marginal, that arm is calibrated at its own $\hat{\lambda}$ rather than ours, which is its best case. It still loses six of the seven columns, including all four capability groups; only PTB is better. What the edges contribute is therefore which *pairs* are costly together, not a per-component budget.

(ii) Sign handling. The co-pruning edges the solver actually acts on are positive: of the 11,325 edges inside the selected set on Llama-3.1-8B-Instruct at $\rho=20\%$, every one is positive, as are 99.3% of the full off-diagonal. Signed, clipped $\max(H_{uv}, 0)$ and $|H_{uv}|$ therefore define the same objective

Table 21: **Interaction granularity** (Llama-3.1-8B-Instruct, $\rho=20\%$). Both rows use the same solver, guard and budget and differ only in whether the off-diagonal acts per unit pair or per component pair; each is calibrated at its own $\hat{\lambda}$ (0.0108 and 0.0078) and realises its target ratio to within 0.2 points. Bold is best in column.

Interaction granularity	WikiText↓	PTB↓	C4↓	Commonsense↑	Science↑	Reading↑	Knowledge↑
unit pair (CoCurve)	12.88	29.63	19.58	63.7	48.0	58.6	38.5
component pair	12.90	28.81	19.84	62.8	47.2	57.9	37.8

on that set, and the choice between them is not an empirical question here. Signed remains the principled default (Section B.4): a negative marginal is a genuine partial cancellation, and on a model with sign-mixed edges clipping discards it.

(iii) **Estimation sensitivity.** Increasing top- r beyond 256 leaves $\mathbf{H}$ nearly unchanged; parameter cost is the cleanest budget; the calibration size $|\mathcal{C}|$ and the logit support r are swept in Table 22.

E.2 SURROGATE FIDELITY: EMPIRICAL STUDY

Same-budget discrimination. A cross-ratio fit can look strong simply because both axes grow with ρ . The decisive question is whether the surrogate distinguishes good from bad pruning sets at one fixed budget. We fix $\rho=20\%$ on Llama-3.1-8B-Instruct and construct a pool of 26 prunings at the same target budget: the CoCurve-greedy set, the diagonal-greedy set, 14 random sets and 10 local-swap perturbations of the CoCurve set, each filled to within 0.75% of the same removed cost. We compare predicted $\frac{1}{2}s^\top \mathbf{H}s$ against measured $D_{\text{KL}}(p_0 \| p_S)$ across the pool (Figure 6). The separation is complete: the twelve structured selections average D_{KL} 0.62 over a range of 0.57–0.71, and *every* random set is worse than *every* structured one, averaging 1.26. Complete separation alone would carry a high rank correlation over the whole pool (0.87), so the number that matters is the one measured *inside* the structured cluster, where all twelve sets lie within 0.15 nats of each other: the surrogate still ranks them at Spearman 0.55, and at 0.58 over the ten local swaps alone, which is the hardest discrimination in the pool and the one the greedy solver actually makes at every step.

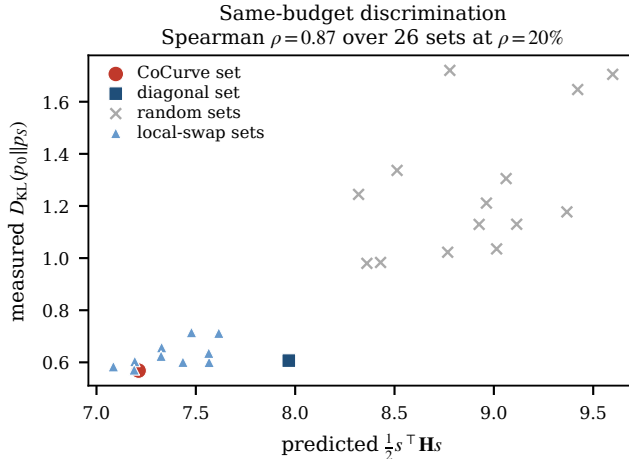


Figure 6: **The surrogate discriminates at one fixed budget.** Predicted $\frac{1}{2}s^\top \mathbf{H}s$ against measured $D_{\text{KL}}(p_0 \| p_S)$ over 26 prunings matched to within 0.75% of the same removed cost on Llama-3.1-8B-Instruct at $\rho=20\%$: the CoCurve and diagonal greedy sets, ten local swaps of the former, and fourteen random sets. Every random set is above every structured one; within the structured cluster the rank correlation is 0.55.

E.2.1 SENSITIVITY TO THE SURROGATE BUDGET

The edge estimator has two finite-budget knobs: the calibration size (number of sampled positions) and the top- r logit support of the Fisher-Gram features. Neither choice drives the result. We re-build $\mathbf{H}$ on the cheapest model (Llama-3.2-3B) at top- $r \in \{128, 256, 512\}$ and calibration size $\in \{64, 128\}$, then run each build through the shipped selector – the guard of Section C.2 and $\hat{\lambda}$ re-chosen on held-out KL for that build – and evaluate (Table 22). Across the builds measured, WikiText-2 perplexity spans 3.1% and MMLU 0.4 points, while the $\hat{\lambda}$ each build selects moves by 20%. Doubling top- r from 256 to 512 is the sharpest case: it shifts $\hat{\lambda}$ from 0.0911 to 0.0909, a different breakpoint of a different path, and returns the *bit-identical* pruning — the same 133 units, the same realised ratio, the same scores. The selection is therefore not sensitive to the surrogate budget within the regime we use, and the insensitivity is not because nothing upstream changed. Figure 7 asks the same question of the selection directly rather than through a benchmark: how many calibration sequences are needed before the chosen units stop changing.

Table 22: **Surrogate-budget sensitivity** (Llama-3.2-3B at $\rho=20\%$, under the deployed guard). Upper block varies top- r at fixed calibration 128; lower block varies calibration at fixed top-256. The default is (128, 256). Each build re-derives its own $\hat{\lambda}$, so the column shows that the knob does move what the rule selects — and the pruned model does not follow.

knob	value	$\hat{\lambda}$	Wiki↓	MMLU↑
top- r	128	0.1094	17.17	.340
top- r	256 (default)	0.0911	16.66	.339
top- r	512	0.0909	16.66	.339
calib	64	0.1050	16.82	.336
calib	128 (default)	0.0911	16.66	.339

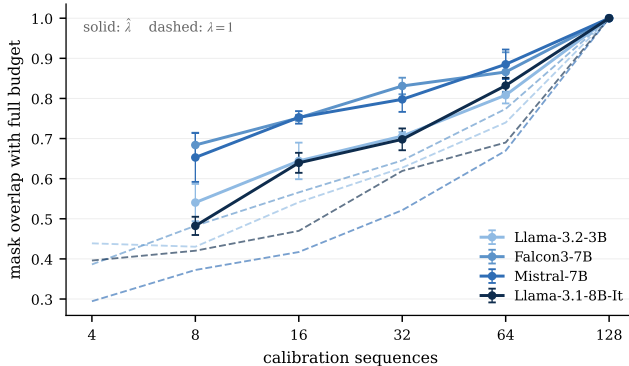


Figure 7: **How much calibration data the selection needs.** Overlap between the units chosen from n sequences and those chosen from the full budget of 128, mean over three draws. Measuring the selection rather than downstream accuracy separates estimator convergence from benchmark resolution. At the calibrated $\hat{\lambda}$ the overlap is consistently higher than at $\lambda=1$: damping the off-diagonal, the part of the matrix that needs the most data, also makes the solution easier to estimate.

Calibration source. C4 is our default, following the SparseGPT/Wanda convention, and the comparison that justifies it rebuilds $\mathbf{H}$ for Llama-3.1-8B-Instruct entirely from 128 WikiText-2 sequences and re-runs the same 20% selection. The two agree on which units matter and differ exactly as source–target alignment predicts. Calibrating on WikiText buys a little WikiText perplexity (12.58 against 12.91) and a little on the other two corpora (−1.0 on PTB, −0.7 on C4), and it costs accuracy on every capability task the two runs score identically: −0.9 on ARC-challenge, −1.3 on ARC-easy, −2.6 on OpenBookQA, −0.7 on PIQA and −3.8 on WinoGrande. The more diverse corpus is the better calibration source for everything except the corpus it is matched against, which is why C4 is the default.

Calibration draw. Sensitivity to the draw itself is tested in two ways. Rebuilding $\mathbf{H}$ from three independent samples of 128 C4 sequences and selecting at $\lambda=1$ — so that the draw is the only thing that varies and the selection cannot absorb it — leaves the result stable: WikiText-2 13.11 ± 0.22 , MMLU 38.1 ± 0.7 , commonsense average 60.0 ± 0.9 over seeds 1/7/123. That isolates the estimator rather than the shipped configuration. The stronger test, rebuilding the curvature from disjoint text and re-running the entire protocol at the model’s own $\hat{\lambda}$, is reported on Llama-3.2-3B in Section E.8.

E.3 MECHANISM ANALYSIS

Bridge units. With $d_u = \mathbf{H}_{uu}$ and $b_u = \sum_{v \neq u} |\mathbf{H}_{uv}|$, *bridge* units (low d_u , high b_u) are retained by CoCurve but removed by diagonal pruning.

A concrete bridge unit. To make the phenomenon tangible, we single out one unit from the real estimated $\mathbf{H}$ of Llama-3.1-8B-Instruct. A representative bridge is an *FFN channel group at layer 1*. Its node saliency $\mathbf{H}_{uu}$ places it in the *bottom* 31% of all units, so a node-first scorer (magnitude, activation, or diagonal Taylor) would rank it as safely prunable. Yet its coupling $b_u = \sum_{v \neq u} |\mathbf{H}_{uv}|$ is in the *top decile*: its strongest co-pruning edges run to *attention* units at layer 0 (a cross-module coupling, $|\mathbf{H}_{uv}| \approx 5.2 \times 10^{-3}$, comparable to its own saliency 7.4×10^{-3}) and to FFN groups at layers 1–3 (cross-layer). This is exactly the configuration CoCurve is built to catch: a unit whose removal is cheap in isolation but costly alongside the early-attention and nearby-FFN units it is coupled to. Diagonal pruning deletes it together with those partners; CoCurve retains it.

Controlled causal test: bridge connectivity, not saliency, drives the damage. The bridge diagnosis above is structural, read off $\mathbf{H}$; we confirm it is causal with a matched-pair intervention, run on every model we build $\mathbf{H}$ for. On each, we force-prune the bridge units (low d_u , high b_u) and, as a control, an equal-size set drawn from the same low-saliency pool but with low connectivity, matched unit-for-unit on saliency d_u , unit type and per-unit cost. The remainder of the budget is one cost-normalised greedy fill, computed once and shared by both arms. The two masks that result hold the same number of units, the same attention/FFN split, the same removed cost and the same realised ratio to the last bit, and differ only in which low-saliency population was forced out, so any gap isolates connectivity with saliency, type, cost and ratio all held fixed. On Llama-3.1-8B-Instruct that is 169 units each, 1.399×10^9 parameters, realised ratio 0.2005, 82 units shared and 87 forced.

Removing the high-connectivity units is far more destructive on capability (Table 23). Nineteen of the twenty capability cells fall, by up to 23.7 points, even though a node-first scorer rates the two forced sets equally prunable. Perplexity is not a reliable witness to it. On three of the five models perplexity moves the *other* way — WikiText improves by 4.5, 5.5 and 9.7 while every capability group in those rows falls — so on those the arm a next-token proxy prefers is the arm that has lost the most capability. On Falcon3-7B the two arms are within 0.8 perplexity of each other, and on Mistral-Small-24B, where the intervention is by far the most damaging, perplexity registers it in full (+425.5). The proxy is therefore free to point either way at a fixed capability cost, which is the property that makes it unsafe to select on. That is the failure mode a node-first criterion is built to walk into, and the one CoCurve avoids by scoring the edges.

How much of the selection the edge term moves. Comparing the calibrated solution against the pure diagonal one at $\rho=20\%$, the edge term moves between 1% and 36% of the removed set across the five models, and the low end is the instructive case. On Mistral-Nemo-12B, $\hat{\lambda} \approx 2.4 \times 10^{-4}$ yields a set differing in three units out of 187: an attention head and an FFN group in layer 19 that the diagonal selector removes and the edge term holds back, and one FFN group in layer 24 taken in their place, at an almost unchanged module split (attention takes 16.2% of the removed cost against 16.5%). Those three substitutions are worth 3.1 paired standard errors of held-out risk: the edge term corrects specific, identifiable units rather than rewriting the selection.

Where the effect is smallest. Falcon3-7B is the one model on which a capability group does not fall (Reading, +0.8), and it is also the model whose off-diagonal is least depth-graded of the ten in Table 12 ($\rho_s(\ell_u, b_u) = -0.54$ against a median of -0.87) and the one whose bridge corner is smallest (30 of 560 units against 5 to 14% elsewhere). Less positional structure in the off-diagonal, less for the edge term to protect, a smaller effect when it is removed by force: the two orderings agree, on the model where they are most exposed.

Table 23: **The bridge test, on every model we build H for** ($\rho=20\%$). Low-saliency units split by off-diagonal connectivity, matched unit-for-unit on saliency, unit type and cost, with one greedy fill shared by both arms. Each pair holds the same number of units at the same removed cost and the same realised ratio. Nineteen of the twenty capability cells fall; on three of the five models perplexity moves the other way, on one it is flat, and on Mistral-Small-24B it agrees with them. The Mistral-Small-24B pair is pruned on the second server and enters from its export (170 units per arm, realised ratio 0.2014).

Model	Forced set	Commonsense $\uparrow$	Science $\uparrow$	Reading $\uparrow$	Knowledge $\uparrow$	Wiki $\downarrow$
Llama-3.2-3B	Control	58.5	42.9	45.7	35.2	31.8
	Bridge	48.8	36.9	41.5	30.7	26.2
	Δ	-9.7	-6.0	-4.2	-4.5	-5.5
Falcon3-7B	Control	60.3	48.8	52.4	38.4	7.3
	Bridge	55.8	43.4	53.1	37.1	8.1
	Δ	-4.5	-5.4	+0.8	-1.3	+0.8
Llama-3.1-8B-It	Control	66.5	53.0	52.2	40.5	22.8
	Bridge	53.6	38.9	45.6	32.0	18.3
	Δ	-12.9	-14.1	-6.6	-8.5	-4.5
Mistral-Nemo-12B	Control	64.5	51.8	57.1	40.1	23.3
	Bridge	53.1	39.3	41.5	31.7	13.6
	Δ	-11.4	-12.5	-15.6	-8.4	-9.7
Mistral-Small-24B	Control	63.9	51.7	57.3	40.7	14.4
	Bridge	40.9	29.3	33.6	25.9	439.9
	Δ	-23.0	-22.4	-23.7	-14.8	+425.5

Layer pattern. Figure 4 gives the per-layer, per-type profile on four models. On Llama-3.1-8B-Instruct at $\rho=20\%$ the solution removes 46 attention units and 105 FFN groups over 26 of 32 layers, at most nine in any one layer, and splits evenly over the second half of the depth alone, 46 attention against 46 FFN.

E.3.1 FIDELITY TO THE DENSE MODEL

The selection criterion is token-level KL to the dense model, so the first thing to check is whether the pruned model actually reproduces the dense model’s decisions more faithfully than the alternatives do, and the second is what that fidelity buys. On Llama-3.1-8B-Instruct at $\rho=30\%$ we measure, per task, the fraction of items on which each pruned model predicts what dense predicts. CoCurve is first or second on all six tasks examined and first on three; LLM-Pruner takes the other three, by 0.1 to 0.4 points. Every other baseline is behind both on every task. The objective does what it claims.

Splitting those items by whether *dense* was right sharpens it: on PIQA, CoCurve reproduces the dense model on 81.8% of the items dense answers correctly, the best of the eight methods. The selection delivers the fidelity its objective asks for, and it does so without the benchmark ever being consulted.

E.4 FEW-SHOT PROMPTING

Zero-shot accuracy asks the model one question at a time. Five-shot prompting puts a long demonstration prefix in front of the same question, which is where a pruned model has to carry more context to answer the same item, the regime in which damage to attention would show first. SlimGPT and 2SSP both report a few-shot number for that reason.

Table 24 gives MMLU at five shots, with demonstrations drawn from the dev split of the subject being asked about and identical for every method. The ordering is the one the zero-shot tables show: CoCurve leads on both models at both reported ratios, by 0.6 to 2.1 points over the strongest baseline. The advantage the selection buys is therefore not an artefact of the zero-shot prompt format.

Table 24: **MMLU at five shots (%)**. Demonstrations come from the dev split of each subject and are identical for every method; dense accuracy in grey.

Model	ρ	LLM-Pruner	SlimLLM	CoCurve
Llama-3.1-8B-Instruct (dense 50.8)	20%	34.9	34.8	37.0
	30%	31.3	27.0	32.7
Llama-3.2-3B (dense 41.5)	20%	33.0	31.0	33.6
	30%	29.3	28.4	30.1

E.5 COMPOSITION WITH QUANTISATION

Pruning and quantisation attack different redundancies: one removes units, the other lowers the precision of what remains. Whether they compose is a deployment question, and it is not settled by either measurement alone. If the two drew on the same slack, stacking them would cost more than their separate costs suggest.

Table 25 pairs each model state with itself at bf16 and at 4 bits, so the cost of quantisation is a within-row difference and the dense row is the reference for what that cost should be. On the dense model 4 bits costs 5.7% of perplexity and 0.8 multiple-choice points. After pruning to $\rho=30\%$ it costs 7.0% and 1.8 points, and at $\rho=40\%$ 13.1% and 1.3 points. The accuracy cost is therefore roughly what the dense model pays at both ratios, while the perplexity cost grows as the remaining slack is spent, which is what a redundancy account predicts, and small enough that the two compressions can be applied together.

The comparison arms turn this from a property of the pipeline into a property of the selection. The two strongest baselines are quantised from their own pruned models under the identical quantiser and load order, and the ordering after 4 bits is the ordering before: CoCurve reaches 23.1 perplexity and 49.9 accuracy, against 29.2 and 48.8 for LLM-Pruner and 80.0 and 44.9 for SlimLLM.

Quantisation is close to additive with the selection, so a later decision to quantise does not make the choice of which units to remove for the practitioner.

Table 25: **Pruning composes with 4-bit quantisation** (Llama-3.1-8B-Instruct). Each row is one model state measured twice; the last two columns are the within-row cost of quantising it. Perplexity on WikiText-2, accuracy the eight-task multiple-choice mean. The quantiser and the load order are identical in every row.

Model state	bf16		4-bit		cost of 4 bits	
	Wiki↓	MC↑	Wiki↓	MC↑	Wiki %	MC pt
Dense	6.4	67.2	6.8	66.4	+5.7	−0.8
$\rho=30\%$						
CoCurve	21.6	51.7	23.1	49.9	+7.0	−1.8
LLM-Pruner	26.6	49.7	29.2	48.8	+9.7	−0.9
SlimLLM	78.2	45.3	80.0	44.9	+2.2	−0.4
$\rho=40\%$						
CoCurve	59.1	43.5	66.9	42.2	+13.1	−1.3

E.6 ROBUSTNESS OF THE SELECTION

Three questions about stability, each answered by a different perturbation: resampling the held-out set, changing the selection protocol, and rebuilding the curvature from independent calibration text. The third costs a full re-estimation, which we paid.

E.7 HOW STABLE IS THE SELECTED λ ?

Every number in this paper comes from one calibration draw and one held-out set of 16 sequences. Two questions follow, and they are different: whether the *selection step* would survive a different

draw of the held-out set, and whether the *curvature it selects over* would survive a different draw of the calibration text. We answer both.

Resampling the held-out set collapses the candidate space. The path files record the per-sequence risk at every path point, so the bootstrap is exact and needs no further computation. Over 26 paths spanning ten models and five ratios, at 2000 resamples each, the resampled criterion reaches a median of 3.3% of the members of the enumerated family, and 2.5% at the three ratios this paper reports. A typical family holds 318 prunings and the whole bootstrap moves among ten of them. The 5–95 percentile of $\hat{\lambda}$ spans a median of 0.016 of $[0, 1]$, and is under 0.05 on 16 of the 26 paths.

What the criterion delivers is a pruning at the minimum rather than a distinguished value of λ : evaluated back on the full held-out set, the mask a resample chooses costs a median of 0.00% additional risk over the one $\hat{\lambda}$ chooses, and every path puts at least 98.5% of resamples within one standard error of the minimum. The solution is piecewise constant and the path is flat near its optimum, so the members the bootstrap moves among are the ones that cannot be told apart.

Does the selected point hold up downstream? The bootstrap judges $\hat{\lambda}$ by held-out risk, the quantity it is selected on. The independent check is to take the selected point and its neighbours and evaluate all of them on the full suite. On Llama-3.2-3B at $\rho=20\%$ we do this for five points: $\hat{\lambda}$ itself, the two members at the edges of its 1-SE band, and the two nearest solutions just outside it. The selected point is within 0.3 points of the best of the five on three of the four capability groups and within 0.91 on the fourth, and it is 2.7 to 2.9 points above the worst on the twelve-task mean. The criterion lands at the favourable end of the region it cannot resolve.

Changing the protocol. A bootstrap only perturbs the sample. We also varied the protocol itself on Llama-3.1-8B-Instruct at $\rho=20\%$, one change at a time: the selection statistic ($\text{KL} \rightarrow \text{NLL}$), the number of held-out sequences ($16 \rightarrow 48$, pooling three independent draws), and the draw itself (two further seeds). *Four of these five variants select the bit-identical solution*, $\hat{\lambda} = 0.0108$. The exception is changing the held-out domain from C4 to WikiText, which moves $\hat{\lambda}$ to 0.0213: a different breakpoint, the same order of magnitude, and the direction one would predict when the evaluation corpus changes.

Holding the curvature fixed. Both perturbations above act on the selection while $\mathbf{H}$ stays as estimated. Perturbing the curvature itself, by rebuilding it from disjoint calibration text, is a separate and more expensive question, and Section E.8 reports it.

E.8 REBUILDING THE CURVATURE FROM INDEPENDENT CALIBRATION TEXT

Section E.7 perturbs the held-out set and the selection protocol but holds $\mathbf{H}$ fixed. The question it cannot reach is whether a curvature estimate built from *different text* would lead anywhere else. Answering it means paying the full estimation cost a second time, which we did. On Llama-3.2-3B we rebuilt $\mathbf{H}$ from scratch (activation collection, all 672 single-unit ablations, Gram assembly) from a disjoint draw of the same sources, with its own held-out file, and re-ran the identical protocol with no parameter changed.

What the selector consumes transfers. Diagonal saliency d_u correlates at 0.99 between the two draws, and total coupling b_u , the quantity the bridge-unit argument rests on, at 0.96. This is the level at which the method reads $\mathbf{H}$: the marginal of Eq. 8 contracts the matrix against the selected set, and Section B.3.2 shows the same contraction is what the split-half reproduction measures. Individual entries transfer less well than their aggregates, and the selector consumes the aggregates.

The selection moves, and the model does not. $\hat{\lambda}$ goes from 0.0911 to 0.0383 and the two prunings share 57.9% of their removed units. At $\rho=20\%$ no capability group moves by more than 0.81 points (commonsense -0.30 , science QA $+0.16$, reading $+0.42$, world knowledge $+0.81$) and the twelve-task mean moves $+0.15$. Perplexity is within 2.1% on two of three corpora (-0.3% WikiText, -2.1% C4) and $+14.0\%$ on PTB, the least stable of the three here as elsewhere.

The three levels together say more than agreement would have. What the selector reads off $\mathbf{H}$ transfers (d_u at 0.99, b_u at 0.96); the tie-breaking among units the marginal cannot separate does

not, and does not need to. The held-out bootstrap shows the same structure from the other direction: mask agreement of 58.7% alongside a median excess risk of 0.00%. What transfers is the aggregate, which is what the method is asked to deliver.

E.9 VISION–LANGUAGE MODELS: PER-BENCHMARK RESULTS

Units and masking. Units are attention heads — grouped by shared key/value head where the tower uses grouped-query attention — and FFN channel groups, sixteen groups per layer, in *both* towers. A unit is removed by zeroing its slice of the input to the projection that writes into the residual stream, exactly as in the text-only pipeline. The layer index is composite, so the per-layer cap $\kappa=1.5\rho$ and the first/last-two-layer protection apply within each tower rather than to a pooled index. Nothing else is added: no per-tower budget, no tower-level weighting, and no new hyperparameter.

Calibration. 128 image–caption pairs from Flickr30k (Young et al., 2014) — 192 for Qwen2-VL-2B and 96 for LLaVA-NeXT-7B, whose matrices were estimated under earlier budgets and left as they were. The difference is a build-time budget on our own estimator and nothing else: every baseline draws its own fixed calibration on every model alike (64 pairs for Wanda-sp and FLAP, 32 for LLM-Pruner’s gradient pass), and the 48 held-out pairs used to choose λ are taken from the same deterministic stream immediately *after* that model’s own calibration draw — pairs 128–175 where the matrix saw 128 or fewer, pairs 192–239 on Qwen2-VL-2B — so on every model they are images the estimator never saw. Neither departure moves a comparison: a draw below 128 makes the off-diagonal noisier and the held-out criterion correspondingly more reluctant to trust it, which costs our arm and no other; and the one model built on more carries the median of our five headline margins, not the largest. The held-out pairs are used only for the λ argmin and enter nothing else. Flickr rather than COCO is deliberate: SEED-Bench and several neighbouring suites are built on COCO images, so a COCO calibration draw could place an evaluation image in the estimator. Scored positions are the caption tokens only — the prompt and the image placeholders are context, and scoring them would let a unit look important for reproducing text the model was handed. The teacher support is top- $r=64$ rather than the 256 used in the text-only experiments, because an image-grounded sequence is expensive; splitting the scored positions into disjoint halves gives two independent estimates of $\mathbf{H}$ that correlate at $r=0.99$ on the off-diagonal, so the reduction costs nothing that matters here.

Evaluation. Answers are scored by comparing the logits of the admissible answer tokens — option letters where the benchmark is multiple choice, the words *yes* and *no* where it is binary — at 1000 items per benchmark. The dense model is measured under this same harness rather than quoted, so every row is commensurable with every other. Retention is floor-corrected, $(\text{arm} - c)/(\text{dense} - c)$, against the chance level c of a selector that has learned nothing: 25 on the four-way multiple-choice suites (MMBench, SEED-Bench, ScienceQA, AI2D, MMStar) and 50 on the two binary ones (POPE, MME), giving $c=32.1$ for the seven-benchmark mean. Without the correction a model at chance would read as a third of dense retained rather than zero.

Baselines. All are training-free and structured, and all receive the identical budget, guard, calibration draw and scoring code. Magnitude scores a unit by the norm of *all* the weights it owns. Wanda-sp and FLAP are computed on the input channels of the output projection, where both are defined; FLAP additionally standardises each layer’s metric distribution to the global mean and standard deviation before the global search, which is the half of that method that decides depth allocation. LLM-Pruner is the first-order Taylor score $\sum |w \odot \partial \mathcal{L} / \partial w|$ accumulated one image–caption pair per backward pass. The layer-drop arm is ShortGPT’s block-influence signal measured on the same calibration pass.

The block-diagonal control. One row of Table 26 is an arm rather than a baseline. It re-runs our own selector on $\mathbf{H}$ with the vision–language blocks set to zero, holding the diagonal, the within-tower off-diagonal, the budget, the guard and the λ protocol fixed — the assumption a per-tower pruner makes without stating it, made explicit. On 8 of the 21 cells the two selectors return the same mask, so within-tower coupling alone already fills the budget there; on the rest the masks differ and the seven-benchmark mean moves by -1.2 to $+3.6$ points, its two largest values on LLaVA-1.5-7B. The cross-tower block is reported because it is measurable, not because the shared budget depends on it: what Section 3.6 asks of the off-diagonal as a whole is the $\lambda=0$ comparison.

Higher budgets. At $\rho=40\%$ no selector in this comparison, ours included, retains a third of its dense score above chance on any model, so that budget appears in Table 26 and nowhere else: the numbers are measured, they are simply not a comparison. Section E.10 reports what one identical LoRA stage does to every arm at that budget.

Table 26: Vision–language models, per benchmark: the complete grid, the configurations the body does not table included. Seven models at three ratios, ten arms each, every arm at 1000 items per benchmark; the block header repeats the ratio and the item count. **Bold** is the best selector in the column, with the CoCurve ablation excluded from that comparison. *cap. PPL* is perplexity on the held-out captions, the quantity the selection was made against.

Model	Selector	MMB	SEED	SQA	AI2D	MMStar	POPE	MME	Mean	cap. PPL	Ret.
<i>Qwen2-VL-2B, $\rho=20\%$, 1000 items</i>											
	Dense	78.6	62.4	80.5	73.0	44.6	88.4	83.6	73.0	15.39	
	CoCurve	65.1	51.9	57.1	50.3	38.2	85.9	66.5	59.3	17.37	66%
	block-diagonal H	65.9	51.9	53.0	46.9	36.0	81.8	74.8	58.6	18.18	65%
	diagonal only ($\lambda=0$)	59.5	50.1	47.2	38.2	34.2	82.9	63.6	53.7	14.99	53%
	LLM-Pruner	57.6	50.2	51.5	48.5	32.3	87.2	71.7	57.0	18.01	61%
	Wanda-sp	59.0	49.6	55.3	39.0	31.4	74.6	51.6	51.5	18.33	47%
	FLAP	34.3	28.8	36.3	26.3	26.6	53.3	50.3	36.6	58.11	11%
	ShortGPT (layer drop)	30.6	27.5	32.7	22.3	24.9	61.0	49.5	35.5	17.10	8%
	Magnitude	43.1	32.8	41.8	28.3	25.9	58.5	51.5	40.3	25.05	20%
	Random	56.3	43.9	41.1	33.8	31.1	70.3	59.7	48.0	18.64	39%
<i>Qwen2-VL-2B, $\rho=30\%$, 1000 items</i>											
	Dense	78.6	62.4	80.5	73.0	44.6	88.4	83.6	73.0	15.39	
	CoCurve	35.4	30.7	36.9	24.0	26.3	70.9	53.7	39.7	23.75	18%
	block-diagonal H	36.0	32.6	36.5	24.0	26.6	66.0	54.4	39.4	23.39	18%
	diagonal only ($\lambda=0$)	34.4	31.1	35.9	24.6	27.1	62.1	49.7	37.8	25.84	14%
	LLM-Pruner	42.3	37.6	41.1	28.5	26.6	76.4	61.6	44.9	25.70	31%
	Wanda-sp	34.3	29.6	36.3	23.9	26.8	71.1	59.9	40.3	27.64	20%
	FLAP	33.4	29.3	36.5	23.9	27.4	51.2	50.8	36.1	138.28	10%
	ShortGPT (layer drop)	33.8	29.6	36.0	24.3	27.3	49.6	50.7	35.9	26.40	9%
	Magnitude	33.9	30.3	34.4	25.9	25.4	58.4	49.0	36.8	77.09	11%
	Random	34.2	29.2	36.4	27.2	26.9	50.8	49.3	36.3	129.57	10%
<i>Qwen2-VL-2B, $\rho=40\%$, 1000 items</i>											
	Dense	78.6	62.4	80.5	73.0	44.6	88.4	83.6	73.0	15.39	
	CoCurve	34.2	30.1	35.3	23.9	27.5	53.2	50.7	36.4	61.60	10%
	block-diagonal H	34.4	29.2	35.4	23.6	27.2	52.2	53.6	36.5	61.06	11%
	diagonal only ($\lambda=0$)	34.4	29.8	35.5	23.6	27.2	54.5	49.9	36.4	73.93	10%
	LLM-Pruner	34.3	29.8	35.4	23.2	27.3	51.1	50.3	35.9	50.78	9%
	Wanda-sp	33.9	29.5	34.9	23.6	27.3	51.0	49.2	35.6	122.79	9%
	FLAP	31.8	27.4	37.2	24.6	28.4	51.2	49.4	35.7	301.08	9%
	ShortGPT (layer drop)	32.7	28.7	34.1	24.7	28.8	52.7	48.3	35.7	165.43	9%
	Magnitude	29.7	24.0	34.8	24.0	29.4	51.0	49.2	34.6	235.66	6%
	Random	29.3	24.7	36.0	23.8	27.3	48.5	49.2	34.1	957.02	5%
<i>Qwen2.5-VL-3B, $\rho=20\%$, 1000 items</i>											
	Dense	83.8	65.2	80.9	80.9	56.4	89.8	85.0	77.4	13.15	
	CoCurve	70.5	56.0	60.5	58.9	42.2	89.7	71.0	64.1	15.64	71%
	block-diagonal H	69.6	56.2	68.0	57.6	44.0	89.3	71.5	65.2	15.05	73%
	diagonal only ($\lambda=0$)	70.0	58.4	65.6	60.2	42.6	81.3	72.7	64.4	16.09	71%
	LLM-Pruner	70.4	57.6	61.6	58.9	43.7	87.8	70.5	64.4	15.38	71%
	Wanda-sp	65.2	52.5	59.1	54.4	38.0	70.9	65.8	58.0	18.01	57%
	FLAP	32.5	27.7	33.4	26.9	30.9	62.3	52.3	38.0	95.87	13%
	ShortGPT (layer drop)	56.4	38.9	49.5	34.5	32.2	88.2	68.8	52.6	15.86	45%
	Magnitude	29.7	28.1	35.3	25.7	25.7	49.3	51.7	35.1	3396.11	6%
	Random	62.4	48.8	55.8	42.0	40.5	74.4	55.2	54.2	20.24	49%
<i>Qwen2.5-VL-3B, $\rho=30\%$, 1000 items</i>											
	Dense	83.8	65.2	80.9	80.9	56.4	89.8	85.0	77.4	13.15	
	CoCurve	57.0	45.1	44.1	34.2	33.1	72.1	64.3	50.0	21.86	39%
	block-diagonal H	53.4	44.5	42.3	33.3	32.8	68.0	62.9	48.2	20.53	35%
	diagonal only ($\lambda=0$)	44.5	35.6	40.2	30.2	26.5	67.8	55.7	42.9	19.90	24%
	LLM-Pruner	51.1	41.5	48.0	31.9	28.3	82.9	65.2	49.8	23.46	39%
	Wanda-sp	47.6	32.4	41.6	30.8	28.0	57.8	52.4	41.5	21.81	21%
	FLAP	30.9	28.0	36.8	26.5	28.1	50.5	50.4	35.9	261.11	8%
	ShortGPT (layer drop)	32.1	24.0	37.0	28.1	24.1	49.8	50.3	35.1	24.78	6%
	Magnitude	31.3	27.4	36.9	24.7	26.2	50.1	49.9	35.2	8909.21	7%
	Random	31.5	30.5	35.8	25.8	23.3	48.2	49.3	34.9	56.61	6%
<i>Qwen2.5-VL-3B, $\rho=40\%$, 1000 items</i>											
	Dense	83.8	65.2	80.9	80.9	56.4	89.8	85.0	77.4	13.15	
	CoCurve	33.8	26.5	36.4	23.9	27.7	56.1	49.1	36.2	55.67	9%

Model	Selector	MMB	SEED	SQA	AI2D	MMStar	POPE	MME	Mean	cap. PPL	Ret.
	block-diagonal H	32.2	28.0	37.3	24.4	27.8	56.8	51.2	36.8	48.49	10%
	diagonal only ($\lambda=0$)	34.2	29.7	38.9	25.5	27.1	57.4	55.4	38.3	59.95	14%
	LLM-Pruner	30.3	27.0	35.1	24.7	26.0	56.7	50.3	35.7	51.45	8%
	Wanda-sp	32.6	26.4	35.5	27.9	25.2	55.0	49.1	36.0	37.39	8%
	FLAP	31.0	25.6	34.9	24.9	23.6	53.7	50.1	34.8	1239.02	6%
	ShortGPT (layer drop)	31.1	27.3	33.9	27.2	25.0	51.5	50.1	35.2	31.37	7%
	Magnitude	32.1	26.8	36.7	25.8	28.8	49.0	48.7	35.4	18629.02	7%
	Random	32.4	27.4	38.0	25.7	26.6	50.7	49.4	35.7	404.45	8%
<i>Qwen2.5-VL-7B, $\rho=20\%$, 1000 items</i>											
	Dense	86.7	70.5	88.5	84.5	63.8	89.6	87.3	81.6	20.53	
	CoCurve	81.9	67.1	77.8	75.2	51.5	86.9	80.5	74.4	17.96	86%
	block-diagonal H	81.4	67.1	78.2	74.2	51.6	86.6	80.2	74.2	18.00	85%
	diagonal only ($\lambda=0$)	81.1	66.3	75.4	70.6	53.9	87.6	79.4	73.5	16.27	84%
	LLM-Pruner	80.5	61.9	78.8	73.6	52.5	84.7	73.5	72.2	19.22	81%
	Wanda-sp	73.8	56.7	72.0	63.5	43.2	74.4	56.5	62.9	17.43	62%
	FLAP	80.4	63.4	78.6	70.5	50.6	88.6	80.9	73.3	15.17	83%
	ShortGPT (layer drop)	31.5	28.9	35.6	24.3	26.6	51.0	49.0	35.3	433.69	6%
	Magnitude	79.9	65.1	73.1	69.9	53.6	87.3	81.7	72.9	14.51	83%
	Random	71.4	61.2	70.1	61.9	43.5	80.1	62.3	64.4	16.34	65%
<i>Qwen2.5-VL-7B, $\rho=30\%$, 1000 items</i>											
	Dense	86.7	70.5	88.5	84.5	63.8	89.6	87.3	81.6	20.53	
	CoCurve	68.1	59.7	59.5	55.3	41.5	90.7	70.1	63.6	19.88	64%
	block-diagonal H	68.3	59.6	59.6	55.3	40.7	90.8	70.9	63.6	19.84	64%
	diagonal only ($\lambda=0$)	67.7	55.8	55.5	52.6	40.6	84.9	70.5	61.1	17.48	59%
	LLM-Pruner	61.5	43.4	59.9	43.6	37.2	82.3	72.2	57.2	18.12	51%
	Wanda-sp	63.4	44.3	54.0	44.6	33.3	71.3	54.2	52.2	19.21	41%
	FLAP	65.6	50.5	58.2	45.1	35.7	83.2	71.5	58.5	15.91	53%
	ShortGPT (layer drop)	32.5	29.5	35.7	23.0	25.6	51.0	49.1	35.2	3811.02	6%
	Magnitude	65.2	54.9	56.0	39.5	38.4	84.4	72.2	58.7	17.94	54%
	Random	58.7	44.3	46.1	33.8	37.6	52.1	50.7	46.2	24.24	28%
<i>Qwen2.5-VL-7B, $\rho=40\%$, 1000 items</i>											
	Dense	86.7	70.5	88.5	84.5	63.8	89.6	87.3	81.6	20.53	
	CoCurve	45.4	35.9	41.9	30.7	27.7	72.7	53.4	44.0	26.68	24%
	block-diagonal H	45.4	35.9	41.9	30.7	27.7	72.7	53.4	44.0	26.68	24%
	diagonal only ($\lambda=0$)	50.5	39.2	43.7	34.1	31.7	69.9	59.5	46.9	29.32	30%
	LLM-Pruner	45.3	35.4	38.2	32.2	29.6	61.0	60.5	43.2	24.86	22%
	Wanda-sp	30.8	26.5	36.8	25.6	27.3	48.7	49.7	35.1	38.08	6%
	FLAP	38.4	30.3	36.2	26.0	25.2	75.3	59.9	41.6	25.25	19%
	ShortGPT (layer drop)	32.7	28.7	36.2	24.3	25.6	49.3	50.4	35.3	1131.56	6%
	Magnitude	38.7	34.1	39.2	29.2	27.8	66.5	56.1	41.7	27.43	19%
	Random	38.1	30.3	37.1	23.2	26.0	48.2	50.3	36.2	32.18	8%
<i>Qwen3-VL-8B, $\rho=20\%$, 1000 items</i>											
	Dense	89.7	69.4	94.4	84.8	66.9	90.4	88.4	83.4	19.41	
	CoCurve	81.1	61.8	76.3	69.2	52.4	88.7	68.0	71.1	18.10	76%
	block-diagonal H	80.4	61.7	75.9	68.9	52.0	88.5	67.8	70.7	18.10	75%
	diagonal only ($\lambda=0$)	72.8	59.1	68.4	59.7	45.4	88.7	71.6	66.5	13.43	67%
	LLM-Pruner	74.3	63.1	74.5	60.4	46.5	81.7	60.4	65.8	19.12	66%
	Wanda-sp	38.3	33.8	46.0	33.7	27.6	52.2	51.8	40.5	26.49	16%
	FLAP	74.9	61.4	69.8	62.4	47.6	88.5	74.5	68.4	15.54	71%
	ShortGPT (layer drop)	30.1	26.1	38.1	26.2	26.0	48.8	51.1	35.2	9093.50	6%
	Magnitude	68.4	57.3	62.6	57.9	43.3	58.0	51.2	57.0	15.82	48%
	Random	59.7	44.5	48.2	34.8	35.6	85.0	68.7	53.8	19.26	42%
<i>Qwen3-VL-8B, $\rho=30\%$, 1000 items</i>											
	Dense	89.7	69.4	94.4	84.8	66.9	90.4	88.4	83.4	19.41	
	CoCurve	62.6	53.2	54.2	47.3	39.3	83.9	63.7	57.7	25.40	50%
	block-diagonal H	63.0	52.8	54.2	47.7	38.4	83.7	63.6	57.6	25.40	50%
	diagonal only ($\lambda=0$)	49.2	43.7	46.5	33.1	33.7	83.3	61.2	50.1	25.43	35%
	LLM-Pruner	43.4	47.1	39.5	25.0	29.5	85.4	59.0	47.0	30.17	29%
	Wanda-sp	34.5	29.6	36.1	23.8	27.5	49.0	50.8	35.9	48.60	7%
	FLAP	42.2	41.3	42.4	28.8	31.4	82.5	57.0	46.5	24.95	28%
	ShortGPT (layer drop)	33.2	27.4	34.8	23.1	28.1	49.5	50.8	35.3	41895.50	6%
	Magnitude	34.9	29.8	35.0	24.5	27.5	59.6	52.4	37.7	55.45	11%
	Random	41.6	36.5	36.9	28.3	30.4	63.0	51.9	41.2	35.15	18%
<i>Qwen3-VL-8B, $\rho=40\%$, 1000 items</i>											
	Dense	89.7	69.4	94.4	84.8	66.9	90.4	88.4	83.4	19.41	
	CoCurve	34.5	30.0	34.9	23.5	27.4	52.1	51.9	36.3	179.42	8%
	block-diagonal H	34.5	30.0	34.9	23.5	27.4	52.1	51.9	36.3	179.42	8%
	diagonal only ($\lambda=0$)	34.7	29.5	34.9	23.5	27.1	50.4	52.7	36.1	238.44	8%
	LLM-Pruner	34.6	29.8	36.1	23.3	26.8	49.0	51.9	35.9	166.53	7%
	Wanda-sp	34.3	29.6	34.9	24.0	27.4	48.2	48.7	35.3	1035.76	6%
	FLAP	34.2	29.8	35.5	23.5	27.2	64.0	51.3	37.9	66.41	11%
	ShortGPT (layer drop)	29.9	26.2	36.5	22.7	24.8	49.3	50.4	34.3	147864.18	4%
	Magnitude	34.1	30.4	36.0	23.1	26.7	50.5	50.8	35.9	787.21	7%
	Random	33.9	29.7	35.9	25.1	27.2	52.3	50.9	36.4	98.81	8%

InternVL3-8B, $\rho=20\%$, 1000 items

Model	Selector	MMB	SEED	SQA	AI2D	MMStar	POPE	MME	Mean	cap. PPL	Ret.
	Dense	88.2	67.4	97.5	83.7	67.1	93.3	88.1	83.6	14.05	
	CoCurve	83.7	64.2	82.7	72.3	53.8	92.5	82.3	75.9	14.04	85%
	block-diagonal H	83.7	64.2	82.7	72.3	53.8	92.5	82.3	75.9	14.04	85%
	diagonal only ($\lambda=0$)	81.0	62.0	77.9	71.0	53.3	91.0	80.8	73.9	15.04	81%
	LLM-Pruner	78.4	61.8	78.7	68.4	52.2	84.0	79.0	71.8	14.53	77%
	Wanda-sp	50.0	36.3	62.6	47.0	31.9	65.8	55.1	49.8	19.17	34%
	FLAP	80.7	62.1	84.8	73.8	52.6	88.3	81.2	74.8	14.72	83%
	ShortGPT (layer drop)	33.8	32.2	61.7	51.5	24.1	51.0	49.2	43.4	20.33	22%
	Magnitude	79.8	57.7	74.6	64.1	48.3	91.1	79.3	70.7	13.46	75%
	Random	72.9	61.0	74.3	62.9	49.7	87.0	67.6	67.9	14.68	69%
<i>InternVL3-8B, $\rho=30\%$, 1000 items</i>											
	Dense	88.2	67.4	97.5	83.7	67.1	93.3	88.1	83.6	14.05	
	CoCurve	67.7	54.0	59.6	52.7	41.6	89.0	71.9	62.4	16.64	59%
	block-diagonal H	67.7	54.0	59.6	52.7	41.6	89.0	71.9	62.4	16.64	59%
	diagonal only ($\lambda=0$)	62.2	49.1	54.0	50.3	36.0	80.0	56.9	55.5	17.30	45%
	LLM-Pruner	65.3	53.2	61.4	52.5	40.4	84.8	64.7	60.3	19.31	55%
	Wanda-sp	49.2	36.3	51.5	34.8	28.6	56.5	58.3	45.0	21.74	25%
	FLAP	74.5	56.1	65.0	54.7	42.1	83.6	53.1	61.3	18.27	57%
	ShortGPT (layer drop)	31.8	29.5	37.8	26.3	25.7	49.7	48.9	35.7	26.04	7%
	Magnitude	47.5	42.2	46.0	32.1	30.2	65.8	52.3	45.2	17.41	25%
	Random	58.9	45.0	48.7	38.6	34.9	62.5	57.9	49.5	19.48	34%
<i>InternVL3-8B, $\rho=40\%$, 1000 items</i>											
	Dense	88.2	67.4	97.5	83.7	67.1	93.3	88.1	83.6	14.05	
	CoCurve	38.3	32.8	37.8	27.2	30.7	51.0	49.4	38.2	29.28	12%
	block-diagonal H	38.3	32.8	37.8	27.2	30.7	51.0	49.4	38.2	29.28	12%
	diagonal only ($\lambda=0$)	36.8	27.7	37.5	24.6	26.7	51.0	49.0	36.2	31.03	8%
	LLM-Pruner	36.1	29.4	37.9	23.7	28.5	57.5	49.2	37.5	66.94	10%
	Wanda-sp	35.0	28.5	34.0	24.9	28.8	51.1	49.3	35.9	56.20	7%
	FLAP	34.6	31.8	35.3	25.1	29.2	51.5	48.0	36.5	101.33	8%
	ShortGPT (layer drop)	31.3	28.3	35.4	25.8	26.6	51.1	50.3	35.5	70.42	7%
	Magnitude	35.7	30.0	36.2	23.1	27.0	52.9	52.2	36.7	33.12	9%
	Random	37.6	30.0	37.8	26.5	29.7	55.6	52.6	38.5	47.66	12%
<i>LLaVA-NeXT-7B, $\rho=20\%$, 1000 items</i>											
	Dense	76.5	64.5	73.5	68.0	38.3	90.2	79.0	70.0	16.70	
	CoCurve	70.5	57.0	61.7	54.7	36.8	90.8	70.9	63.2	15.96	82%
	block-diagonal H	70.5	57.0	61.7	54.7	36.8	90.8	70.9	63.2	15.96	82%
	diagonal only ($\lambda=0$)	70.5	57.0	61.7	54.7	36.8	90.8	70.9	63.2	15.96	82%
	LLM-Pruner	71.8	60.0	61.2	55.3	36.8	85.5	73.5	63.4	17.46	83%
	Wanda-sp	41.5	36.1	37.2	27.8	25.4	57.6	55.2	40.1	23.13	21%
	FLAP	70.9	61.6	61.5	54.0	38.0	89.2	71.9	63.9	17.89	84%
	ShortGPT (layer drop)	34.3	29.8	35.5	23.6	27.1	51.0	49.2	35.8	38085.10	10%
	Magnitude	67.8	58.4	62.4	51.7	36.3	91.2	72.9	63.0	18.97	81%
	Random	66.5	53.5	59.6	48.7	32.8	88.9	74.4	60.6	18.49	75%
<i>LLaVA-NeXT-7B, $\rho=30\%$, 1000 items</i>											
	Dense	76.5	64.5	73.5	68.0	38.3	90.2	79.0	70.0	16.70	
	CoCurve	56.3	45.5	44.5	36.1	33.0	89.6	66.2	53.0	18.71	55%
	block-diagonal H	56.3	45.5	44.5	36.1	33.0	89.6	66.2	53.0	18.71	55%
	diagonal only ($\lambda=0$)	49.3	41.8	42.4	36.7	31.7	88.8	64.4	50.7	17.73	49%
	LLM-Pruner	65.7	56.1	53.6	47.1	34.0	88.5	56.9	57.4	21.59	67%
	Wanda-sp	36.3	29.7	35.5	25.1	25.9	49.0	50.6	36.0	24.74	10%
	FLAP	39.4	35.3	43.2	30.1	27.3	66.5	61.5	43.3	25.22	30%
	ShortGPT (layer drop)	36.4	30.3	35.9	28.2	25.6	59.6	54.6	38.7	27.54	17%
	Magnitude	57.1	46.8	45.7	35.7	32.1	88.2	62.9	52.6	26.41	54%
	Random	62.2	49.0	53.9	42.7	30.0	87.9	62.5	55.5	21.38	62%
<i>LLaVA-NeXT-7B, $\rho=40\%$, 1000 items</i>											
	Dense	76.5	64.5	73.5	68.0	38.3	90.2	79.0	70.0	16.70	
	CoCurve	34.5	29.8	35.5	23.9	26.5	80.6	53.3	40.6	34.34	22%
	block-diagonal H	40.2	32.2	38.6	23.4	27.4	79.4	51.5	41.8	38.06	26%
	diagonal only ($\lambda=0$)	34.3	30.0	35.4	23.6	27.4	51.4	49.2	35.9	37.54	10%
	LLM-Pruner	34.4	30.7	35.2	23.6	27.1	51.3	49.2	35.9	57.16	10%
	Wanda-sp	33.7	29.3	37.2	24.4	26.2	57.8	51.9	37.2	98.60	13%
	FLAP	34.2	29.8	35.3	24.3	26.8	47.5	47.4	35.0	77.18	8%
	ShortGPT (layer drop)	34.4	29.8	35.4	23.5	26.9	52.8	48.4	35.9	67.88	10%
	Magnitude	34.2	29.0	35.5	23.8	26.7	52.3	49.1	35.8	91.65	10%
	Random	38.3	33.2	38.3	24.7	26.8	80.9	66.2	44.1	34.20	31%
<i>LLaVA-1.5-7B, $\rho=20\%$, 1000 items</i>											
	Dense	73.2	56.8	64.6	52.7	36.1	88.5	77.1	64.1	11.86	
	CoCurve	68.1	51.0	53.9	38.9	32.4	85.8	69.5	57.1	15.06	78%
	block-diagonal H	68.0	51.4	54.4	40.5	31.4	81.6	52.7	54.3	15.13	69%
	diagonal only ($\lambda=0$)	56.0	39.6	46.5	33.1	31.2	82.6	61.1	50.0	12.42	56%
	LLM-Pruner	63.7	49.3	54.5	41.5	32.5	80.0	54.4	53.7	13.75	67%
	Wanda-sp	34.2	29.4	36.2	25.7	26.0	51.0	49.2	36.0	17.86	12%
	FLAP	48.2	31.4	43.2	29.6	31.3	55.1	49.2	41.1	14.63	28%
	ShortGPT (layer drop)	32.7	28.7	35.0	23.2	22.9	51.2	48.6	34.6	324.80	8%
	Magnitude	55.9	39.9	42.5	29.0	29.4	81.0	58.9	48.1	18.52	50%

Model	Selector	MMB	SEED	SQA	AI2D	MMStar	POPE	MME	Mean	cap. PPL	Ret.
	Random	62.2	48.2	57.2	36.9	31.5	83.0	50.4	52.8	16.05	64%
<i>LLaVA-1.5-7B, $\rho=30\%$, 1000 items</i>											
	Dense	73.2	56.8	64.6	52.7	36.1	88.5	77.1	64.1	11.86	
	CoCurve	46.7	32.2	44.5	28.4	29.1	72.1	51.9	43.6	26.92	36%
	block-diagonal H	35.1	28.2	37.6	24.5	28.0	69.0	57.1	39.9	27.21	24%
	diagonal only ($\lambda=0$)	34.8	29.9	35.2	26.6	28.0	51.0	49.4	36.4	31.38	13%
	LLM-Pruner	57.7	45.6	46.4	35.2	27.6	68.7	53.0	47.7	25.88	49%
	Wanda-sp	34.4	30.5	36.3	24.7	26.7	51.0	49.2	36.1	30.22	12%
	FLAP	33.8	30.0	34.9	23.9	27.1	51.0	49.1	35.7	36.03	11%
	ShortGPT (layer drop)	35.4	30.6	35.7	23.9	27.2	51.0	49.2	36.1	48.41	12%
	Magnitude	34.4	29.8	35.4	23.6	27.2	51.0	49.2	35.8	232.81	11%
	Random	43.3	31.6	42.9	28.0	26.5	51.0	49.2	38.9	25.08	21%
<i>LLaVA-1.5-7B, $\rho=40\%$, 1000 items</i>											
	Dense	73.2	56.8	64.6	52.7	36.1	88.5	77.1	64.1	11.86	
	CoCurve	34.4	29.8	35.5	23.6	27.2	51.0	49.2	35.8	2303.06	11%
	block-diagonal H	34.4	29.8	35.5	23.6	27.2	51.0	49.2	35.8	2302.96	11%
	diagonal only ($\lambda=0$)	34.4	29.8	35.5	23.6	27.2	51.0	49.2	35.8	2404.90	11%
	LLM-Pruner	33.0	30.0	35.2	23.6	26.8	51.0	49.2	35.5	237.95	11%
	Wanda-sp	32.8	29.2	35.7	24.3	25.0	51.1	49.4	35.4	471.24	10%
	FLAP	34.0	29.7	35.5	24.2	27.1	48.3	48.8	35.4	72.08	10%
	ShortGPT (layer drop)	34.3	29.8	35.8	23.4	27.4	51.0	49.2	35.8	172.17	12%
	Magnitude	34.4	29.8	35.5	23.7	27.2	51.0	49.2	35.8	3057.86	12%
	Random	34.5	29.6	35.9	23.4	27.0	51.1	49.2	35.8	513.86	11%

E.10 VISION–LANGUAGE MODELS: WHAT A RECOVERY STAGE CHANGES

Every number in Section 3.6 is selection only: a mask is chosen, installed, and scored, and nothing is trained. This subsection reports what happens when one identical recovery stage is added to every arm, because it is the comparison a reader of the pruning literature expects and because what a recovery budget does to a pruned model is worth reporting in its own right.

The stage. LoRA (Hu et al., 2022) of rank 128 and $\alpha=256$ on the language tower, the vision encoder frozen and the multimodal projector trained full-rank at 2×10^{-5} , which is the arrangement LLaVA’s own recovery script uses. The learning rate is LLaVA’s 2×10^{-4} scaled linearly to this run’s global batch of 16, giving 2.5×10^{-5} . The corpus is 3000 image–instruction samples, half long-form conversation from LLaVA’s own instruction data (Liu et al., 2023) (COCO *train* and Visual Genome images) and half short-answer VQA from TextVQA *train* (Singh et al., 2019) (OpenImages), carrying the same instruction suffixes the benchmarks are scored under. Both halves are training splits, and no evaluation image of the seven suites appears in either. The mask stays installed throughout, and every arm of a cell gets this recipe unchanged — the only thing that differs between two rows is which units were removed. Each run re-scores its own pruned model before training and prints the comparison against the selection-only row it will be compared with, so a recovered number and the number above it are the same harness, not two.

What it does. On three of the four cells recovery lifts every arm — by 6.2 to 21.7 points of seven-benchmark mean — and pulls them together: the spread across the four arms falls from 6.4 to 2.2 points on Qwen2.5-VL-7B at $\rho=30\%$, from 2.4 to 0.5 at $\rho=40\%$, and from 20.1 to 11.2 on Qwen3-VL-8B at $\rho=30\%$. CoCurve finishes first in two of those three and second by 1.2 points in the other; two runs of one configuration on a contended card differ by 3.2 points, so that second place is a tie rather than a loss.

The fourth cell behaves differently and is the more informative one. At $\rho=40\%$ on Qwen3-VL-8B the four masks start within 2.0 points of each other and end 24.2 apart: LLM-Pruner’s gains 23.9, ours 15.8, and Magnitude’s gains nothing at all (−0.3, its held-out perplexity still above 90). Recovery is not a fixed increment added to whatever selection produced. It is a second optimisation whose reachable endpoint depends on what the mask left behind, and at a budget where no arm retains a quarter of its dense score above chance before training, that dependence is what decides the ordering — not the selection-only scores, which are indistinguishable there.

Two things follow. Where a recovery budget is spent, the ranking it produces is its own: it agrees with the selection-only ranking on the model that still holds most of its dense score and departs from it on the one that does not. Selection and recovery are therefore optimising different things at the highest budget, and mixing them would report neither cleanly — which is why the body measures selection alone, the quantity this paper’s estimator predicts, and why this table is reported separately

rather than folded into Table 6. The second reading is the practical one: with recovery, $\rho=40\%$ becomes usable at all. On Qwen3-VL-8B every recovered arm but Magnitude ends above 46.99, the best selection-only score any baseline reaches at $\rho=30\%$, and the strongest recovered 40% model — LLM-Pruner’s, at 59.8 — edges past the strongest unrecovered 30% one, ours at 57.7.

Table 28: One identical LoRA recovery stage on every arm, 1000 items per benchmark. *sel. only* repeats the selection-only row; +LoRA is the same model after the stage.

Selector	Qwen2.5-VL-7B, $\rho=30\%$		Qwen2.5-VL-7B, $\rho=40\%$		Qwen3-VL-8B, $\rho=30\%$		Qwen3-VL-8B, $\rho=40\%$	
	sel. only	+LoRA	sel. only	+LoRA	sel. only	+LoRA	sel. only	+LoRA
CoCurve	63.6	69.7	44.0	61.9	57.7	69.5	36.3	52.1
LLM-Pruner	57.2	70.9	43.2	61.4	47.0	68.6	35.9	59.8
FLAP	58.5	68.7	41.6	61.5	46.5	63.9	37.9	54.5
Magnitude	58.7	69.2	41.7	61.7	37.7	58.3	35.9	35.6

F DISCUSSION

F.1 EXTENDED RELATED WORK

Section 5 names the three closest structured methods; the remaining near neighbours differ from CoCurve on a defining property of Table 9. SOSP (Nonnenmacher et al., 2022) is the closest prior work on the interaction axis and deserves a precise statement. Its SOSP-I variant does build a full pairwise structure–structure matrix and selects individual structures with it; its scalable SOSP-H variant reduces this to one scalar per structure and, in the authors’ words, cannot account for individual pairwise correlations. Both require a backward pass — Jacobians for SOSP-I, a Hessian-vector product for SOSP-H — both linearise a *supervised training loss* rather than a label-free divergence to the dense model, both take structures of one kind (filters, neurons, whole layers) rather than heterogeneous attention and FFN units, and both fine-tune after pruning. Its Appendix C also derives, for interpretation, an identity of the same shape as our Eq. 5: a per-structure output feature contracted through the softmax Fisher. That identity is obtained from the positive homogeneity of ReLU networks and is stated for networks *without* normalisation layers, so it does not hold for a Transformer, and it is not the route SOSP computes — the algorithm evaluates $\nabla_{\theta} f_{\theta}$. Our construction instead measures δz_u by finite-difference ablation, which assumes nothing about the activation and needs no gradient; that is what makes the same shape available at 24B. The Combinatorial Brain Surgeon (Yu et al., 2022) is the honest ancestor of the argument in Section 1: it observes that the damage from pruning one weight depends on which others go, and asks for removals whose effects cancel. It does so between individual *weights*, from backpropagated gradients, and for large models falls back to the same layer-wise diagonal chunks as WoodFisher (Singh & Alistarh, 2020), so no cross-module or cross-layer term survives; its full form also updates the retained weights. What CoCurve contributes is not the cancellation argument but a form of it that is computable at structured-unit granularity across module types, from forward passes alone, with nothing retrained. Two data points from that literature explain why the granularity matters: Frantar et al. (2021) report that for *unstructured* pruning the effect of correlation is small and the gains come mostly from the weight update, and Yu et al. (2022) need a mixed-integer solve to exploit it at all. At structured-unit granularity, with weight updates forbidden, the interaction is the only lever left — and Section 3.3 is what it is worth. Optimal Brain Connection (Chen et al., 2025), despite the name, assumes $J^{\top} J$ is block diagonal so that only parameters *inside* one filter or channel interact — the cross-unit terms we keep are the ones it drops — and it needs Jacobians, fine-tuning and extra autoencoder parameters. It should not be confused with Optimal Brain Compression (Frantar & Alistarh, 2022). On the attention/FFN axis, AMP (Mugnaini et al., 2025) co-prunes both module types yet scores each unit independently and LoRA-recovers, while 2SSP (Sandri et al., 2025) prunes FFN width then attention in two disjoint stages. The concurrent D²Prune (Xiong et al., 2026) shares the “dual/Taylor” name but discards the off-diagonal cross terms and is unstructured.

Beyond Section 5, we note three further connections. (i) Unstructured/semi-structured pruning (SparseGPT (Frantar & Alistarh, 2023), Wanda (Sun et al., 2024), OBC (Frantar & Alistarh, 2022)) optimises per-layer reconstruction at the weight level; CoCurve instead optimises a global, output-distribution objective over structured units, and is hardware-friendly, requiring no sparse kernels.

(ii) Depth/width search methods (DarwinLM (Tang et al., 2025), CFSP (Wang et al., 2024b)) explore architectures by repeated evaluation; CoCurve is single-shot with a closed-form surrogate. Bonsai (Kolawole et al., 2024) belongs with neither group and is the closest work to ours on the estimation axis: it too scores structured units from forward passes only, with no gradients and no weight update, and it too reasons globally across layers rather than layer by layer. What it fits to those passes is an *additive* linear model of the mask, one coefficient per module, so no pairwise term exists to tell it that two units are costly together; and recovering usable coefficients takes on the order of a thousand perturbation evaluations plus several rounds in which the priors are re-estimated. Our M ablations are one per unit, and what they assemble is the pairwise matrix itself. (iii) Recovery-based pipelines (Minitron (Muralidharan et al., 2024)) restore ability via distillation; they are complementary and orthogonal to the no-recovery regime we study, and could be stacked on top of CoCurve.

Rotation/low-rank residual pruning (SliceGPT). SliceGPT (Ashkboos et al., 2024) compresses by rotating the residual stream into a variance-ordered basis (a computational-invariance transform) and slicing its least-significant dimensions, a fundamentally different mechanism from removing structured units. It is the canonical member of the rotation family and complements our unit-removal baselines. Two of its three headline models improve with recovery fine-tuning, which the selection-only protocol here sets aside, and a faithful no-recovery form of it requires the official per-layer rotation with residual adapters; the single global rotation is a substantially weaker variant, and a number produced from it would understate the method. We therefore position SliceGPT by mechanism rather than by a number its authors would not recognise, and note that its residual-rotation idea is orthogonal to cross-module co-pruning curvature and could in principle be composed with it.

Curvature-based structured pruning (LLM Surgeon). The closest method in spirit is LLM Surgeon (van der Ouderaa et al., 2024), which also uses second-order (Kronecker-factored Fisher) curvature for structured pruning. Two differences are decisive under our regime. (a) Locality: its Fisher factors model correlations among the weights inside one layer, whereas $\mathbf{H}$ here is an interaction between structured units anywhere in the network. That difference is measured rather than assumed: the attention $\leftrightarrow$ FFN block is 0.74–0.98 times the mean of the two within-module blocks on all five models, and coupling persists far from the diagonal band (Figure 3b–d), so a within-layer factorisation discards mass of the same order as the mass it keeps. Table 3 then measures what the off-diagonal is worth: the $\lambda=0$ arm loses on ten models, and the fixed-allocation control shows the gain is not a re-split of the budget between module types. (b) Recovery: its defining component is a closed-form correlated weight update that repairs the remaining weights after each removal, a form of weight repair that our selection-only protocol disables for every method (as we do for SlimGPT’s OBS update). Its repair step is orthogonal to our selection objective and could be stacked on top of CoCurve’s selection, but it is the method, so we do not report a version of it with the method removed. What its absence leaves untested is whether modelling interactions at all is what helps, and that is tested twice without it: by the $\lambda=0$ arm of Table 3, which loses on eight of ten models and is returned unchanged on the other two, and by the block-mean arm of Table 21, which keeps the cross-component block but collapses each block to its scalar mean — the interaction granularity of SAViT — so that the interaction decides how much of each component type to remove and never which unit. That arm, given its own calibrated $\hat{\lambda}$, loses six of seven columns and all four capability groups to the full matrix, so the benefit is not reproduced by a component-level budget.

Cross-structure joint importance in vision (SAViT). The closest precedent in spirit is SAViT (Zheng et al., 2022), which first argued that pruning should account for interactions among heterogeneous structural components rather than scoring them independently, realized through a collaborative Taylor importance over ViT heads/channels/embeddings. Three differences are concrete and empirically anchored. (a) Domain/objective: SAViT targets vision Transformers with a task-supervised Taylor importance; CoCurve derives its edges from a label-free token-level KL self-distillation on text. (b) Training: SAViT realizes its pruning ratios through collaborative optimisation or fine-tuning, whereas CoCurve is strictly one-shot and selection-only, under the no-recovery protocol of Section 3.1. (c) Mechanism: SAViT’s interaction is a global importance budget over components; CoCurve forms the explicit attention $\leftrightarrow$ FFN off-diagonal Fisher block and uses it directly as co-pruning saliency. Empirically, the value of that block over a no-interaction baseline is

exactly the diagonal-only ($\lambda=0$) row of Table 3: CoCurve recovers the cross-structure benefit SAViT motivates, but with no optimisation or fine-tuning.

Concurrent “dual Taylor” sparsification (D²Prune). The concurrent D²Prune (Xiong et al., 2026) shares the “Taylor” vocabulary but is mechanistically orthogonal, and the distinction lands on a row already in our tables. Its “dual” expansion is over weights and activations for unstructured and $N:M$ reconstruction, and it discards the weight–activation cross block as higher-order negligible, keeping an OBS saliency built on a dense inverse Hessian with an attention-distribution KL regulariser. The term it drops is on a different axis from ours, and the two are best read as sharing a vocabulary rather than an object: the gap between the diagonal-only ($\lambda=0$) selector and full CoCurve in Table 3 quantifies, at structured granularity under a global token-KL objective, exactly what discarding the cross term forgoes. Moreover, Proposition 5 shows this omission is a second-order mass scaled by the mean off-diagonal correlation, not the higher-order remainder D²Prune assumes it to be.

F.2 LIMITATIONS

Selection cost. Estimating $\mathbf{H}$ costs hours rather than the minutes an activation-norm criterion needs. It is a one-off: the matrix depends on the model and the calibration set, not on the compression ratio or on λ , so a single estimate serves every ratio and the entire regularisation path (Section D.4.1).

Selection only. This paper studies which units to remove. Weight reconstruction and fine-tuning are orthogonal, and we hold them fixed, disabled for every method, so that the comparison isolates the selection. Section D.3 reports what happens when they are enabled.

Operator coverage. Turning a mask into a smaller model requires a slicing path per attention and FFN operator. We provide one for the architectures studied here and check every sliced model by its realised parameter count (Section A.2); a new operator family needs a new path.

Dense attention stacks. We target dense Transformers whose layers all have attention heads and an FFN, which is what makes a removable unit well defined. Two directions the field is moving in break that assumption rather than the method. Under mixture-of-experts routing the units are already sparsely activated, so what a removal costs depends on the router; and architectures that interleave full attention with linear-recurrent or state-space layers raise the question one level earlier, since a layer without heads has no unit of the kind $\mathcal{U}$ collects. Co-pruning curvature is defined by Eq. 1 in both cases — the risk is a property of the masked model, not of its layer types — but the unit inventory it is minimised over is not, and we do not propose one here.

What a pruned model is for. At the ratios studied here, training-free structured pruning does not yield a model that competes with a smaller *pretrained* checkpoint on aggregate accuracy — not for CoCurve and not for any method compared here. That is a property of the regime rather than of any criterion, and it is why this literature measures pruners against pruners at a matched budget. Our objective states the intended use directly: the risk minimised is KL to *this* parent, so what the procedure delivers is a cheaper model that reproduces a particular parent’s behaviour. A different checkpoint cannot do that by construction, which is what makes the setting worth studying whenever the parent has been aligned, fine-tuned or audited, or when no pretrained checkpoint exists at the target size.

F.3 BROADER IMPACT

Structured, training-free compression lowers the compute, memory, and energy cost of deploying LLMs, broadening access and reducing inference footprint. Because CoCurve requires no labels or fine-tuning, it does not introduce new task-specific data dependencies. As with any compression, practitioners should re-evaluate safety-relevant behaviour on the compressed model, since pruning shifts capabilities unevenly across tasks.

F.4 ASSETS, LICENCES AND ETHICS

All models are public pretrained checkpoints used under their respective licences: Llama-3.1 and Llama-3.2 (Llama Community License), Mistral-Nemo and Mistral-Small (Apache-2.0), Falcon3

(Falcon LLM License), Yi-1.5 (Apache-2.0) and Qwen-2.5 (Qwen License). Evaluation datasets are used under their published terms. Baseline implementations follow the authors' released code where available and their published descriptions otherwise; Section C.3 records every deviation. No model is trained or fine-tuned in the selection-only setting, and the recovery stage of Section D.3 uses the cleaned Alpaca corpus under its stated licence.

F.5 REPRODUCIBILITY CHECKLIST

We will release: the unit registry and runtime-mask/physical-prune implementation with the $< 10^{-3}$ equivalence test; the single-unit ablation and blockwise-Gram estimator (Algorithm 1); the greedy solver with anti-collapse guards (Algorithm 3); per-model pruning solutions and configurations (Table 14); and the evaluation harness (length-normalised log-likelihood) with seeds. The method uses no training, labels, or recovery, so a pruned model is fully determined by (model, calibration seed, ratio, config).